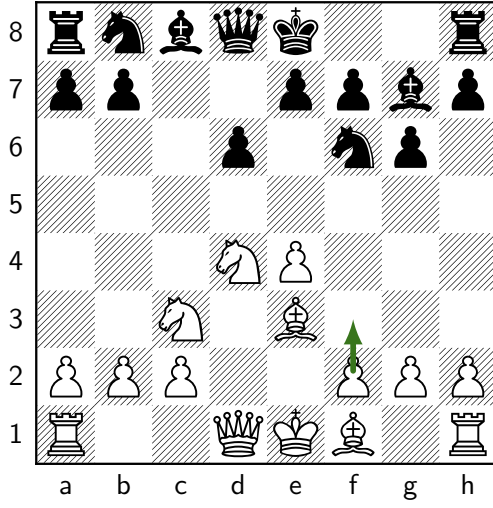


Melanie-BOTS (CCRL engines) — Sicilian, Dragon

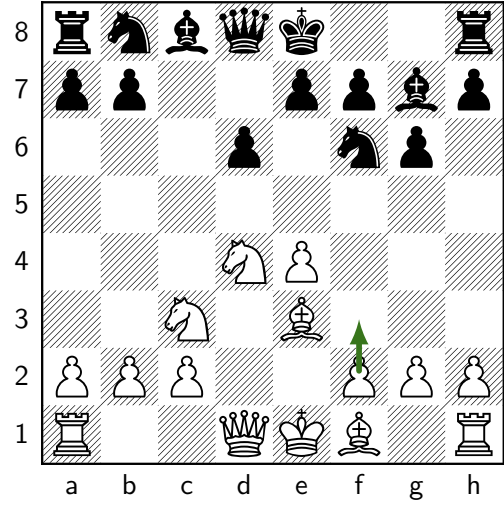
Melanie (Deep_Vision)

Abstract

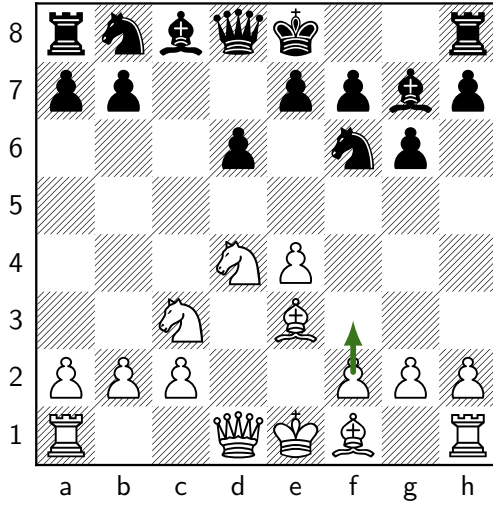
Every move below was produced by relational analogy to named master games (human and/or top-engine – the exact source is named per move). Each figure shows our position with Melanie’s move (a) and the top retrieved analog positions (b–d) with the master move played there; the caption names the exact source and the piece-role correspondence under μ – cross-type maps (a rook playing a queen’s role) are called out, as no material/eval engine explains a move that way.



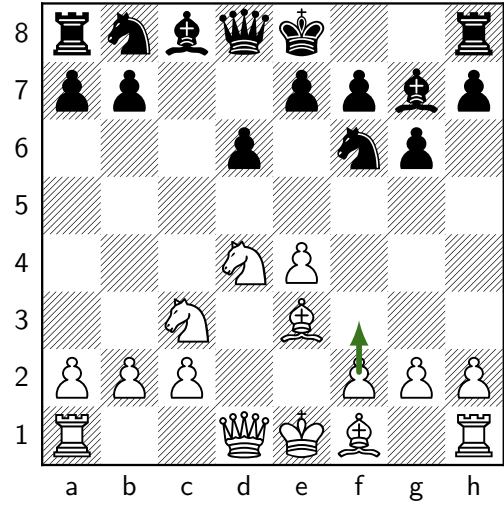
(a)



(b)

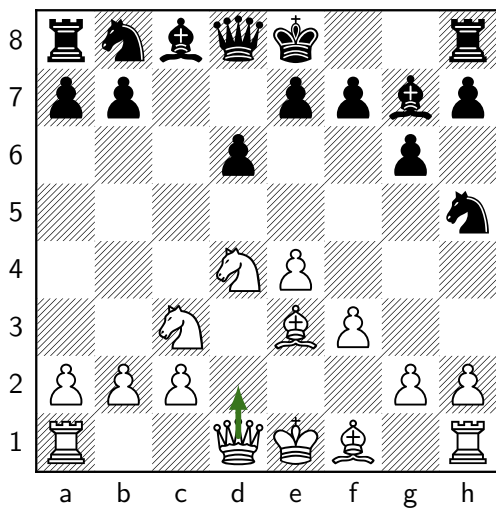


(c)

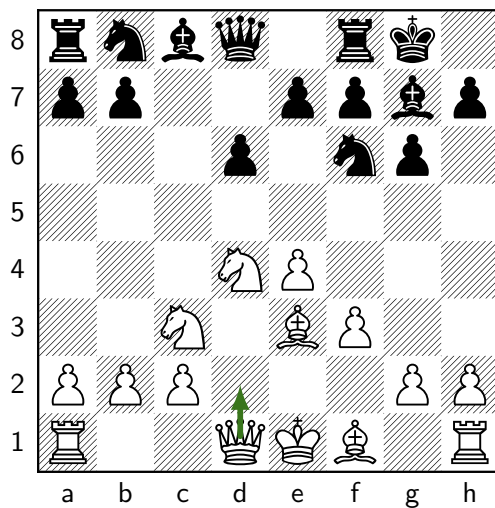


(d)

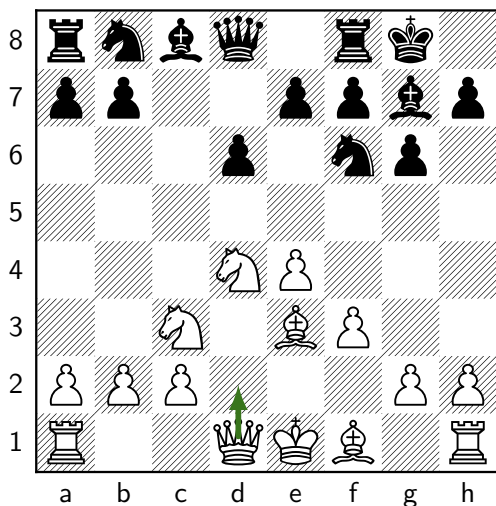
Figure 1: **Move 7, White: f3.** (a) Our position; Melanie plays f3 (f2f3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#212` (ply 12) [structural distance $d=0$, QAP obj=30]: the master played f3; under μ our *Pf2* plays the role of that game’s *Pf2*; cross-type roles: our *bc8* as their *Ra1*, our *Nc3* as their *Bf1*; transferred move f2f3 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#213` (ply 12) [structural distance $d=0$, QAP obj=30]: the master played f3; under μ our *Pf2* plays the role of that game’s *Pf2*; cross-type roles: our *bc8* as their *Ra1*, our *Nc3* as their *Bf1*; transferred move f2f3 (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#298` (ply 12) [structural distance $d=0$, QAP obj=30]: the master played f3; under μ our *Pf2* plays the role of that game’s *Pf2*; cross-type roles: our *bc8* as their *Ra1*, our *Nc3* as their *Bf1*; transferred move f2f3 (**endorses this move**). Synthesis: 12 retrieved analogs transferred to f3 (summed QAP objective 365, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



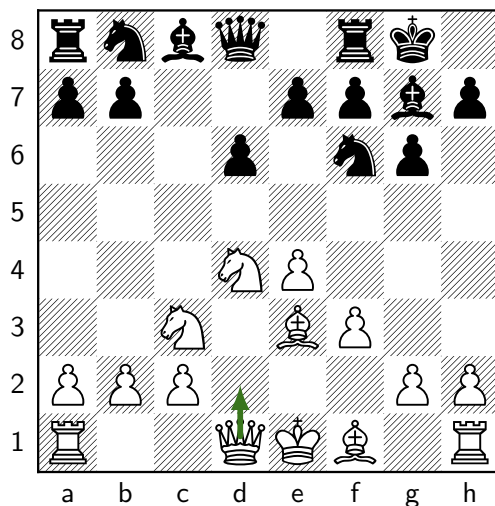
(a)



(b)

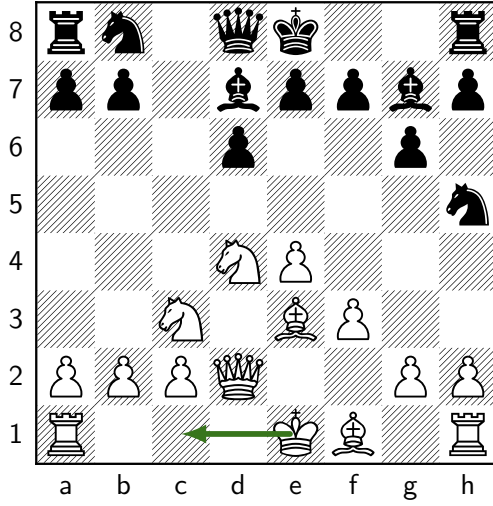


(c)

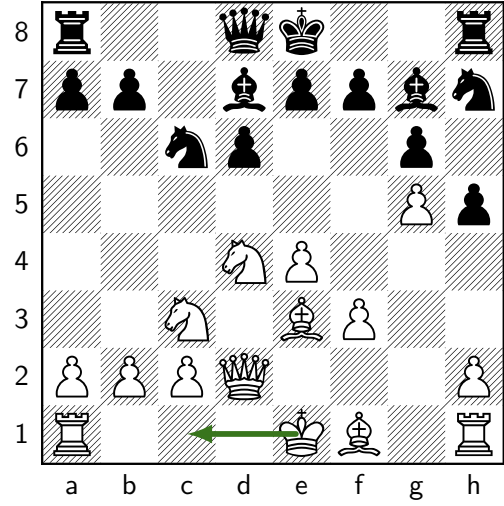


(d)

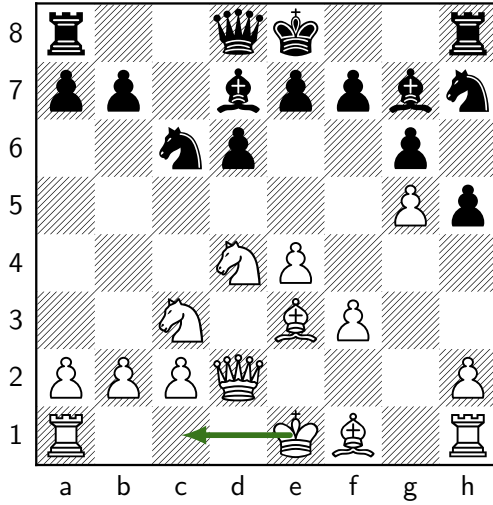
Figure 2: **Move 8, White: Qd2.** (a) Our position; Melanie plays Qd2 (d1d2). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#212` (ply 14) [structural distance $d=26$, QAP obj=16]: the master played Qd2; under μ our Qd1 plays the role of that game’s Qd1; cross-type roles: our Pb2 as their Rh1, our Ra1 as their Pa2; transferred move d1d2 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#213` (ply 14) [structural distance $d=26$, QAP obj=16]: the master played Qd2; under μ our Qd1 plays the role of that game’s Qd1; cross-type roles: our Pb2 as their Rh1, our Ra1 as their Pa2; transferred move d1d2 (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#514` (ply 14) [structural distance $d=26$, QAP obj=16]: the master played Qd2; under μ our Qd1 plays the role of that game’s Qd1; cross-type roles: our Pb2 as their Rh1, our Ra1 as their Pa2; transferred move d1d2 (**endorses this move**). Synthesis: 16 retrieved analogs transferred to Qd2 (summed QAP objective 252, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



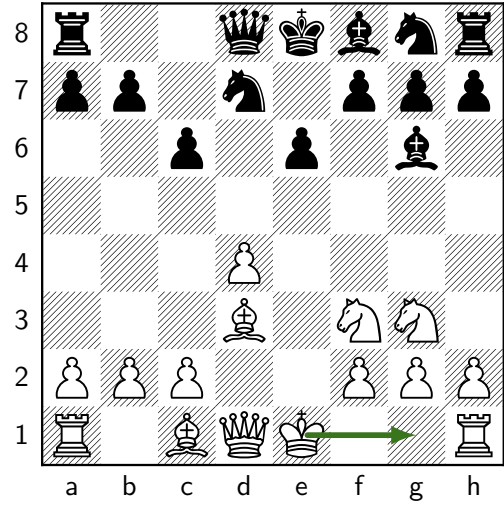
(a)



(b)

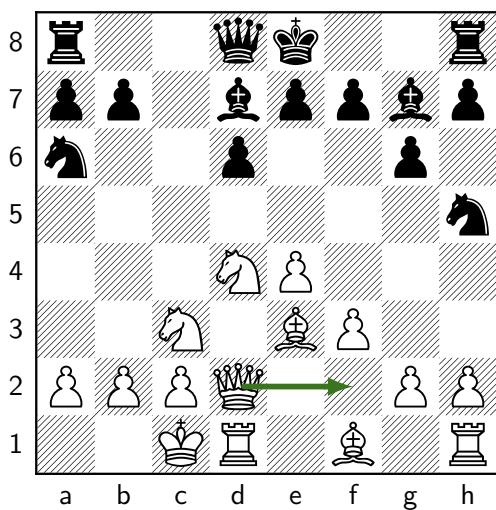


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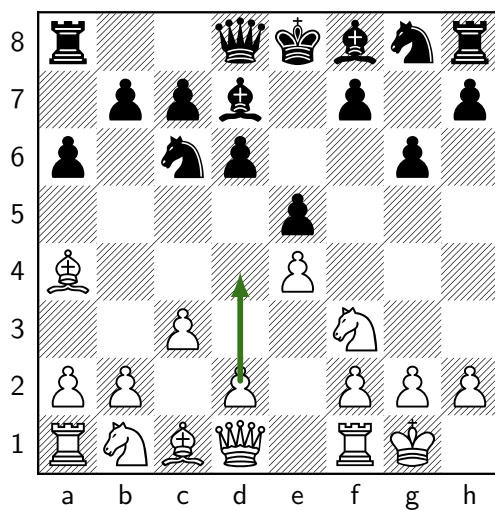


(d)

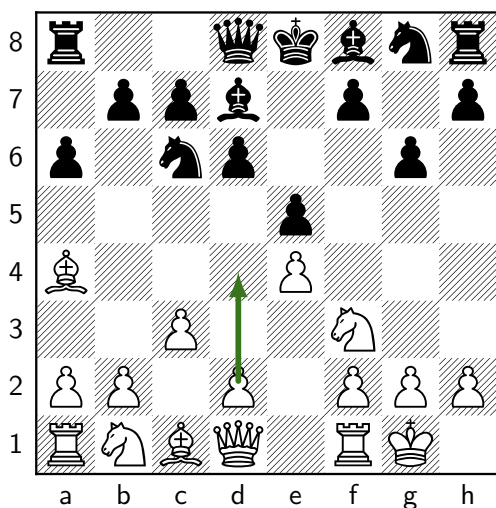
Figure 3: **Move 9, White: 0-0-0.** (a) Our position; Melanie plays 0-0-0 (e1c1). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#2545` (ply 20) [structural distance $d=53$, QAP obj=35]: the master played 0-0-0; under μ our *Ke1* plays the role of that game’s *Ke1*; cross-type roles: our *bd7* as their *Rh1*, our *Ph2* as their *Nc3*; transferred move e1c1 (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#2546` (ply 20) [structural distance $d=53$, QAP obj=35]: the master played 0-0-0; under μ our *Ke1* plays the role of that game’s *Ke1*; cross-type roles: our *bd7* as their *Rh1*, our *Ph2* as their *Nc3*; transferred move e1c1 (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#1222` (ply 14) [structural distance $d=55$, QAP obj=29]: the master played 0-0; under μ our *Ke1* plays the role of that game’s *Ke1*; cross-type roles: our *Pg2* as their *Ra1*, our *bd7* as their *Pf2*; transferred move e1c1 (**endorses this move**). Synthesis: 7 retrieved analogs transferred to 0-0-0 (summed QAP objective 208, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



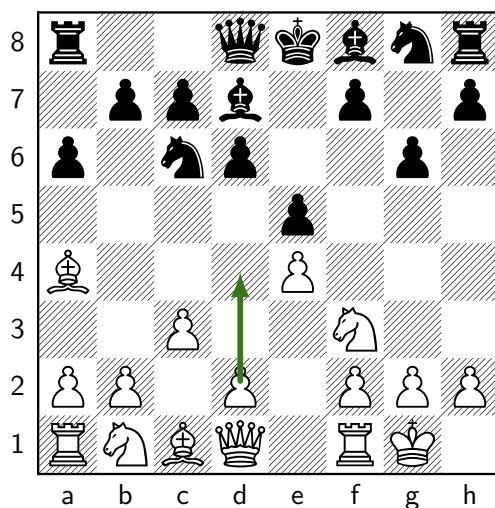
(a)



(b)

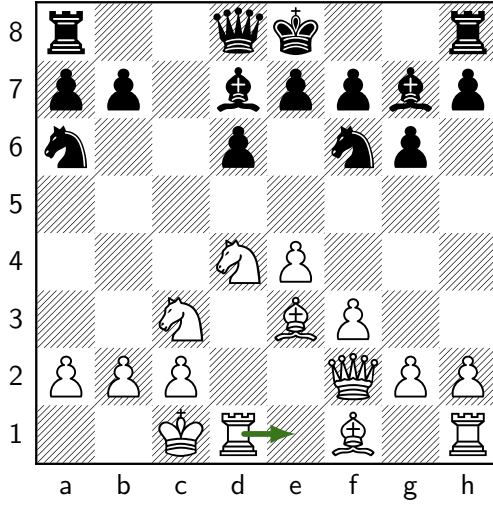


(c)

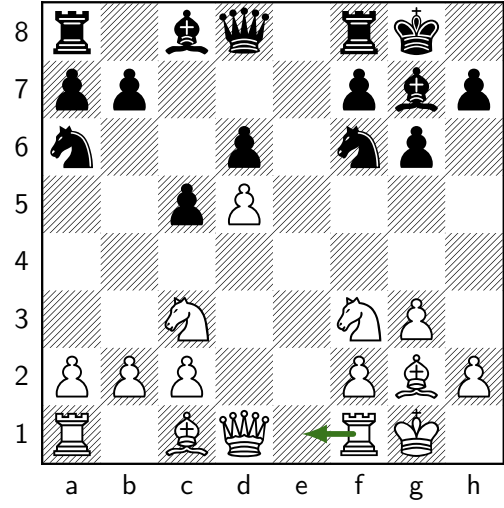


(d)

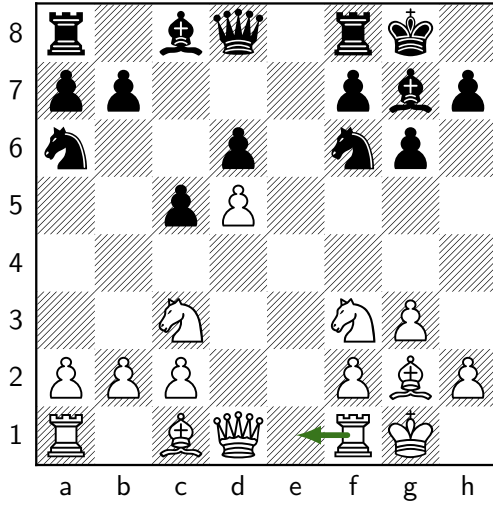
Figure 4: **Move 10, White: Qf2**. (a) Our position; Melanie plays Qf2 (d2f2). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#2289` (ply 12) [structural distance $d=68$, QAP obj=24]: the master played d4; under μ our Qd2 plays the role of that game's Pd2 (a **cross-type** correspondence); cross-type roles: our pg6 as their Ra1, our Pg2 as their Nb1; transferred move d2f2 (**endorses this move**). (c) Retrieved from `ccr1_top_p0001#2290` (ply 12) [structural distance $d=68$, QAP obj=24]: the master played d4; under μ our Qd2 plays the role of that game's Pd2 (a **cross-type** correspondence); cross-type roles: our pg6 as their Ra1, our Pg2 as their Nb1; transferred move d2f2 (**endorses this move**). (d) Retrieved from `ccr1_top_p0002#879` (ply 12) [structural distance $d=68$, QAP obj=24]: the master played d4; under μ our Qd2 plays the role of that game's Pd2 (a **cross-type** correspondence); cross-type roles: our pg6 as their Ra1, our Pg2 as their Nb1; transferred move d2f2 (**endorses this move**). Synthesis: 4 retrieved analogs transferred to Qf2 (summed QAP objective 97, the strongest correspondence among the candidate moves); Melanie committed it via the **search selector**.



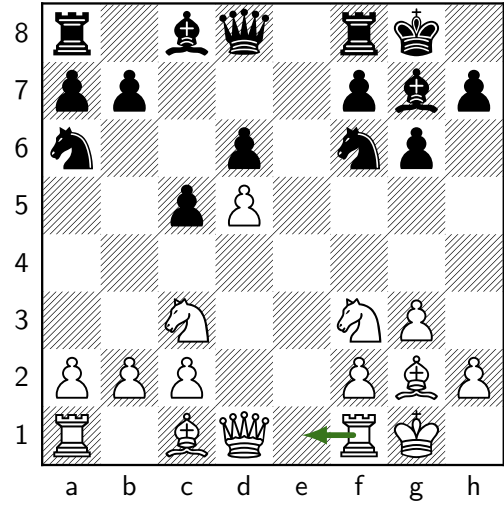
(a)



(b)

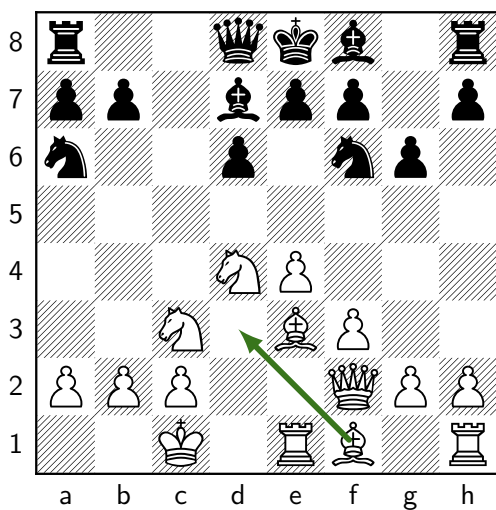


(c)

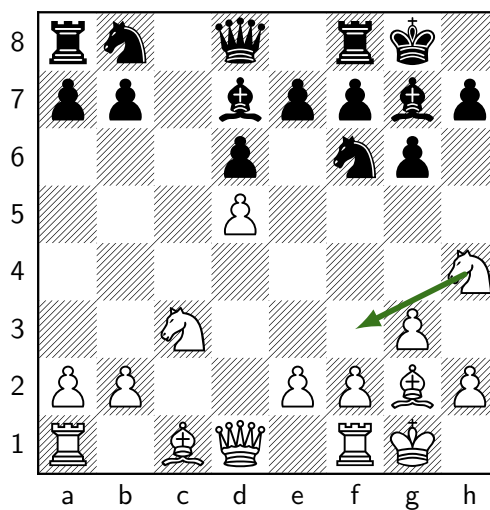


(d)

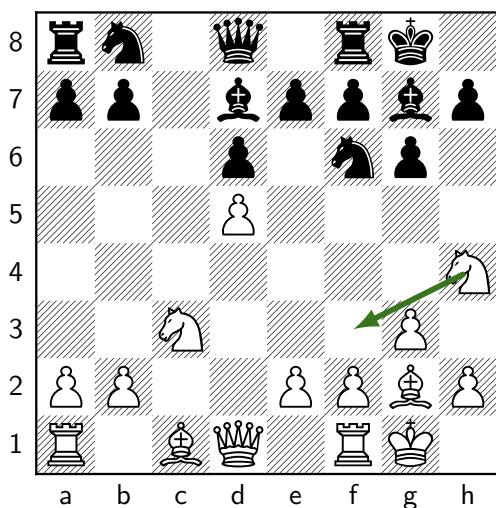
Figure 5: **Move 11, White: Re1.** (a) Our position; Melanie plays **Re1** (**d1e1**). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#2113` (ply 18) [structural distance $d=48$, QAP obj=26]: the master played **Re1**; under μ our *Rd1* plays the role of that game’s *Rf1*; cross-type roles: our *pd6* as their *Ra1*, our *Rh1* as their *Bc1*; transferred move **d1e1** (**endorses this move**). (c) Retrieved from `ccr1_top_p0001#2114` (ply 18) [structural distance $d=48$, QAP obj=26]: the master played **Re1**; under μ our *Rd1* plays the role of that game’s *Rf1*; cross-type roles: our *pd6* as their *Ra1*, our *Rh1* as their *Bc1*; transferred move **d1e1** (**endorses this move**). (d) Retrieved from `ccr1_top_p0003#2376` (ply 18) [structural distance $d=48$, QAP obj=26]: the master played **Re1**; under μ our *Rd1* plays the role of that game’s *Rf1*; cross-type roles: our *pd6* as their *Ra1*, our *Rh1* as their *Bc1*; transferred move **d1e1** (**endorses this move**). Synthesis: 4 retrieved analogs transferred to **Re1** (summed QAP objective 103, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



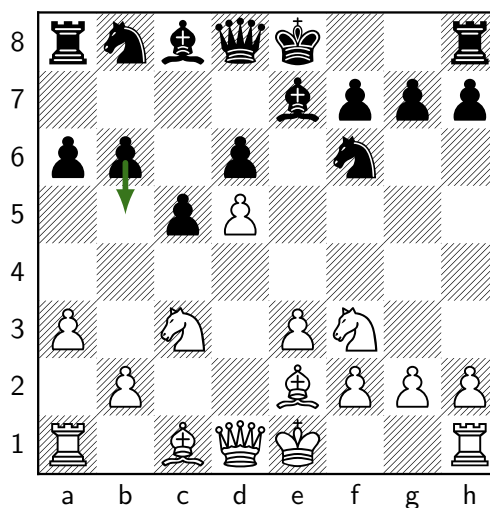
(a)



(b)

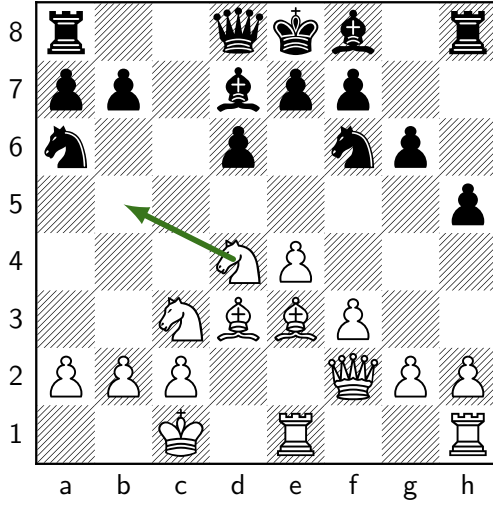


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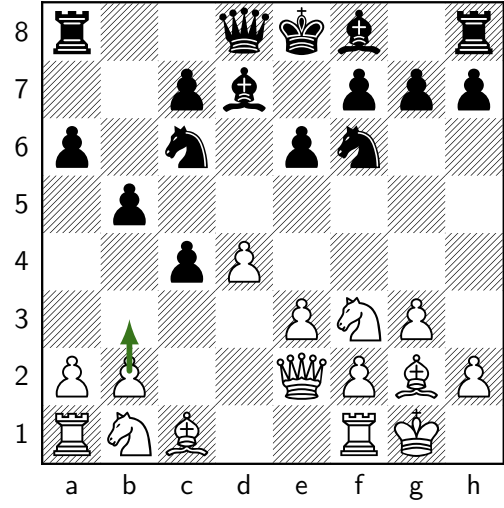


(d)

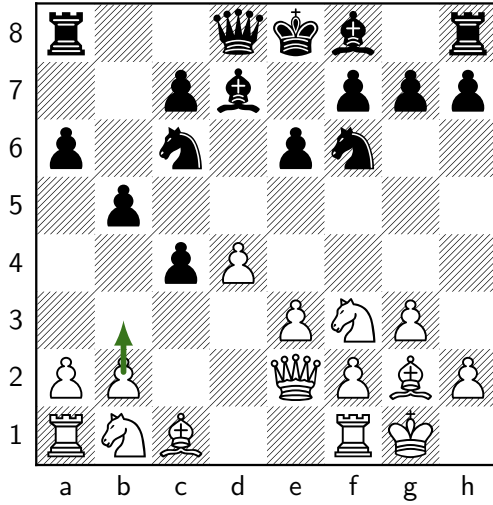
Figure 6: **Move 12, White: Bd3.** (a) Our position; Melanie plays Bd3 (f1d3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0000#1671` (ply 20) [structural distance $d=78$, QAP obj=38]: the master played Nf3; under μ our Bf1 plays the role of that game’s Nh4 (a **cross-type** correspondence); cross-type roles: our pd6 as their Bc1, our Pf3 as their Qd1; transferred move f1d3 (**endorses this move**). (c) Retrieved from `ccrl_top_p0000#1672` (ply 20) [structural distance $d=78$, QAP obj=38]: the master played Nf3; under μ our Bf1 plays the role of that game’s Nh4 (a **cross-type** correspondence); cross-type roles: our pd6 as their Bc1, our Pf3 as their Qd1; transferred move f1d3 (**endorses this move**). (d) Retrieved from `ccrl_top_p0000#3547` (ply 17) [structural distance $d=80$, QAP obj=31]: the master played b5; under μ our Rh1 plays the role of that game’s Pb3 (a **cross-type** correspondence); cross-type roles: our Pc2 as their Ra8, our pd6 as their Nb8; transferred move h1g1. Synthesis: 2 retrieved analogs transferred to Bd3 (summed QAP objective 77, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



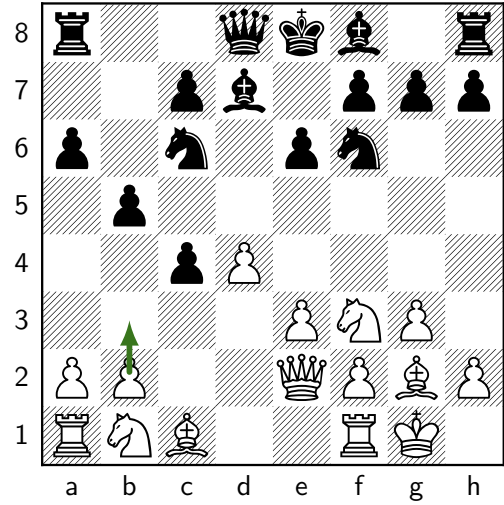
(a)



(b)

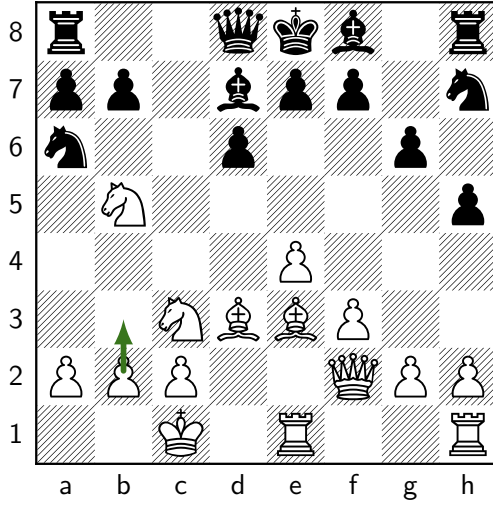


(c)

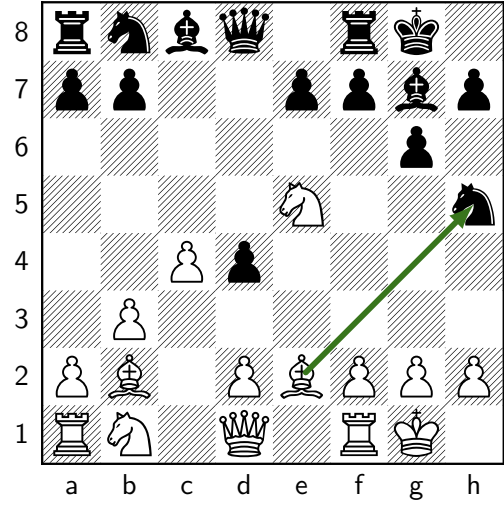


(d)

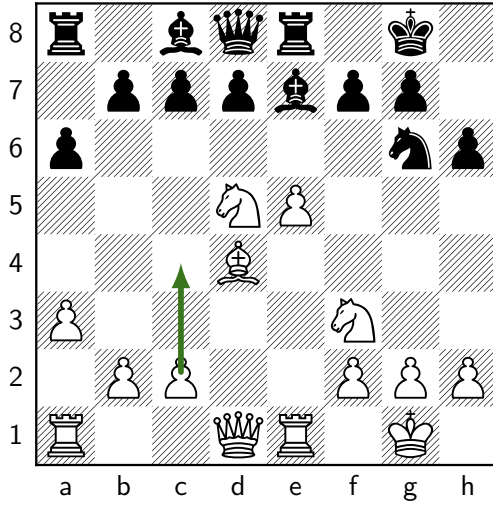
Figure 7: **Move 13, White: Ndb5.** (a) Our position; Melanie plays Ndb5 (d4b5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#578` (ply 16) [structural distance $d=65$, QAP obj=28]: the master played b3; under μ our $Nd4$ plays the role of that game’s $Pb2$ (a **cross-type** correspondence); cross-type roles: our $bf8$ as their $Ra1$, our $Pg2$ as their $Nb1$; transferred move d4b5 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#579` (ply 16) [structural distance $d=65$, QAP obj=28]: the master played b3; under μ our $Nd4$ plays the role of that game’s $Pb2$ (a **cross-type** correspondence); cross-type roles: our $bf8$ as their $Ra1$, our $Pg2$ as their $Nb1$; transferred move d4b5 (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#3372` (ply 16) [structural distance $d=65$, QAP obj=28]: the master played b3; under μ our $Nd4$ plays the role of that game’s $Pb2$ (a **cross-type** correspondence); cross-type roles: our $bf8$ as their $Ra1$, our $Pg2$ as their $Nb1$; transferred move d4b5 (**endorses this move**). Synthesis: 9 retrieved analogs transferred to Ndb5 (summed QAP objective 251, the strongest correspondence among the candidate moves); Melanie committed it via the **search selector**.



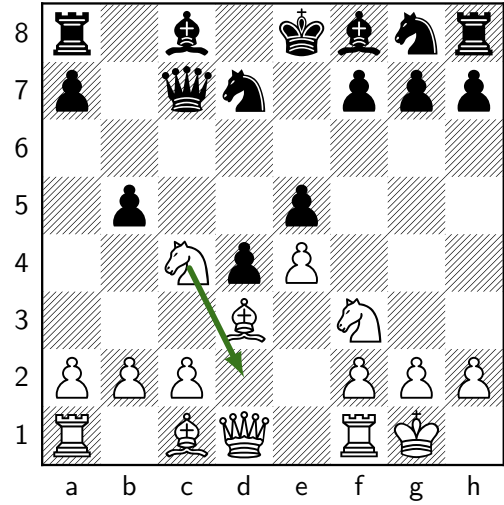
(a)



(b)

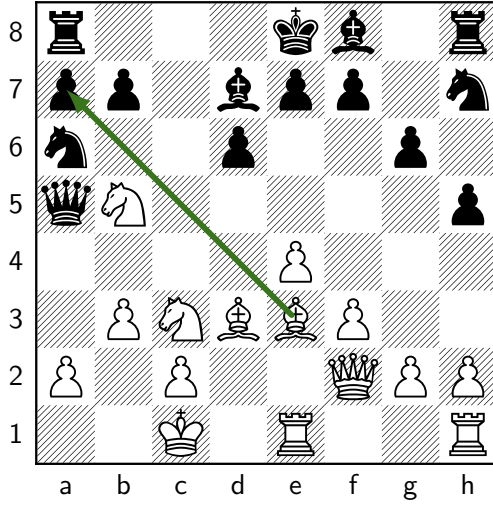


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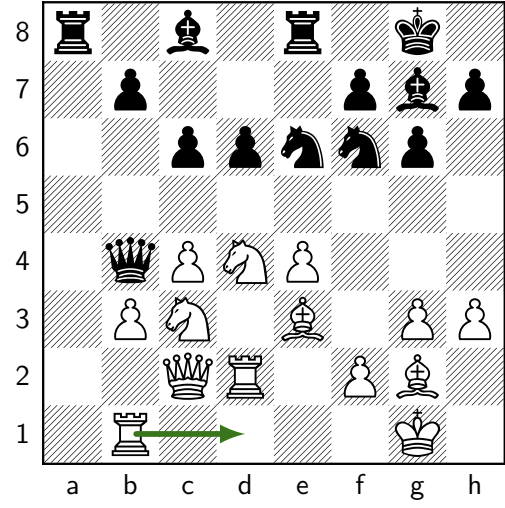


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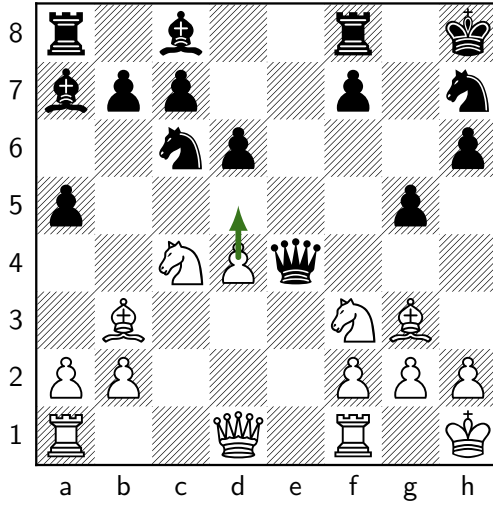
Figure 8: **Move 14, White: b3.** (a) Our position; Melanie plays b3 (b2b3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#2067` (ply 18) [structural distance $d=81$, QAP obj=68]: the master played $Bxh5$; under μ our $Nc3$ plays the role of that game’s $Be2$ (a **cross-type** correspondence); cross-type roles: our $pb7$ as their $Nb1$, our $Nb5$ as their $Bb2$; transferred move $c3a4$. (c) Retrieved from `ccr1_top_p0000#930` (ply 30) [structural distance $d=90$, QAP obj=34]: the master played $c4$; under μ our $Pb2$ plays the role of that game’s $Pc2$; cross-type roles: our $Pg2$ as their $Qd1$, our $Qf2$ as their $Ph2$; transferred move $b2b4$. (d) Retrieved from `ccr1_top_p0000#3096` (ply 20) [structural distance $d=83$, QAP obj=33]: the master played $Ncd2$; under μ our $Nb5$ plays the role of that game’s $Nc4$; cross-type roles: our $Pg2$ as their $Ra1$, our $Bd3$ as their $Qd1$; transferred move $b5a3$. Synthesis: no retrieved analog transferred to b3 – the **search** selector chose it on value (material / mate); panels (b)–(d) show the strongest nearby analogies.



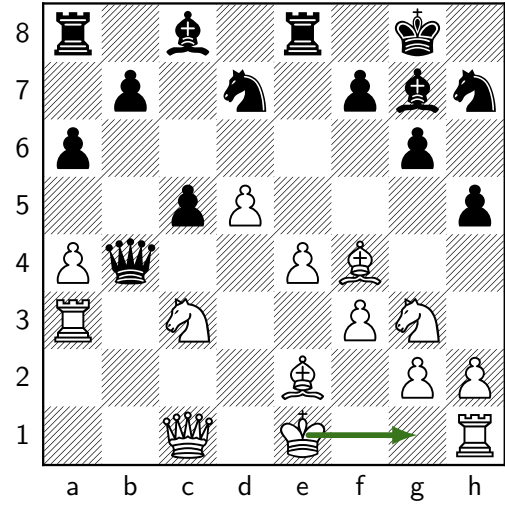
(a)



(b)

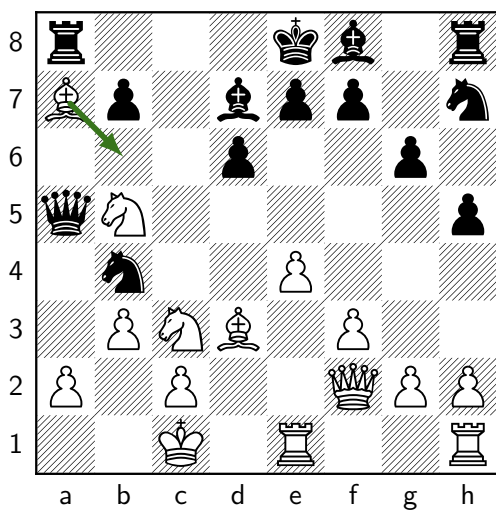


(c)

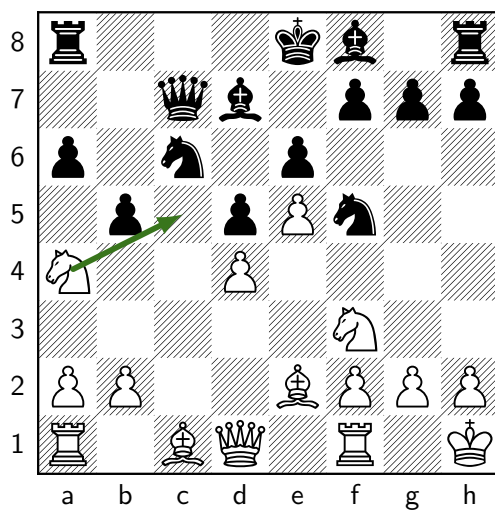


(d)

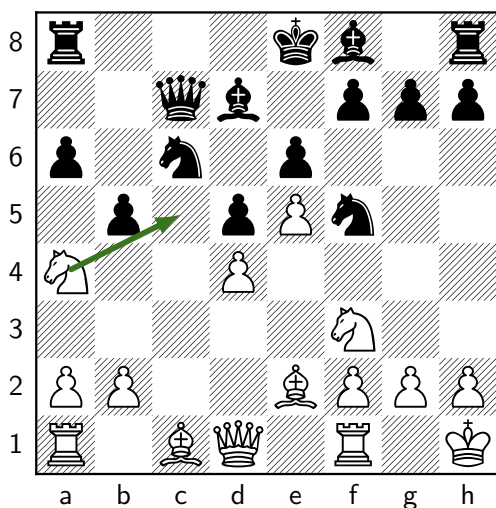
Figure 9: **Move 15, White: Bxa7.** (a) Our position; Melanie plays Bxa7 (e3a7). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0003#2578` (ply 44) [structural distance $d=105$, QAP obj=79]: the master played Rbd1; under μ our *Pb3* plays the role of that game’s *Rb1* (a **cross-type** correspondence); cross-type roles: our *Be3* as their *Qc2*, our *Qf2* as their *Rd2*; transferred move b3b4. (c) Retrieved from `ccr1_top_p0002#1271` (ply 30) [structural distance $d=114$, QAP obj=69]: the master played d5; under μ our *Nc3* plays the role of that game’s *Pd4* (a **cross-type** correspondence); cross-type roles: our *Pg2* as their *Ra1*, our *pb7* as their *Qd1*; transferred move c3d5. (d) Retrieved from `ccr1_top_p0000#1141` (ply 34) [structural distance $d=110$, QAP obj=64]: the master played 0-0; under μ our *Kc1* plays the role of that game’s *Ke1*; cross-type roles: our *Be3* as their *Qc1*, our *Pb3* as their *Be2*; transferred move c1b1. Synthesis: no retrieved analog transferred to Bxa7 – the **search** selector chose it on value (material / mate); panels (b)–(d) show the strongest nearby analogies.



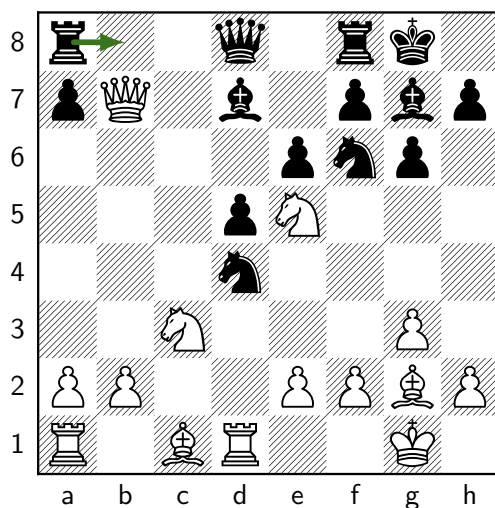
(a)



(b)

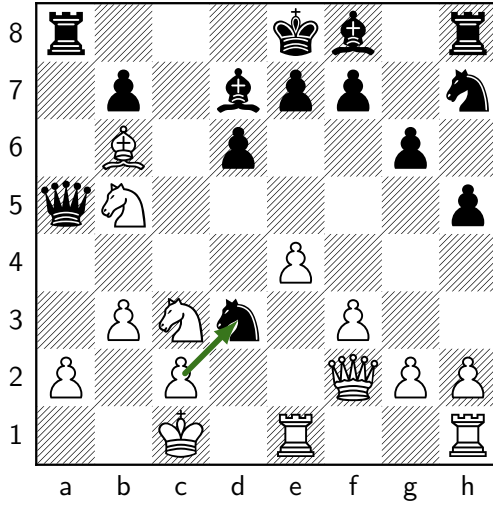


(c)

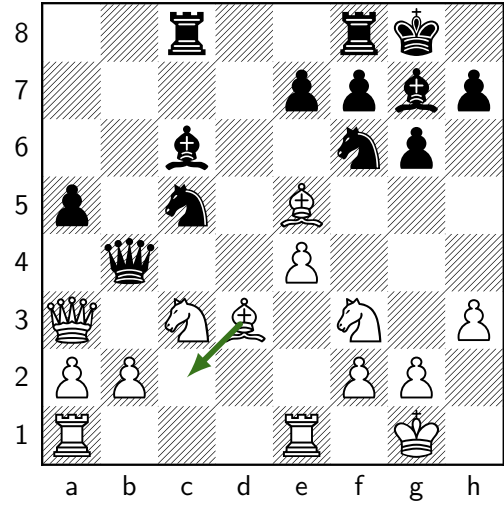


(d)

Figure 10: **Move 16, White: Bb6**. (a) Our position; Melanie plays Bb6 (a7b6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0000#3898` (ply 24) [structural distance $d=112$, QAP obj=35]: the master played Nc5; under μ our Ba7 plays the role of that game’s Na4 (a **cross-type** correspondence); cross-type roles: our Bd3 as their Ra1, our pg6 as their Bc1; transferred move a7b6 (**endorses this move**). (c) Retrieved from `ccrl_top_p0001#1492` (ply 24) [structural distance $d=112$, QAP obj=35]: the master played Nc5; under μ our Ba7 plays the role of that game’s Na4 (a **cross-type** correspondence); cross-type roles: our Bd3 as their Ra1, our pg6 as their Bc1; transferred move a7b6 (**endorses this move**). (d) Retrieved from `ccrl_top_p0001#88` (ply 23) [structural distance $d=132$, QAP obj=49]: the master played Rb8; under μ our Nb5 plays the role of that game’s Ra1 (a **cross-type** correspondence); cross-type roles: our Nc3 as their Qd8, our pg6 as their Bg7; transferred move b5a3. Synthesis: 2 retrieved analogs transferred to Bb6 (summed QAP objective 71, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.

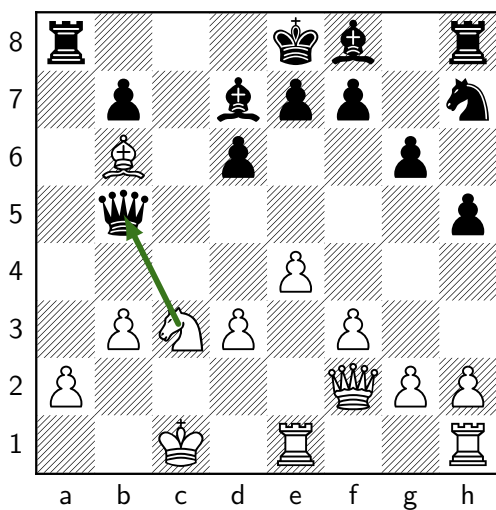


(a)

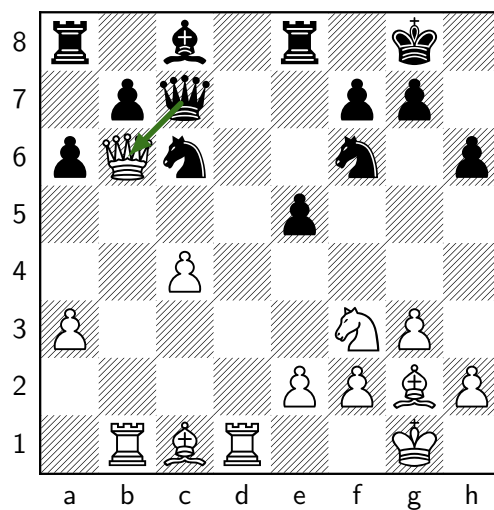


(b)

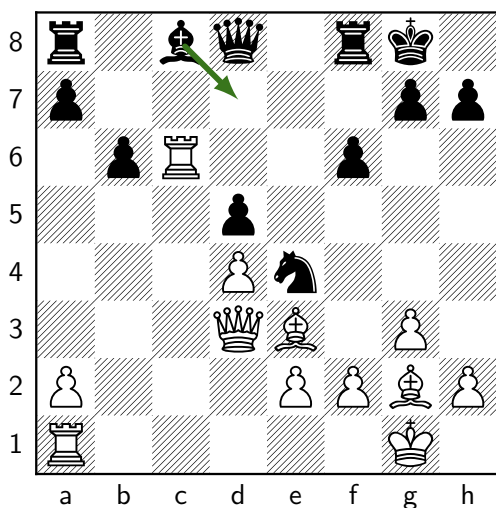
Figure 11: **Move 17, White: cxd3.** (a) Our position; Melanie plays *cx**d*3 (*c*2*d*3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from *ccr1_top-p0002#3167* (ply 38) [structural distance $d=131$, QAP obj=45]: the master played *Bc*2; under μ our *Pc*2 plays the role of that game’s *Bd*3 (a **cross-type** correspondence); cross-type roles: our *Pb*3 as their *Ra*1, our *Qf*2 as their *Pa*2; transferred move *c*2*d*3 (**endorses this move**). Synthesis: 1 retrieved analog transferred to *cx**d*3 (summed QAP objective 45, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



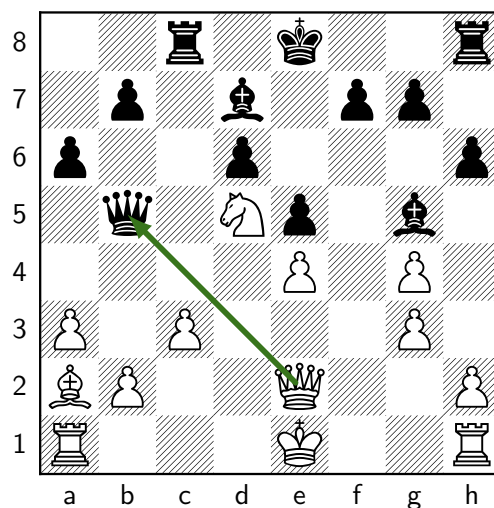
(a)



(b)

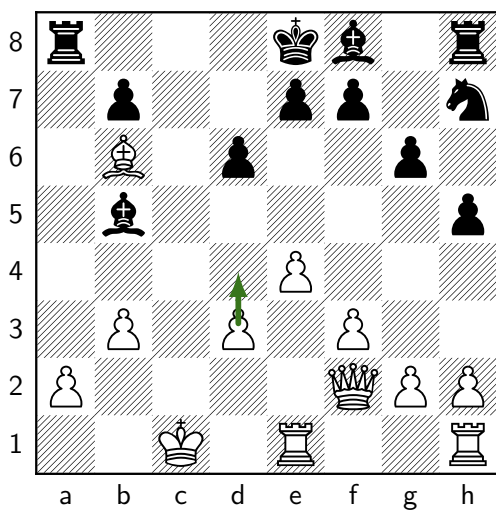


(c)

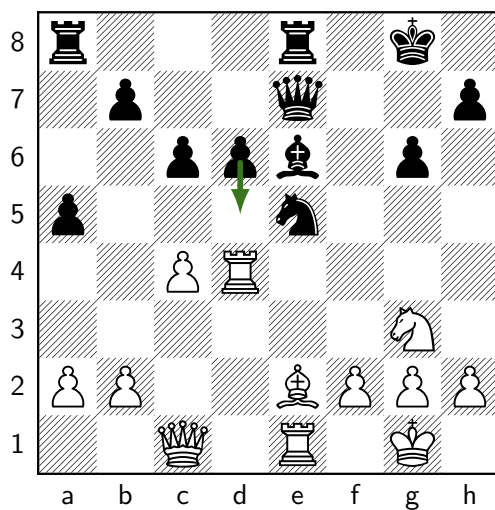


(d)

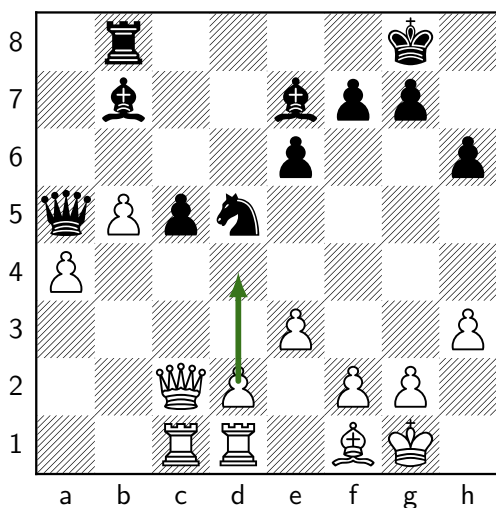
Figure 12: **Move 18, White: Nxb5.** (a) Our position; Melanie plays Nxb5 (c3b5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#1411` (ply 31) [structural distance $d=102$, QAP obj=21]: the master played Qxb6; under μ our *Nc3* plays the role of that game’s *Qc2* (a **cross-type** correspondence); cross-type roles: our *Qf2* as their *Bc8*, our *Bb6* as their *Nc6*; transferred move c3b5 (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#1163` (ply 33) [structural distance $d=106$, QAP obj=64]: the master played Bd7; under μ our *Bb6* plays the role of that game’s *Bc1*; cross-type roles: our *pb7* as their *Rf8*, our *ra8* as their *Pb6*; transferred move b6d4. (d) Retrieved from `ccr1_top_p0000#689` (ply 50) [structural distance $d=107$, QAP obj=29]: the master played Qxb5; under μ our *Nc3* plays the role of that game’s *Qe2* (a **cross-type** correspondence); cross-type roles: our *Pd3* as their *Rh1*, our *Qf2* as their *Ba2*; transferred move c3a4. Synthesis: 1 retrieved analog transferred to Nxb5 (summed QAP objective 21, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



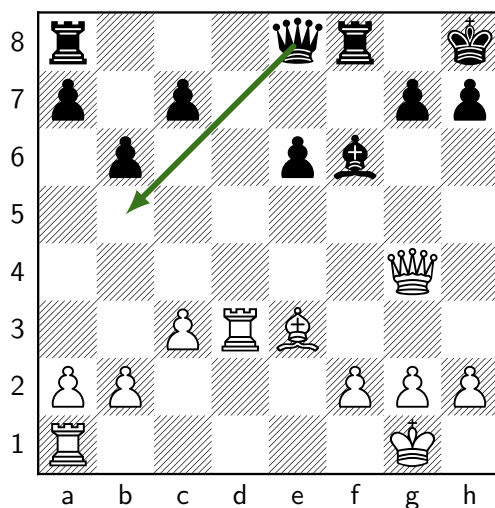
(a)



(b)

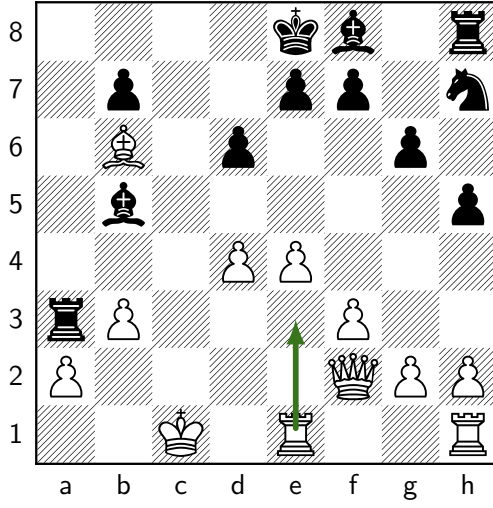


(c)

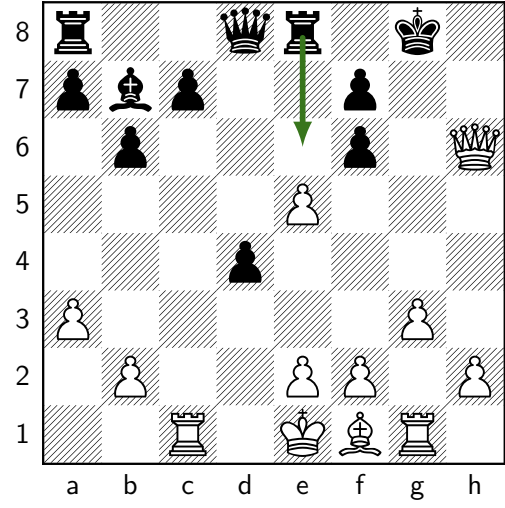


(d)

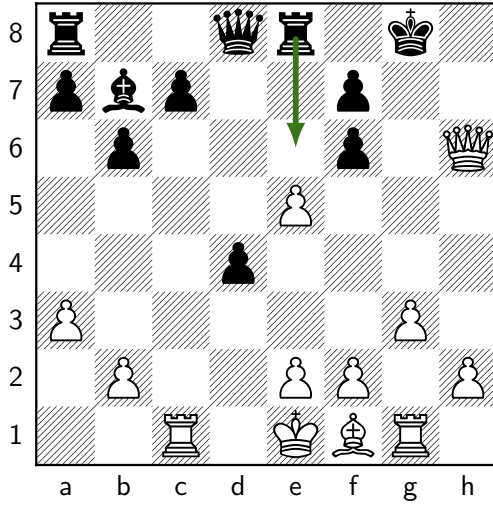
Figure 13: **Move 19, White: d4**. (a) Our position; Melanie plays d4 (d3d4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#812` (ply 41) [structural distance $d=98$, QAP obj=44]: the master played d5; under μ our $Pd3$ plays the role of that game’s $Pd3$; cross-type roles: our $Pb3$ as their $Qe7$, our $bb5$ as their $Ne5$; transferred move d3d4 (**endorses this move**). (c) Retrieved from `ccr1_top_p0001#2373` (ply 46) [structural distance $d=104$, QAP obj=18]: the master played d4; under μ our $Pd3$ plays the role of that game’s $Pd2$; cross-type roles: our $Qf2$ as their $Rd1$, our $Pb3$ as their $Qc2$; transferred move d3d4 (**endorses this move**). (d) Retrieved from `ccr1_top_p0001#982` (ply 41) [structural distance $d=106$, QAP obj=40]: the master played Qb5; under μ our $Qf2$ plays the role of that game’s $Qe1$; cross-type roles: our $Pe4$ as their $Ra8$, our $ra8$ as their $Bf6$; transferred move f2h4. Synthesis: 2 retrieved analogs transferred to d4 (summed QAP objective 62, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



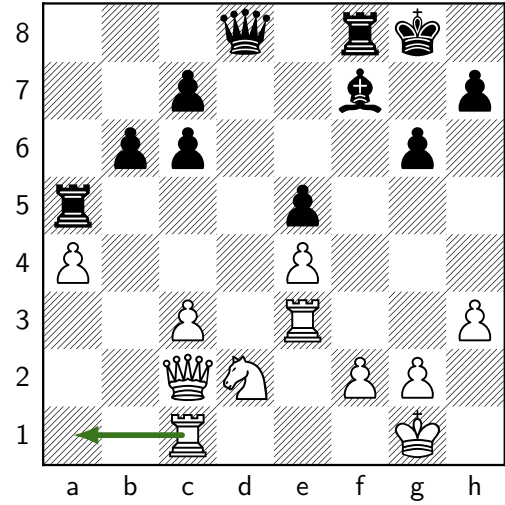
(a)



(b)

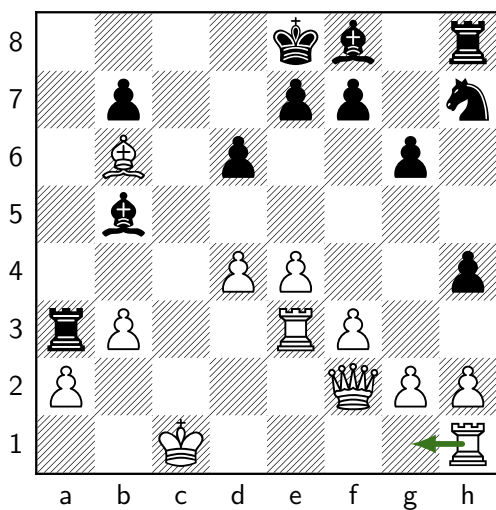


(c)

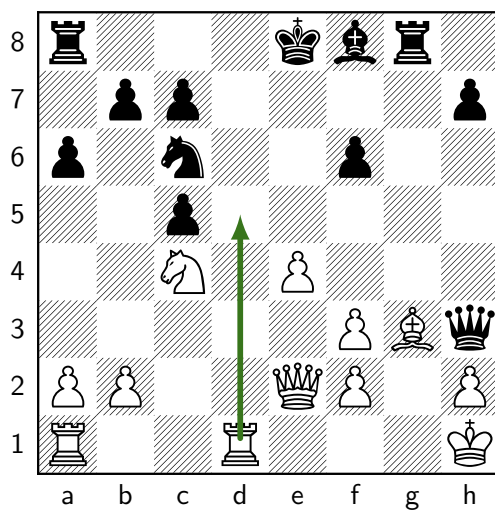


(d)

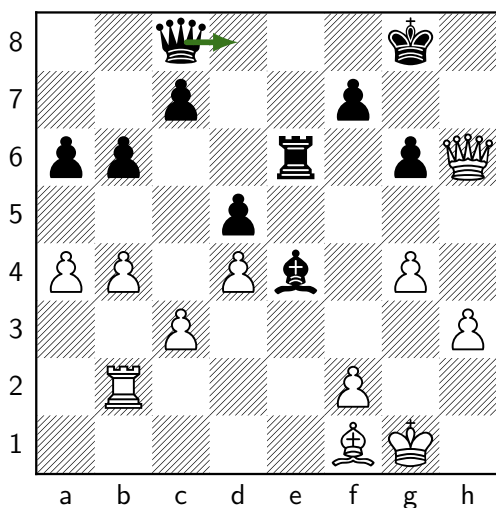
Figure 14: **Move 20, White: Re3.** (a) Our position; Melanie plays Re3 (e1e3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#651` (ply 33) [structural distance $d=92$, QAP obj=29]: the master played Re6; under μ our Re1 plays the role of that game’s Re1; cross-type roles: our Pf3 as their Bb7, our ra3 as their qh6; transferred move e1e3 (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#701` (ply 33) [structural distance $d=92$, QAP obj=29]: the master played Re6; under μ our Re1 plays the role of that game’s Re1; cross-type roles: our Pf3 as their Bb7, our ra3 as their qh6; transferred move e1e3 (**endorses this move**). (d) Retrieved from `ccr1_top_p0003#959` (ply 50) [structural distance $d=96$, QAP obj=28]: the master played Ra1; under μ our Re1 plays the role of that game’s Rc1; cross-type roles: our Pb3 as their Nd2, our Rh1 as their Pf2; transferred move e1e3 (**endorses this move**). Synthesis: 3 retrieved analogs transferred to Re3 (summed QAP objective 86, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



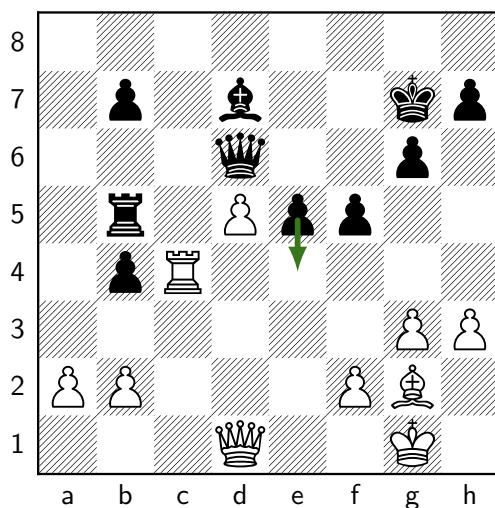
(a)



(b)

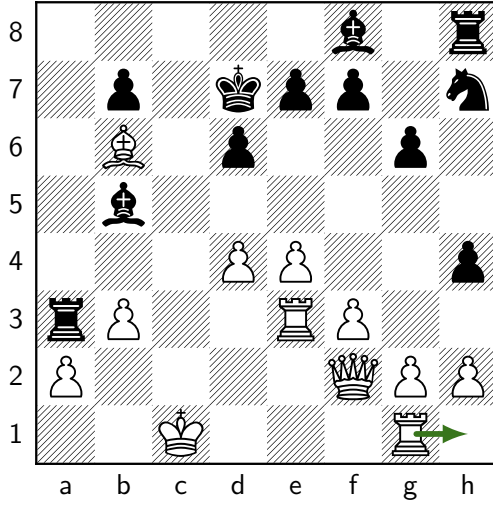


(c)

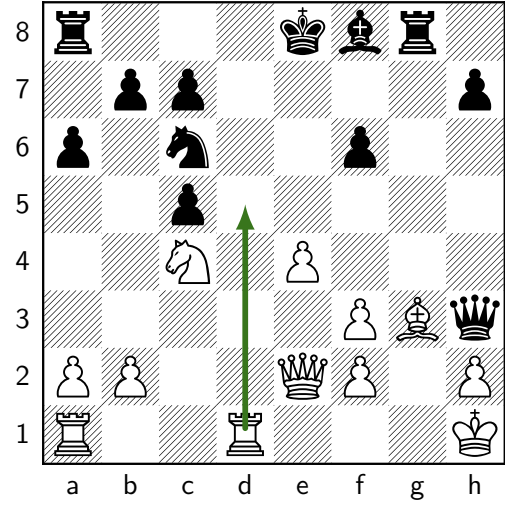


(d)

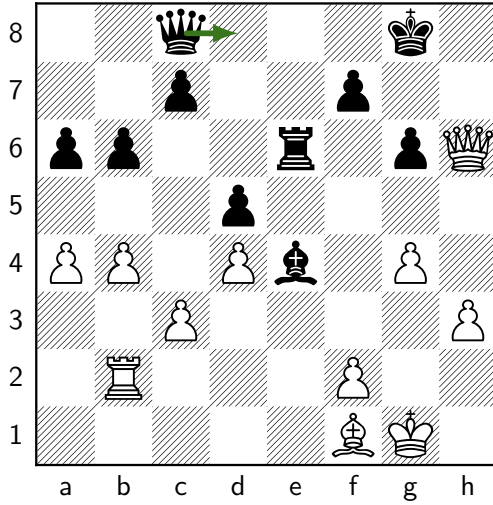
Figure 15: **Move 21, White: Rg1.** (a) Our position; Melanie plays Rg1 (h1g1). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0003#2077` (ply 36) [structural distance $d=114$, QAP obj=54]: the master played Rd5; under μ our *Rh1* plays the role of that game’s *Rd1*; cross-type roles: our *Bb6* as their *Qe2*, our *Re3* as their *Pf2*; transferred move h1g1 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#1276` (ply 53) [structural distance $d=112$, QAP obj=46]: the master played Qd8; under μ our *Rh1* plays the role of that game’s *Qc1* (a **cross-type** correspondence); cross-type roles: our *ra3* as their *qh6*, our *Qf2* as their *Be4*; transferred move h1g1 (**endorses this move**). (d) Retrieved from `ccr1_top_p0002#572` (ply 55) [structural distance $d=114$, QAP obj=59]: the master played e4; under μ our *Re3* plays the role of that game’s *Pe4* (a **cross-type** correspondence); cross-type roles: our *Pd4* as their *Bd7*, our *Rh1* as their *Ph7*; transferred move e3d3. Synthesis: 2 retrieved analogs transferred to Rg1 (summed QAP objective 100, the strongest correspondence among the candidate moves); Melanie committed it via the **search selector**.



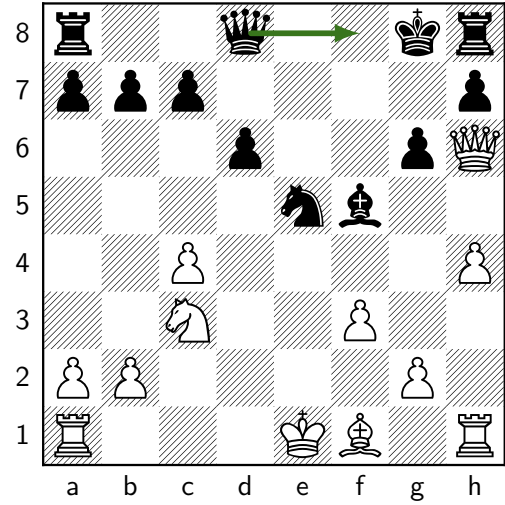
(a)



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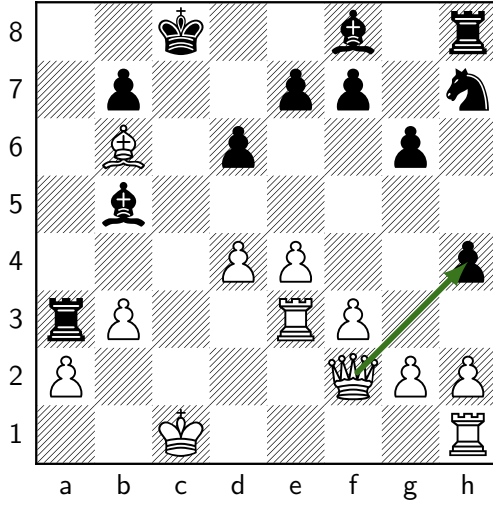


(c)

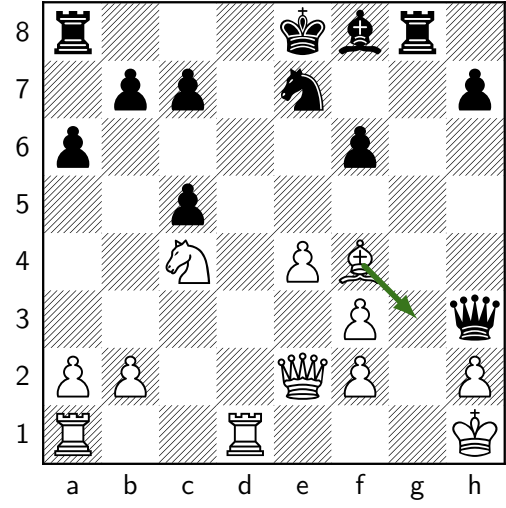


(d)

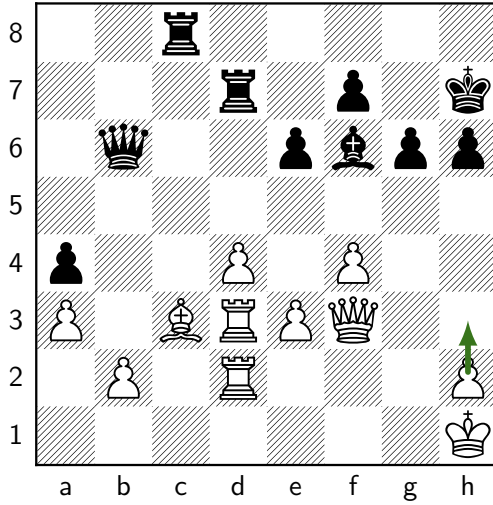
Figure 16: **Move 22, White: Rh1**. (a) Our position; Melanie plays Rh1 (**g1h1**). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0003#2077` (ply 36) [structural distance $d=92$, QAP obj=53]: the master played Rd5; under μ our *Rg1* plays the role of that game’s *Rd1*; cross-type roles: our *ph4* as their *Ra1*, our *bf8* as their *Qe2*; transferred move **g1h1 (endorses this move)**. (c) Retrieved from `ccr1_top_p0000#1276` (ply 53) [structural distance $d=105$, QAP obj=46]: the master played Qd8; under μ our *Rg1* plays the role of that game’s *Qc1* (a **cross-type** correspondence); cross-type roles: our *ra3* as their *qh6*, our *Bb6* as their *pd4*; transferred move **g1h1 (endorses this move)**. (d) Retrieved from `ccr1_top_p0001#1547` (ply 41) [structural distance $d=106$, QAP obj=107]: the master played Qf8; under μ our *Qf2* plays the role of that game’s *Qd1*; cross-type roles: our *Bb6* as their *Pd6*, our *ra3* as their *qh6*; transferred move **f2d2**. Synthesis: 2 retrieved analogs transferred to Rh1 (summed QAP objective 99, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



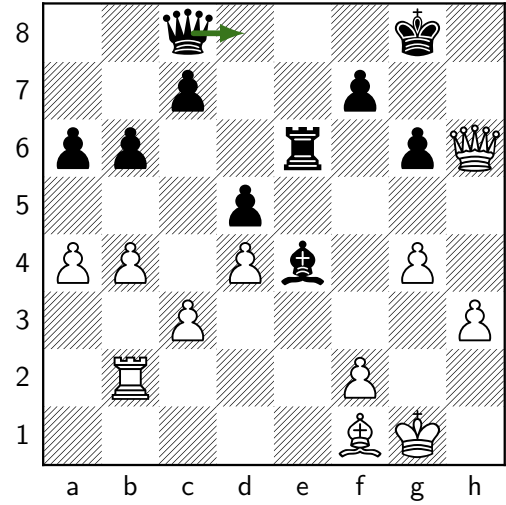
(a)



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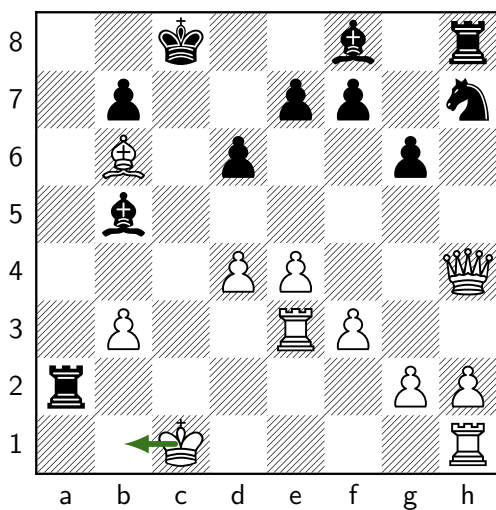


(c)

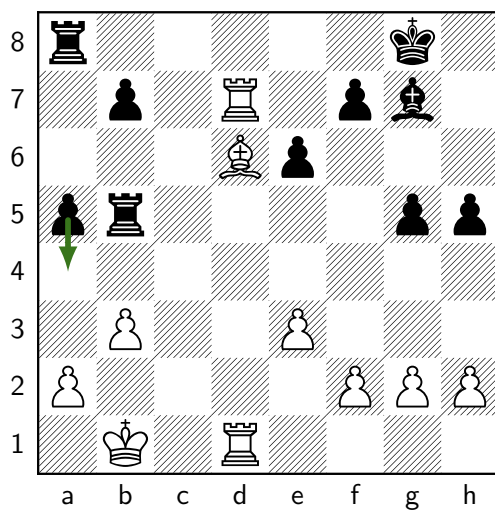


(d)

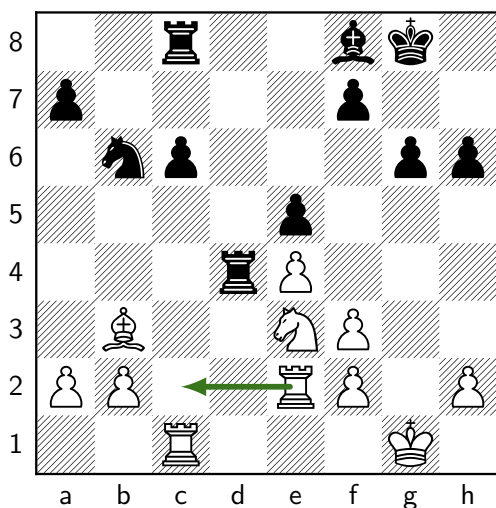
Figure 17: **Move 23, White: Qxh4.** (a) Our position; Melanie plays Qxh4 (f2h4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0003#2077` (ply 34) [structural distance $d=111$, QAP obj=73]: the master played Bg3; under μ our Qf2 plays the role of that game's Bf4 (a **cross-type** correspondence); cross-type roles: our Pd4 as their Ra1, our pe7 as their Qe2; transferred move f2e2. (c) Retrieved from `ccr1_top_p0002#2636` (ply 58) [structural distance $d=112$, QAP obj=53]: the master played h3; under μ our Rh1 plays the role of that game's Ph2 (a **cross-type** correspondence); cross-type roles: our bf8 as their Rd2, our Bb6 as their Pa3; transferred move h1g1. (d) Retrieved from `ccr1_top_p0000#1276` (ply 53) [structural distance $d=113$, QAP obj=46]: the master played Qd8; under μ our Rh1 plays the role of that game's Qc1 (a **cross-type** correspondence); cross-type roles: our ra3 as their qh6, our bb5 as their pg4; transferred move h1g1. Synthesis: no retrieved analog transferred to Qxh4 – the search selector chose it on value (material / mate); panels (b)–(d) show the strongest nearby analogies.



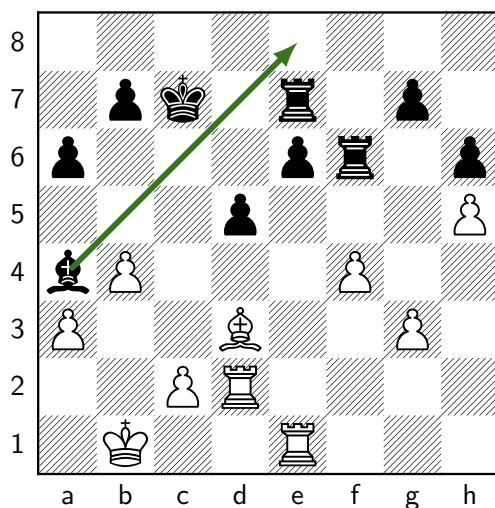
(a)



(b)

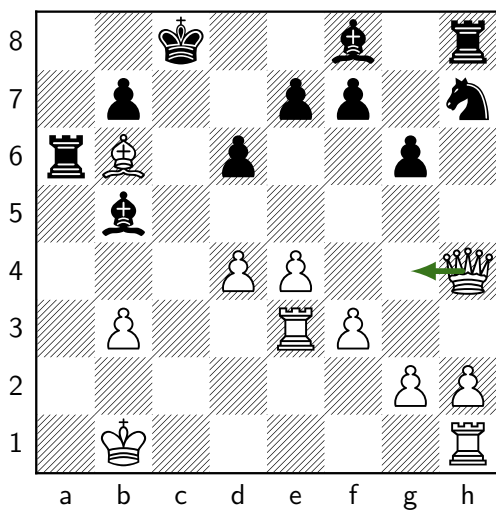


(c)

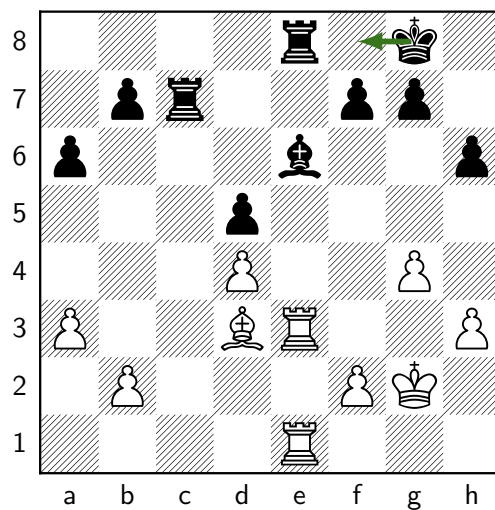


(d)

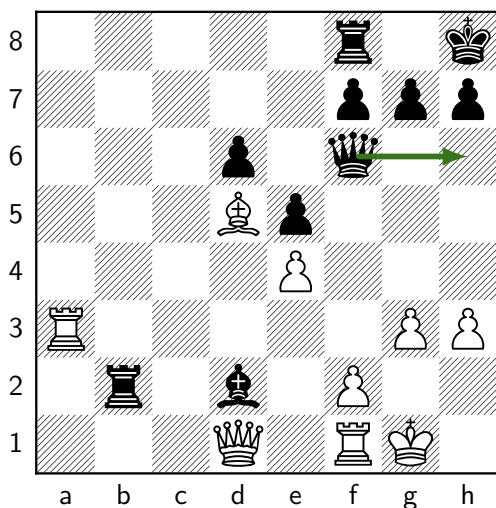
Figure 18: **Move 24, White: Kb1.** (a) Our position; Melanie plays Kb1 (c1b1). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0003#2149` (ply 45) [structural distance $d=85$, QAP obj=41]: the master played a4; under μ our *Ph2* plays the role of that game’s *Pa4*; cross-type roles: our *Qh4* as their *rd7*, our *nh7* as their *Pf7*; transferred move h2h3. (c) Retrieved from `ccrl_top_p0003#2015` (ply 42) [structural distance $d=89$, QAP obj=24]: the master played Rec2; under μ our *Pg2* plays the role of that game’s *Re2* (a **cross-type** correspondence); cross-type roles: our *ra2* as their *Pa2*, our *Re3* as their *Ph2*; transferred move g2g4. (d) Retrieved from `ccrl_top_p0000#827` (ply 59) [structural distance $d=94$, QAP obj=16]: the master played Be8; under μ our *Qh4* plays the role of that game’s *Ba5* (a **cross-type** correspondence); cross-type roles: our *Qh4* as their *Rf6*, our *Re3* as their *Ph6*; transferred move h4g3. Synthesis: no retrieved analog transferred to Kb1 – the **search** selector chose it on value (material / mate); panels (b)–(d) show the strongest nearby analogies.



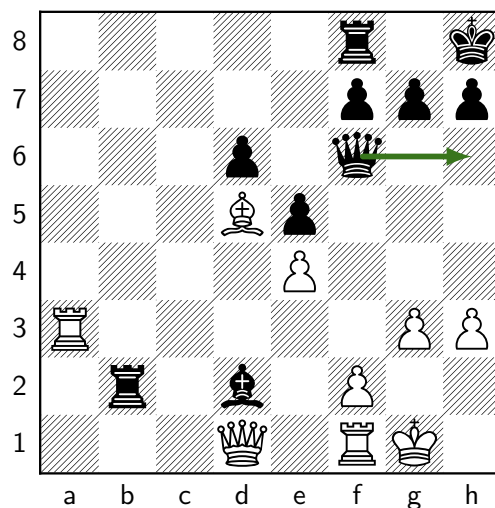
(a)



(b)

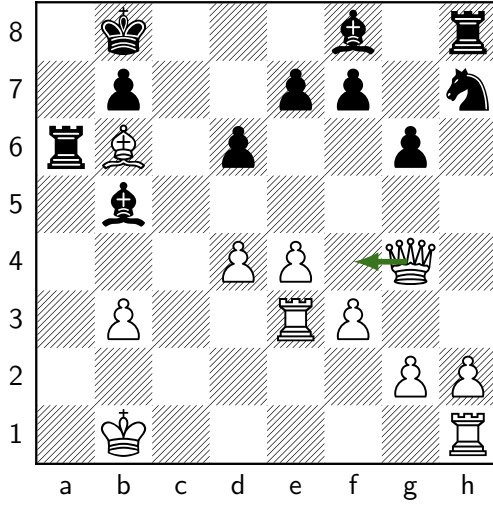


(c)

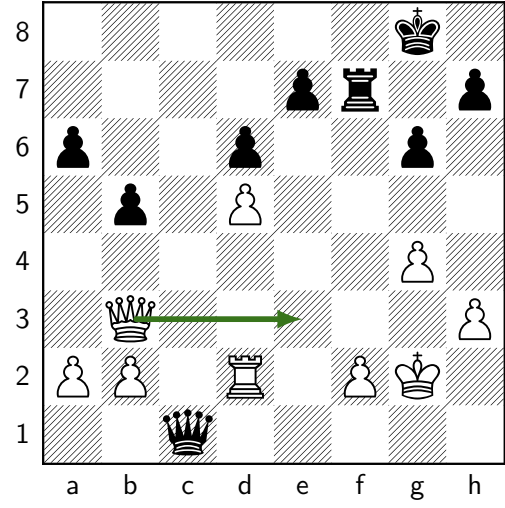


(d)

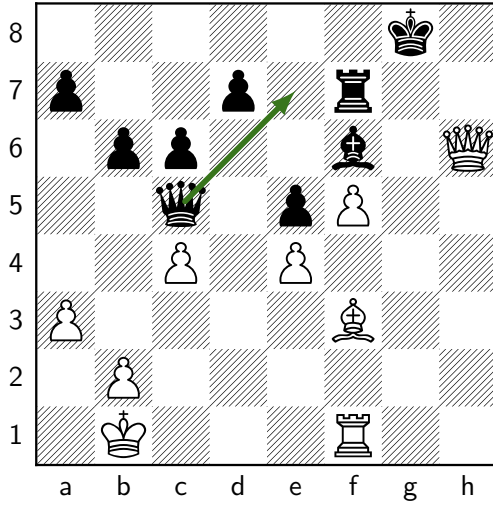
Figure 19: **Move 25, White: Qg4+.** (a) Our position; Melanie plays Qg4+ (h4g4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0003#2467` (ply 55) [structural distance $d=95$, QAP obj=38]: the master played Kf8; under μ our Kb1 plays the role of that game's Kg1; cross-type roles: our pd6 as their Rc7, our bf8 as their Pf7; transferred move b1c1. (c) Retrieved from `ccr1_top_p0001#3187` (ply 47) [structural distance $d=88$, QAP obj=32]: the master played Qh6; under μ our Qh4 plays the role of that game's Qf3; cross-type roles: our Rh1 as their Ph7, our pb7 as their ra3; transferred move h4g3. (d) Retrieved from `ccr1_top_p0003#2310` (ply 47) [structural distance $d=88$, QAP obj=32]: the master played Qh6; under μ our Qh4 plays the role of that game's Qf3; cross-type roles: our Rh1 as their Ph7, our pb7 as their ra3; transferred move h4g3. Synthesis: no retrieved analog transferred to Qg4+ – the search selector chose it on value (material / mate); panels (b)–(d) show the strongest nearby analogies.



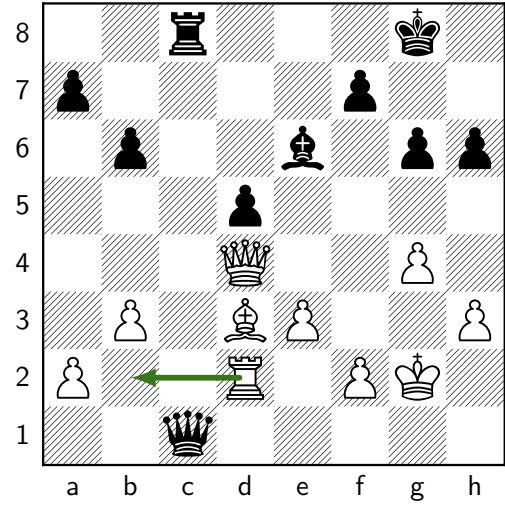
(a)



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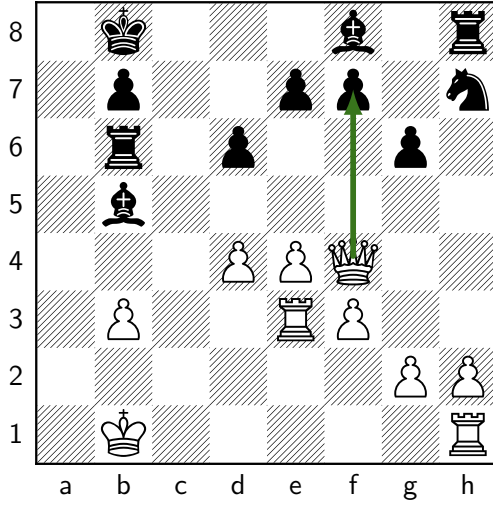


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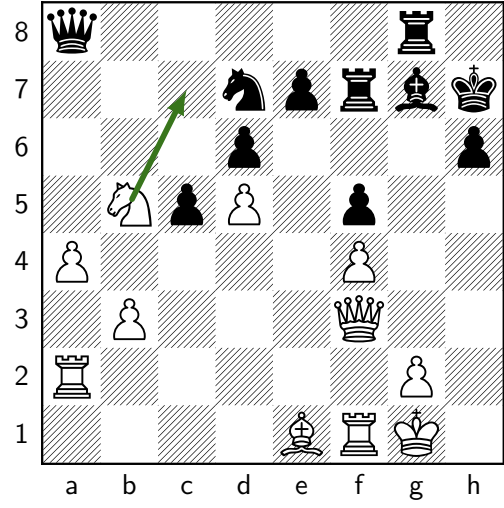


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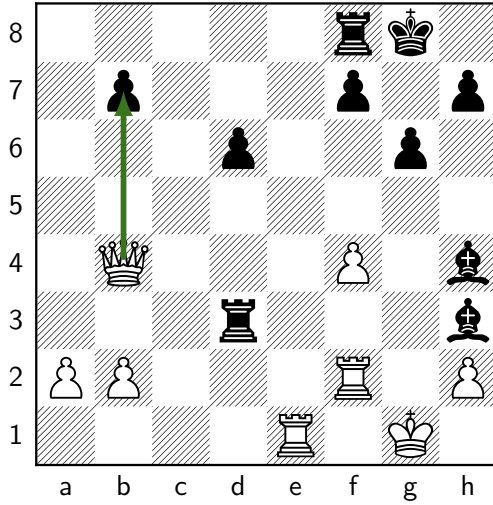
Figure 20: **Move 26, White: Qf4.** (a) Our position; Melanie plays Qf4 (g4f4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0002#873` (ply 60) [structural distance $d=102$, QAP obj=18]: the master played Qe3; under μ our Qg4 plays the role of that game’s Qb3; cross-type roles: our ra6 as their qc1, our Bb6 as their Rd2; transferred move g4f4 (**endorses this move**). (c) Retrieved from `ccrl_top_p0000#1939` (ply 65) [structural distance $d=102$, QAP obj=17]: the master played Qe7; under μ our Qg4 plays the role of that game’s Qc4; cross-type roles: our bf8 as their Pa7, our Pb3 as their Rf7; transferred move g4g3. (d) Retrieved from `ccrl_top_p0000#1142` (ply 52) [structural distance $d=109$, QAP obj=16]: the master played Rb2; under μ our Bb6 plays the role of that game’s Rd2 (a **cross-type** correspondence); cross-type roles: our ra6 as their qc1, our Rh1 as their Pf2; transferred move b6c5. Synthesis: 1 retrieved analog transferred to Qf4 (summed QAP objective 18, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



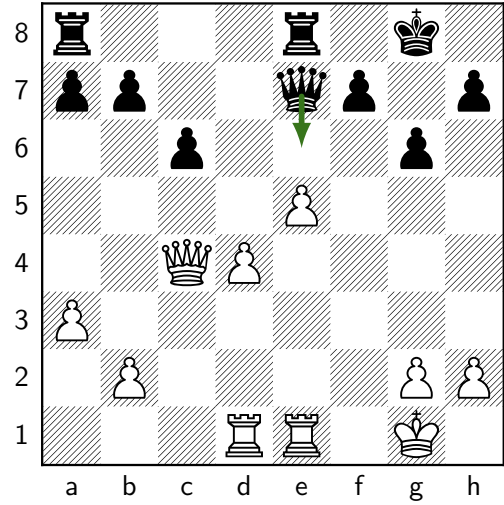
(a)



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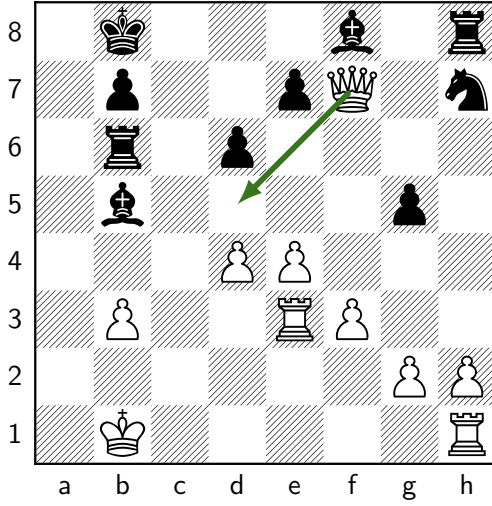


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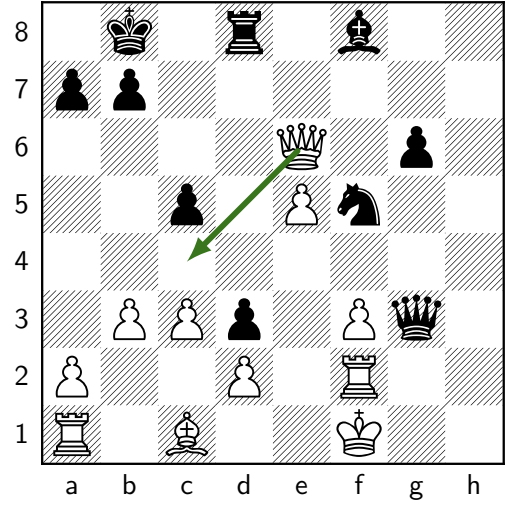


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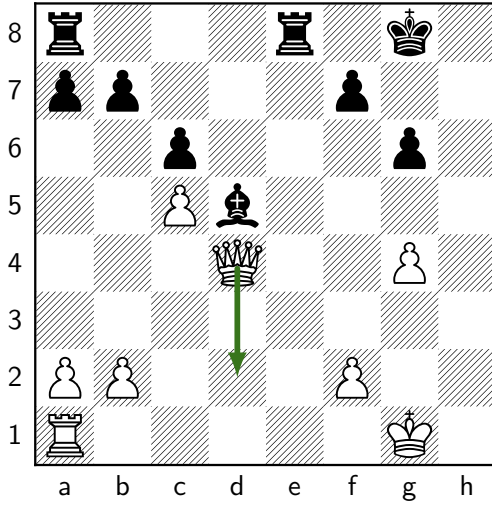
Figure 21: **Move 27, White: Qxf7**. (a) Our position; Melanie plays Qxf7 (f4f7). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#97` (ply 58) [structural distance $d=103$, QAP obj=88]: the master played Nc7; under μ our Qf4 plays the role of that game's Nb5 (a **cross-type** correspondence); cross-type roles: our Pb3 as their Be1, our nh7 as their Ra2; transferred move f4f7 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#3380` (ply 60) [structural distance $d=107$, QAP obj=40]: the master played Qxb7; under μ our Qf4 plays the role of that game's Qb4; cross-type roles: our Rh1 as their Pa2, our Ph2 as their Rf2; transferred move f4f7 (**endorses this move**). (d) Retrieved from `ccr1_top_p0003#2741` (ply 51) [structural distance $d=110$, QAP obj=107]: the master played Qe6; under μ our Qf4 plays the role of that game's Qe2; cross-type roles: our Pf3 as their Re8, our Re3 as their Pb7; transferred move f4e5. Synthesis: 2 retrieved analogs transferred to Qxf7 (summed QAP objective 128, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



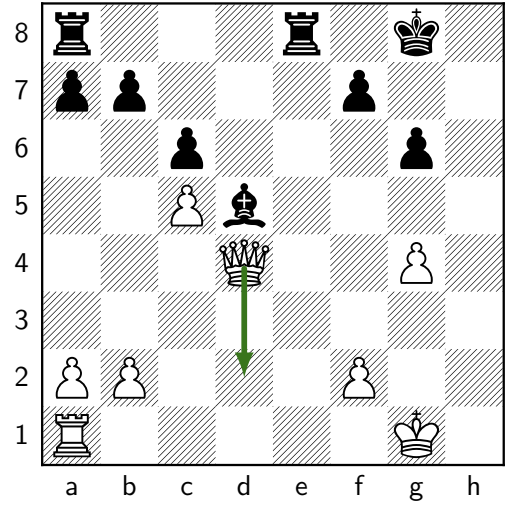
(a)



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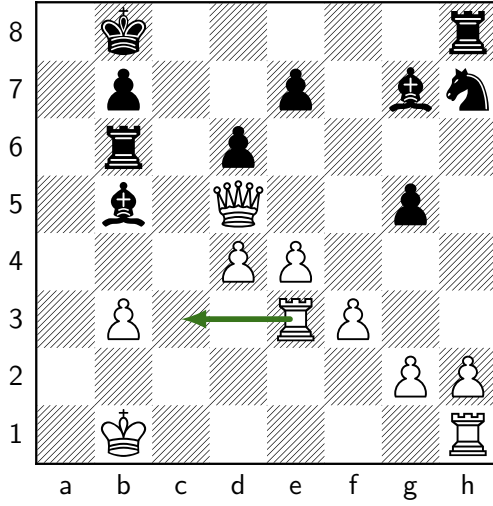


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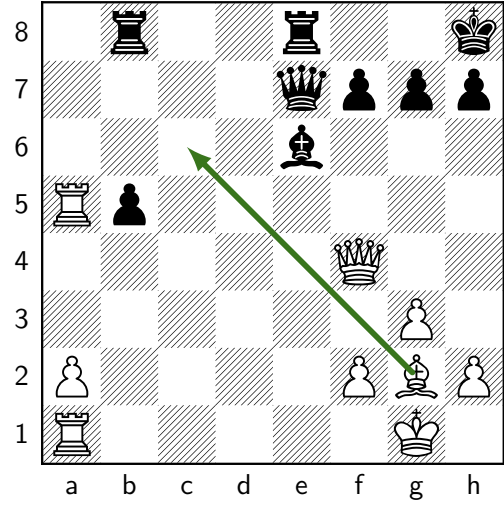


(d)

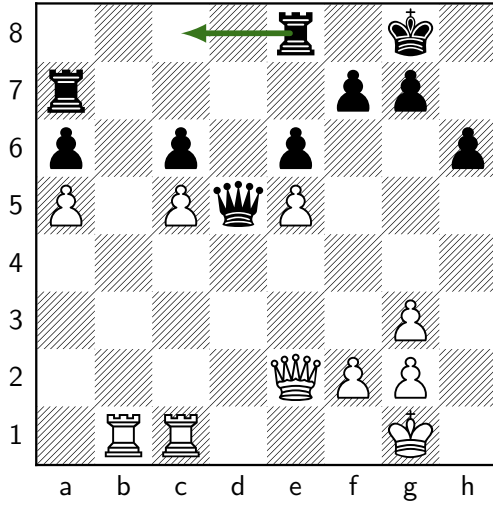
Figure 22: **Move 28, White: Qd5.** (a) Our position; Melanie plays Qd5 (f7d5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#90` (ply 40) [structural distance $d=95$, QAP obj=24]: the master played Qc4; under μ our Qf7 plays the role of that game’s Qe6; cross-type roles: our pb7 as their Bc1, our bf8 as their Rf2; transferred move f7d5 (**endorses this move**). (c) Retrieved from `ccr1_top_p0001#353` (ply 48) [structural distance $d=105$, QAP obj=23]: the master played Qd2; under μ our Qf7 plays the role of that game’s Qd4; cross-type roles: our Re3 as their Pa2, our nh7 as their pc6; transferred move f7h7. (d) Retrieved from `ccr1_top_p0003#2043` (ply 48) [structural distance $d=105$, QAP obj=23]: the master played Qd2; under μ our Qf7 plays the role of that game’s Qd4; cross-type roles: our Re3 as their Pa2, our nh7 as their pc6; transferred move f7h7. Synthesis: 1 retrieved analog transferred to Qd5 (summed QAP objective 24, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



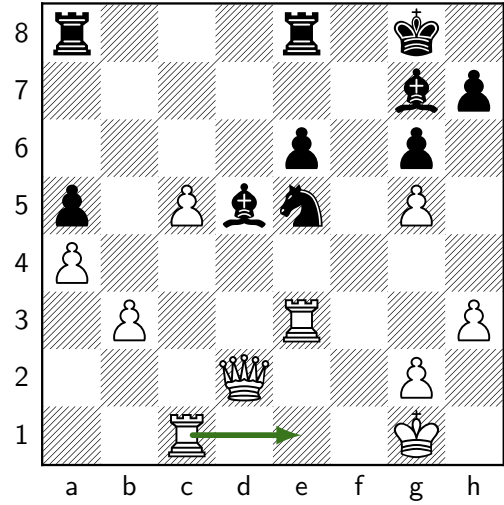
(a)



(b)

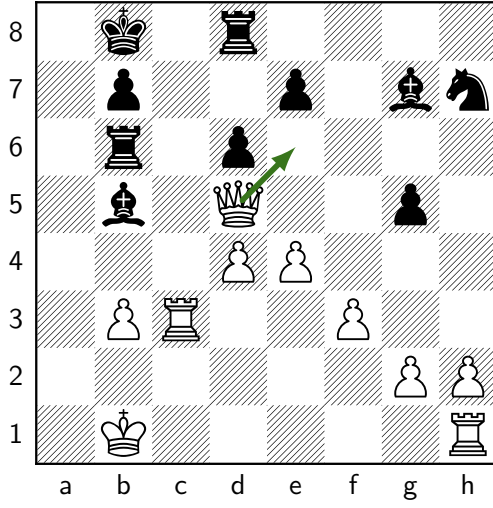


(c)

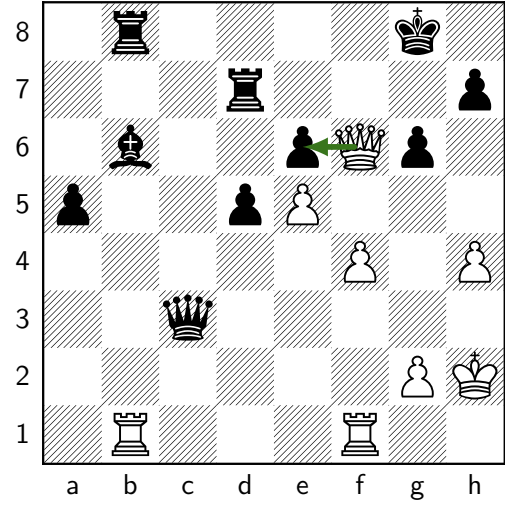


(d)

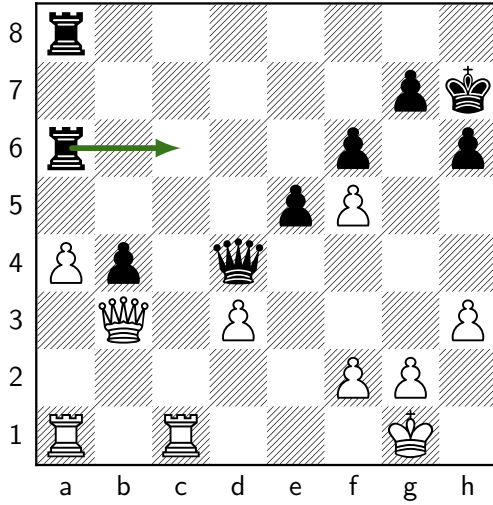
Figure 23: **Move 29, White: Rc3.** (a) Our position; Melanie plays Rc3 (e3c3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#1090` (ply 54) [structural distance $d=104$, QAP obj=38]: the master played Bc6; under μ our *Re3* plays the role of that game’s *Bg2* (a **cross-type** correspondence); cross-type roles: our *Qd5* as their *Ra5*, our *rb6* as their *be6*; transferred move e3c3 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#1562` (ply 53) [structural distance $d=81$, QAP obj=36]: the master played Rc8; under μ our *Rh1* plays the role of that game’s *Re1*; cross-type roles: our *Pg2* as their *Ra7*, our *nh7* as their *Pg7*; transferred move h1g1. (d) Retrieved from `ccr1_top_p0001#3983` (ply 62) [structural distance $d=87$, QAP obj=36]: the master played Rce1; under μ our *Rh1* plays the role of that game’s *Rc1*; cross-type roles: our *Qd5* as their *Re3*, our *bg7* as their *Pa4*; transferred move h1e1. Synthesis: 1 retrieved analog transferred to Rc3 (summed QAP objective 38, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



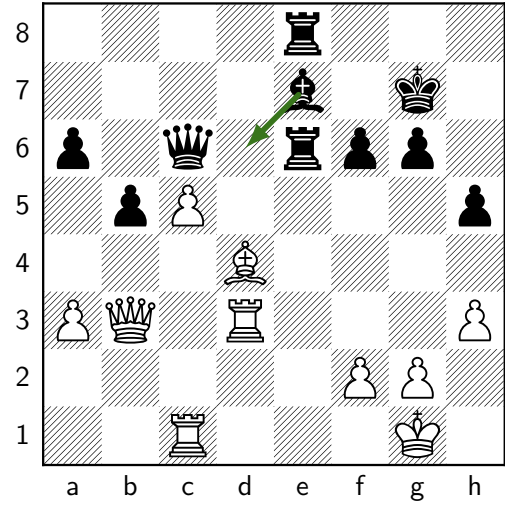
(a)



(b)

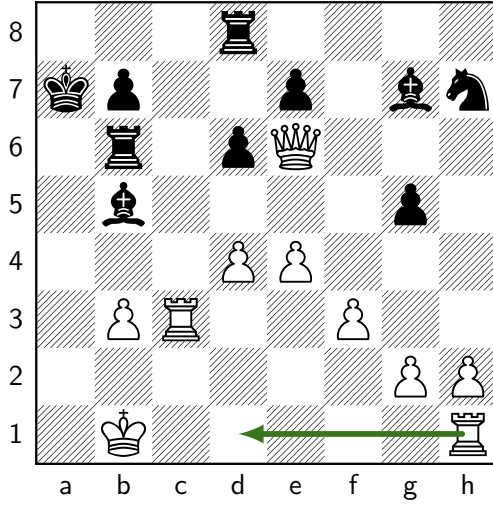


(c)

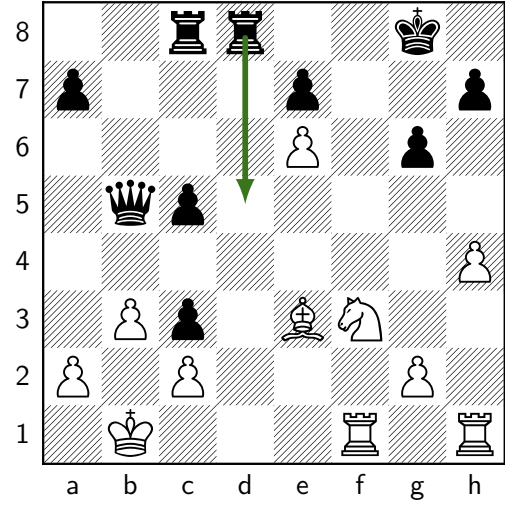


(d)

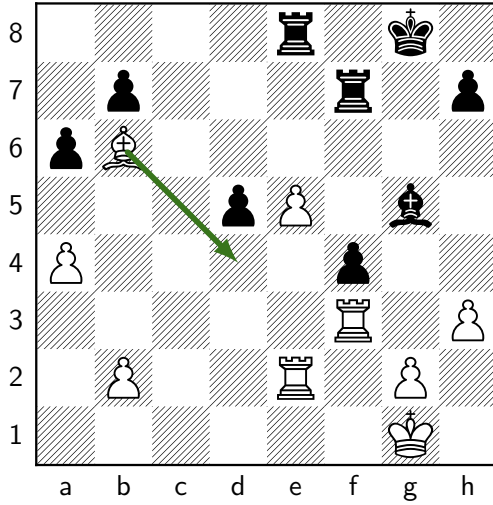
Figure 24: **Move 30, White: Qe6.** (a) Our position; Melanie plays Qe6 (d5e6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0000#1297` (ply 68) [structural distance $d=75$, QAP obj=78]: the master played Qxe6+; under μ our Qd5 plays the role of that game’s Qf6; cross-type roles: our Qd5 as their Rb1, our Pe4 as their Rf1; transferred move d5e6 (**endorses this move**). (c) Retrieved from `ccrl_top_p0001#160` (ply 63) [structural distance $d=86$, QAP obj=73]: the master played Rc6; under μ our Qd5 plays the role of that game’s Ra3 (a **cross-type** correspondence); cross-type roles: our Pd4 as their Ra8, our Rh1 as their Ph6; transferred move d5e5. (d) Retrieved from `ccrl_top_p0003#1957` (ply 63) [structural distance $d=92$, QAP obj=67]: the master played Bd6; under μ our Qd5 plays the role of that game’s Be2 (a **cross-type** correspondence); cross-type roles: our rb6 as their Pa6, our bb5 as their Re6; transferred move d5c5. Synthesis: 2 retrieved analogs transferred to Qe6 (summed QAP objective 117, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



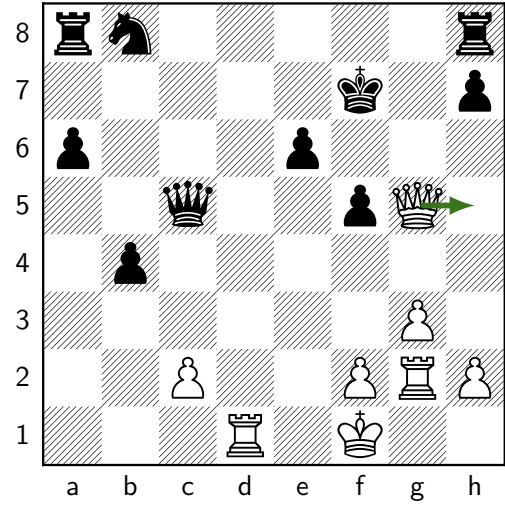
(a)



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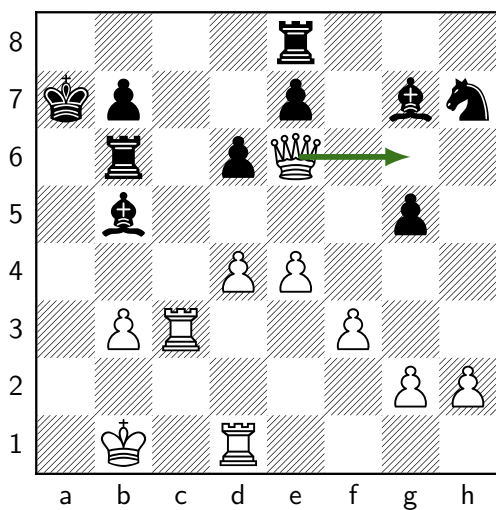


(c)

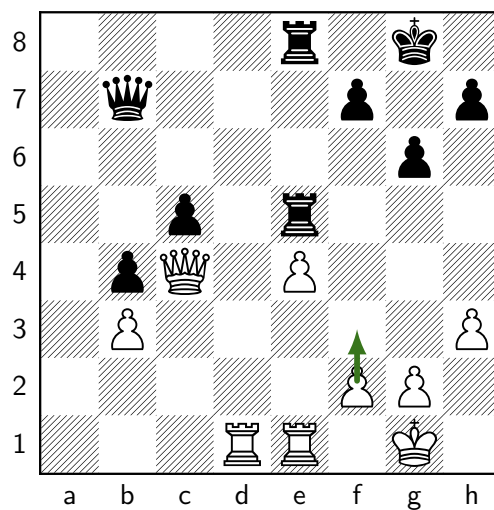


(d)

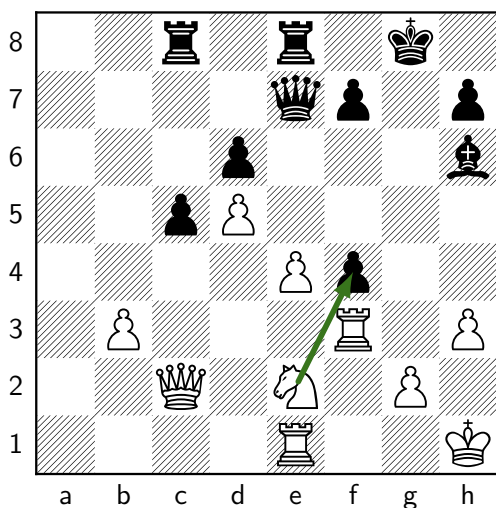
Figure 25: **Move 31, White: Rd1.** (a) Our position; Melanie plays Rd1 (**h1d1**). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#1721` (ply 53) [structural distance $d=66$, QAP obj=50]: the master played Rd5; under μ our *Rh1* plays the role of that game’s *Rd1*; cross-type roles: our *pb7* as their *Rc8*, our *bb5* as their *Pe7*; transferred move **h1d1** (**endorses this move**). (c) Retrieved from `ccr1_top_p0001#1080` (ply 60) [structural distance $d=74$, QAP obj=58]: the master played Bd4; under μ our *Rh1* plays the role of that game’s *Bb6* (a **cross-type** correspondence); cross-type roles: our *Qe6* as their *Rf3*, our *rb6* as their *bg5*; transferred move **h1g1**. (d) Retrieved from `ccr1_top_p0002#2845` (ply 48) [structural distance $d=79$, QAP obj=55]: the master played Qh5+; under μ our *Qe6* plays the role of that game’s *Qg5*; cross-type roles: our *bg7* as their *pa6*, our *rd8* as their *pe6*; transferred move **e6g6**. Synthesis: 1 retrieved analog transferred to Rd1 (summed QAP objective 50, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



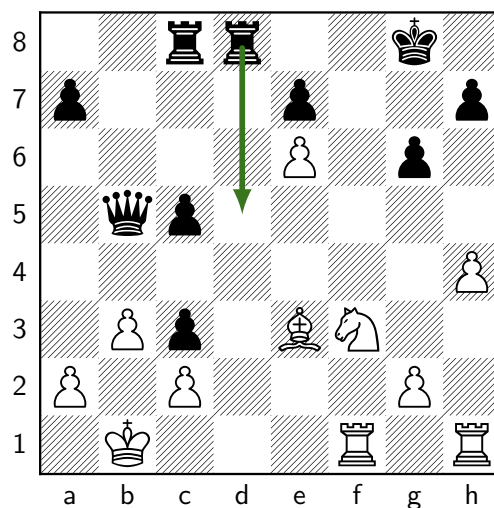
(a)



(b)



(c)



(d)

Figure 26: **Move 32, White: Qg6.** (a) Our position; Melanie plays Qg6 (e6g6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0000#3024` (ply 56) [structural distance $d=87$, QAP obj=112]: the master played f3; under μ our Pg2 plays the role of that game’s Pf2; cross-type roles: our bb5 as their Rd1, our Qe6 as their re5; transferred move g2g3. (c) Retrieved from `ccrl_top_p0000#854` (ply 54) [structural distance $d=80$, QAP obj=52]: the master played Nx f4; under μ our Qe6 plays the role of that game’s Ne2 (a **cross-type** correspondence); cross-type roles: our nh7 as their Pg2, our Qe6 as their Rf3; transferred move e6g4. (d) Retrieved from `ccrl_top_p0000#1721` (ply 53) [structural distance $d=72$, QAP obj=50]: the master played Rd5; under μ our Rd1 plays the role of that game’s Rd1; cross-type roles: our pb7 as their Rc8, our bb5 as their Pe7; transferred move d1e1. Synthesis: no retrieved analog transferred to Qg6 – the **search** selector chose it on value (material / mate); panels (b)–(d) show the strongest nearby analogies.

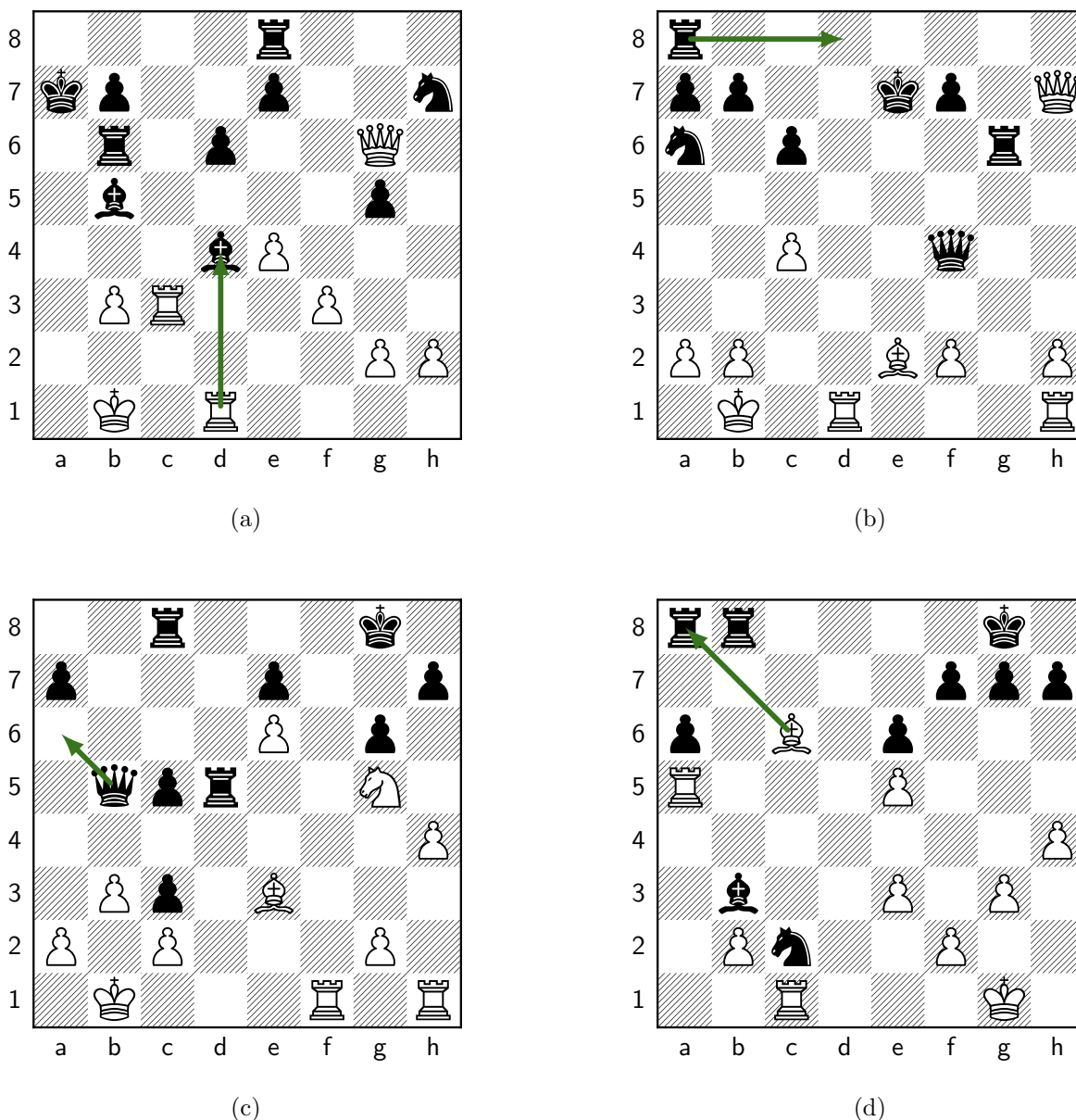


Figure 27: **Move 33, White: Rxd4**. (a) Our position; Melanie plays Rxd4 (d1d4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#1946` (ply 53) [structural distance $d=56$, QAP obj=38]: the master played Rd8; under μ our Pg2 plays the role of that game’s Ra1 (a **cross-type** correspondence); cross-type roles: our Rc3 as their Na6, our bb5 as their pc4; transferred move g2g3. (c) Retrieved from `ccr1_top_p0000#1721` (ply 55) [structural distance $d=79$, QAP obj=33]: the master played Qa6; under μ our Qg6 plays the role of that game’s Qb4; cross-type roles: our Rc3 as their Pe7, our bd4 as their Ph7; transferred move g6h6. (d) Retrieved from `ccr1_top_p0002#583` (ply 52) [structural distance $d=81$, QAP obj=24]: the master played Bxa8; under μ our Qg6 plays the role of that game’s Bc6 (a **cross-type** correspondence); cross-type roles: our bd4 as their Rc1, our Rc3 as their nc2; transferred move g6e8. Synthesis: no retrieved analog transferred to Rxd4 – the **search** selector chose it on value (material / mate); panels (b)–(d) show the strongest nearby analogies.

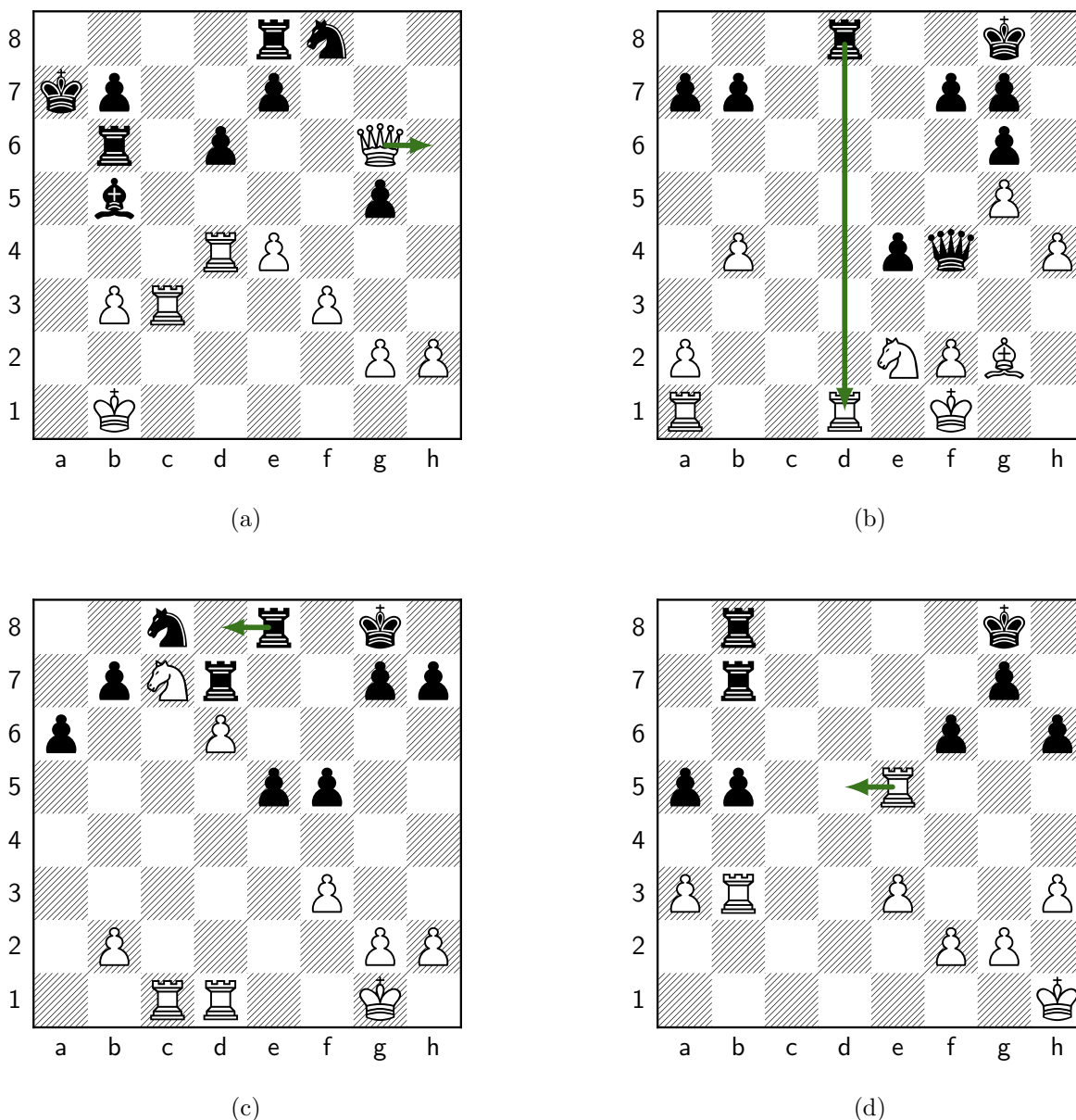
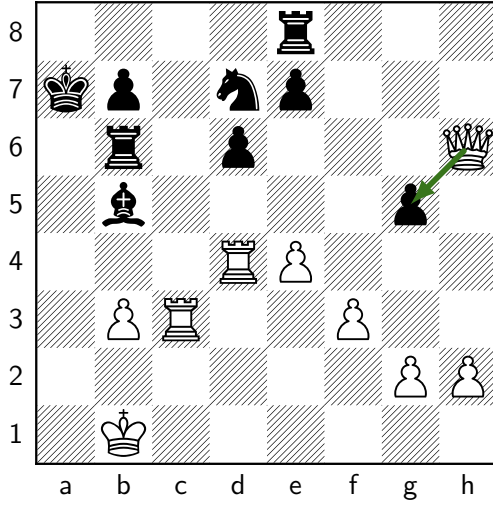
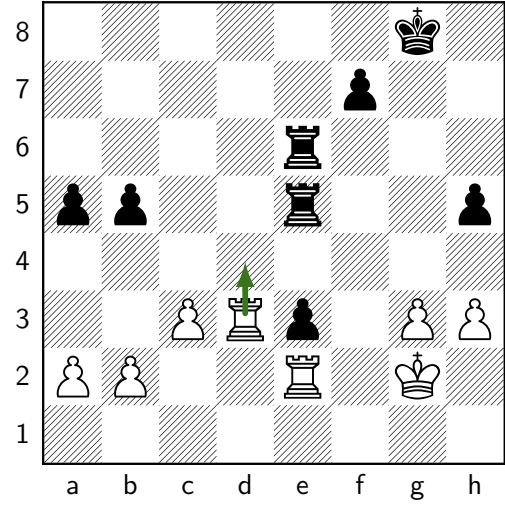


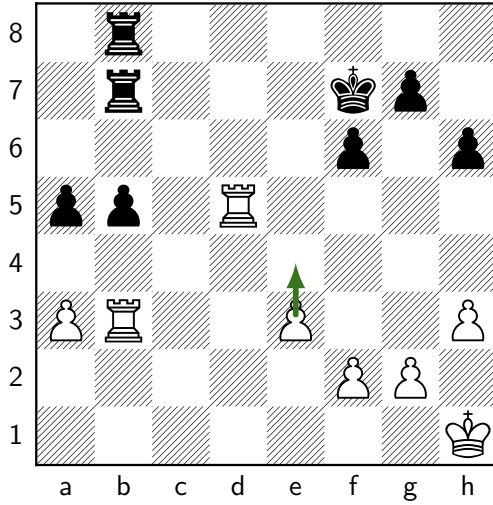
Figure 28: **Move 34, White: Qh6.** (a) Our position; Melanie plays **Qh6** (**g6h6**). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#2248` (ply 57) [structural distance $d=73$, QAP obj=57]: the master played **Rxd1+**; under μ our *Rc3* plays the role of that game’s *Rd1*; cross-type roles: our *Rd4* as their *Pg6*, our *re8* as their *ph4*; transferred move **c3c8**. (c) Retrieved from `ccr1_top_p0002#914` (ply 59) [structural distance $d=71$, QAP obj=41]: the master played **Red8**; under μ our *Ph2* plays the role of that game’s *Re1* (a **cross-type** correspondence); cross-type roles: our *Rd4* as their *Nc8*, our *re8* as their *nc7*; transferred move **h2h3**. (d) Retrieved from `ccr1_top_p0001#50` (ply 72) [structural distance $d=62$, QAP obj=33]: the master played **Rd5**; under μ our *Qg6* plays the role of that game’s *Re5* (a **cross-type** correspondence); cross-type roles: our *Qg6* as their *Rb3*, our *bb5* as their *pa5*; transferred move **g6g5**. Synthesis: no retrieved analog transferred to **Qh6** – the **search** selector chose it on value (material / mate); panels (b)–(d) show the strongest nearby analogies.



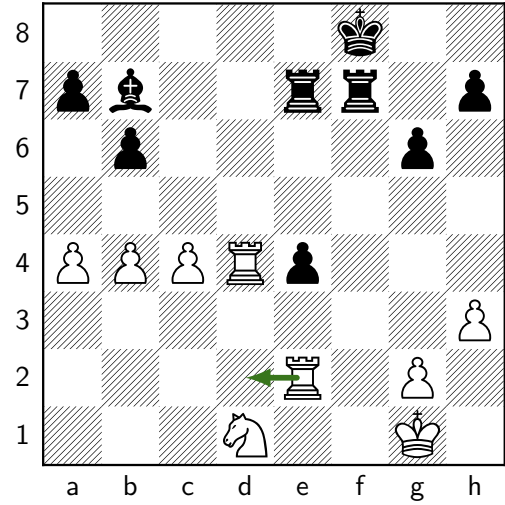
(a)



(b)

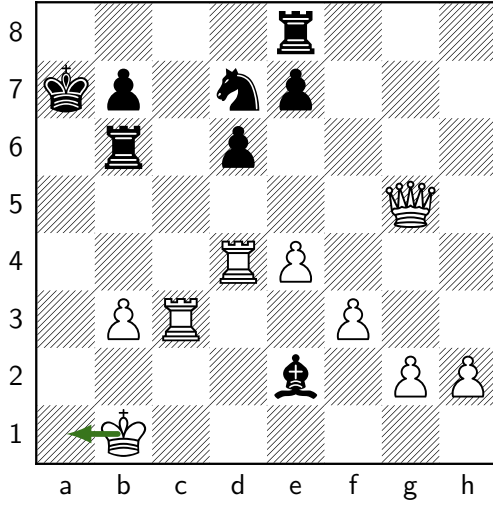


(c)

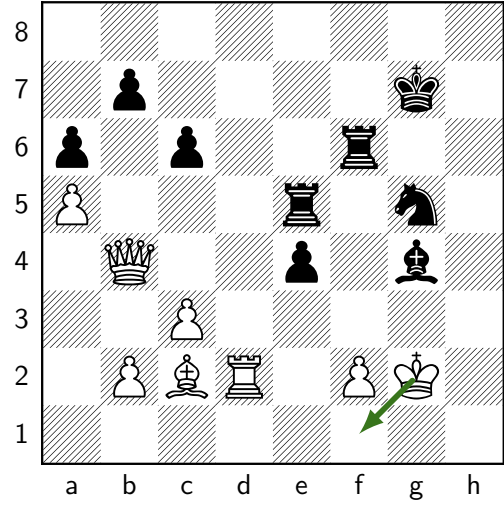


(d)

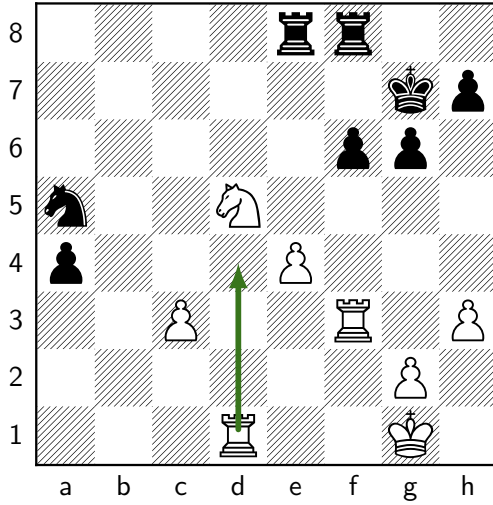
Figure 29: **Move 35, White: Qxg5.** (a) Our position; Melanie plays Qxg5 (h6g5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#1751` (ply 70) [structural distance $d=59$, QAP obj=36]: the master played Rd4; under μ our Qh6 plays the role of that game’s Rd3 (a **cross-type** correspondence); cross-type roles: our Qh6 as their Re2, our Rc3 as their Ph3; transferred move h6g5 (**endorses this move**). (c) Retrieved from `ccr1_top_p0001#50` (ply 74) [structural distance $d=52$, QAP obj=48]: the master played e4; under μ our Ph2 plays the role of that game’s Pe3; cross-type roles: our Rc3 as their Ph3, our nd7 as their pa5; transferred move h2h4. (d) Retrieved from `ccr1_top_p0000#1104` (ply 74) [structural distance $d=60$, QAP obj=46]: the master played Red2; under μ our Rd4 plays the role of that game’s Re2; cross-type roles: our Pb3 as their Nd1, our re8 as their Ph3; transferred move d4d2. Synthesis: 1 retrieved analog transferred to Qxg5 (summed QAP objective 36, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



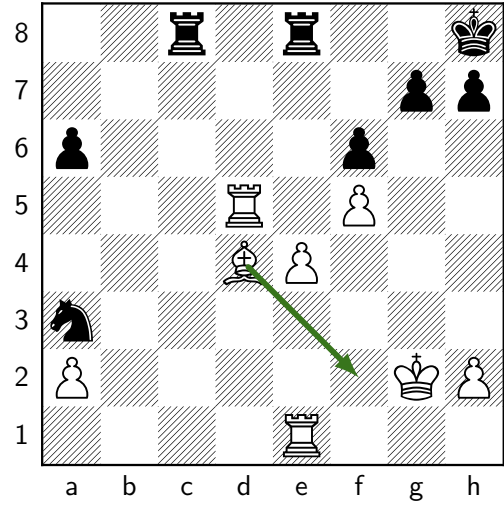
(a)



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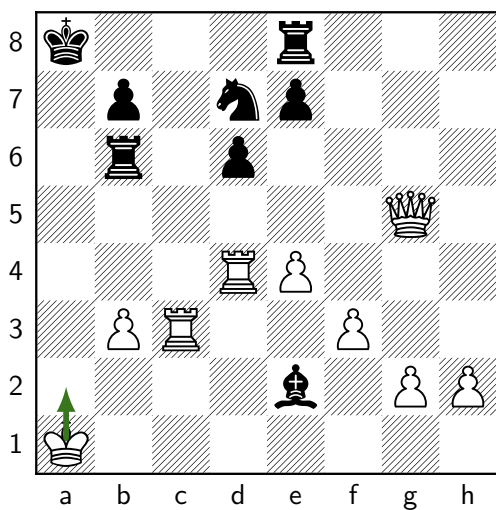


(c)

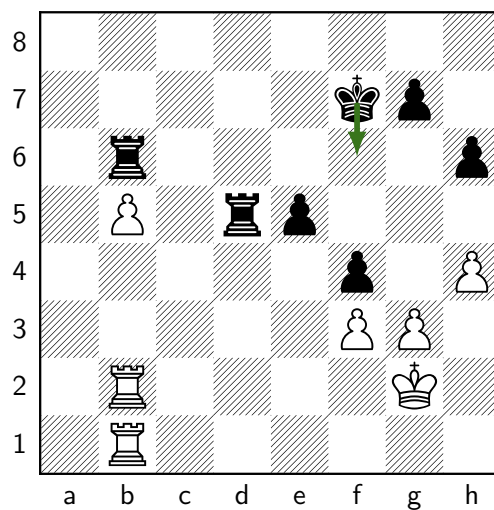


(d)

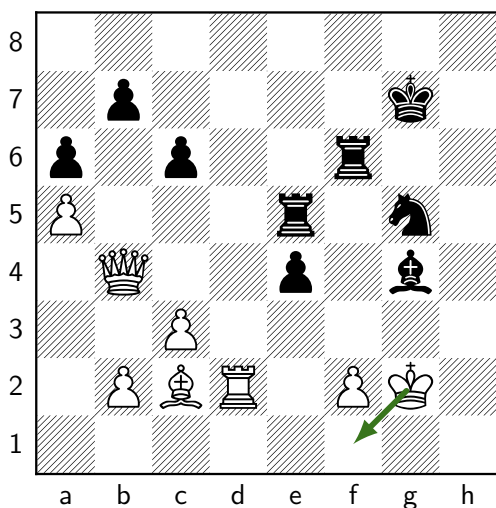
Figure 30: **Move 36, White: Ka1.** (a) Our position; Melanie plays Ka1 (b1a1). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0001#1206` (ply 74) [structural distance $d=56$, QAP obj=45]: the master played Kf1; under μ our Kb1 plays the role of that game’s Kg2; cross-type roles: our Rd4 as their Bc2, our Rd4 as their Qb4; transferred move b1a1 (**endorses this move**). (c) Retrieved from `ccrl_top_p0002#2378` (ply 72) [structural distance $d=59$, QAP obj=33]: the master played Rd4; under μ our Pg2 plays the role of that game’s Rd1 (a **cross-type** correspondence); cross-type roles: our Rc3 as their na5, our Rd4 as their Nd5; transferred move g2g3. (d) Retrieved from `ccrl_top_p0003#2167` (ply 78) [structural distance $d=61$, QAP obj=30]: the master played Bf2; under μ our Rd4 plays the role of that game’s Bd4 (a **cross-type** correspondence); cross-type roles: our pb7 as their na3, our Qg5 as their Rd5; transferred move d4a4. Synthesis: 1 retrieved analog transferred to Ka1 (summed QAP objective 45, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



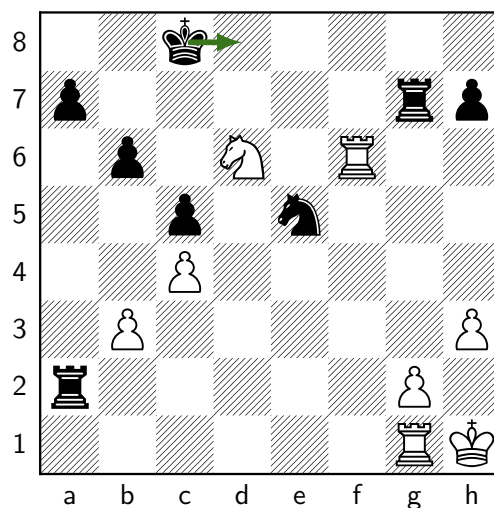
(a)



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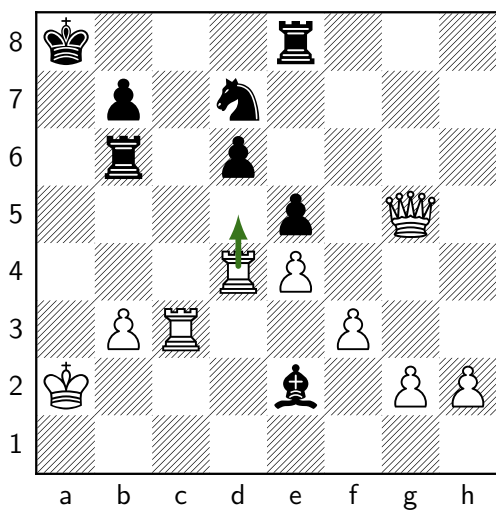


(c)

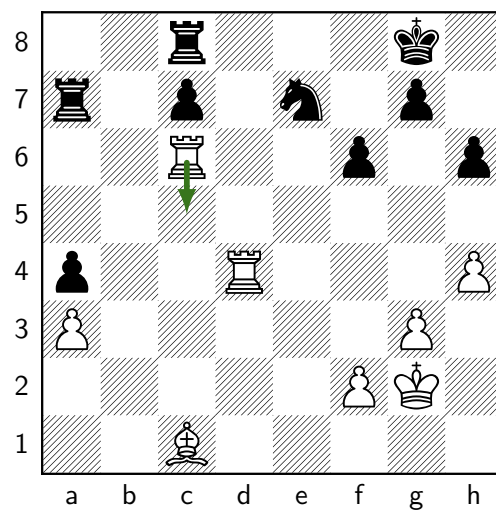


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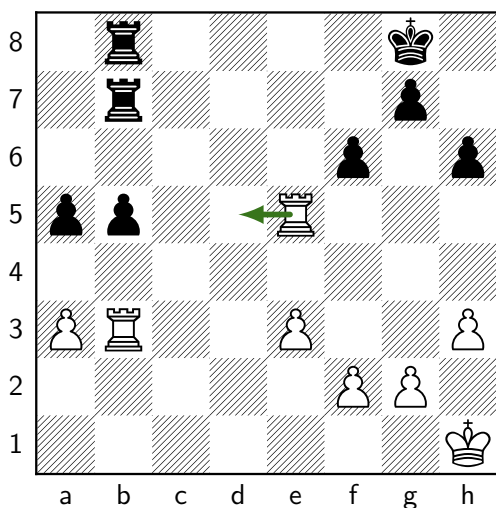
Figure 31: **Move 37, White: Ka2.** (a) Our position; Melanie plays Ka2 (a1a2). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0003#1590` (ply 85) [structural distance $d=58$, QAP obj=46]: the master played Kf6; under μ our Ka1 plays the role of that game’s Kf2; cross-type roles: our Qg5 as their Ph6, our be2 as their Pf4; transferred move a1a2 (**endorses this move**). (c) Retrieved from `ccr1_top_p0001#1206` (ply 74) [structural distance $d=51$, QAP obj=45]: the master played Kf1; under μ our Ka1 plays the role of that game’s Kg2; cross-type roles: our Rd4 as their Bc2, our Rd4 as their Qb4; transferred move a1a2 (**endorses this move**). (d) Retrieved from `ccr1_top_p0002#1022` (ply 59) [structural distance $d=56$, QAP obj=12]: the master played Kd8; under μ our Ka1 plays the role of that game’s Kc1; cross-type roles: our be2 as their Pb6, our re8 as their nd6; transferred move a1a2 (**endorses this move**). Synthesis: 3 retrieved analogs transferred to Ka2 (summed QAP objective 102, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



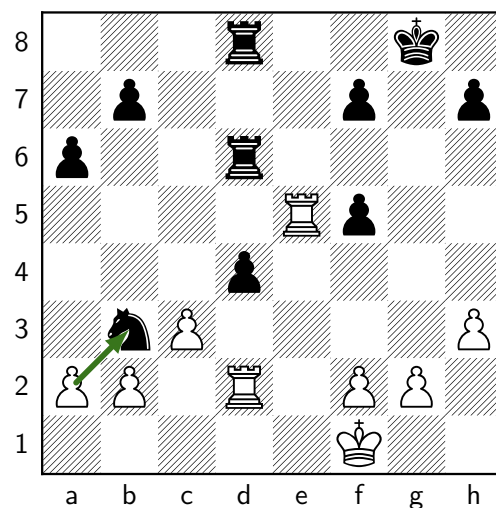
(a)



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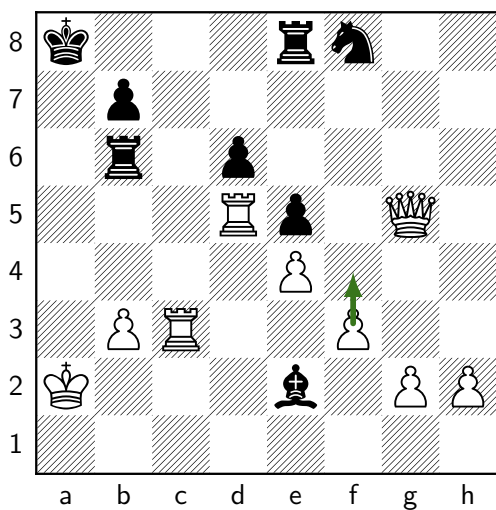


(c)

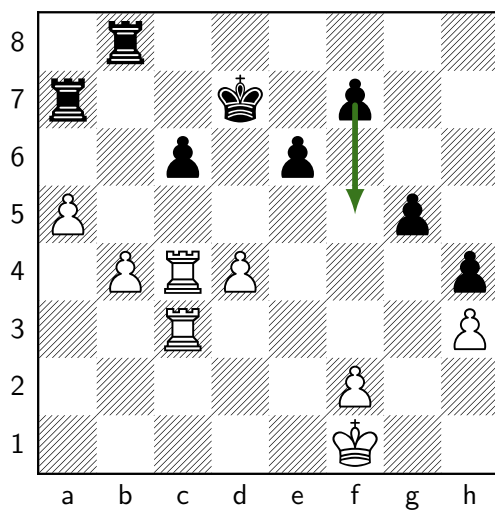


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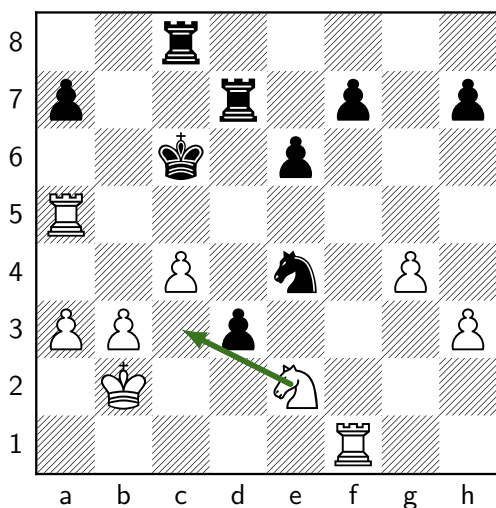
Figure 32: **Move 38, White: Rd5.** (a) Our position; Melanie plays Rd5 (d4d5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#1947` (ply 66) [structural distance $d=62$, QAP obj=45]: the master played Rc5; under μ our $Rd4$ plays the role of that game’s $Rc6$; cross-type roles: our $Pe4$ as their $Bc1$, our $be2$ as their $Pf2$; transferred move d4d5 (**endorses this move**). (c) Retrieved from `ccr1_top_p0001#50` (ply 72) [structural distance $d=55$, QAP obj=43]: the master played Rd5; under μ our $Rd4$ plays the role of that game’s $Re5$; cross-type roles: our $be2$ as their $Pg2$, our $Rc3$ as their $Ph3$; transferred move d4d5 (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#646` (ply 62) [structural distance $d=56$, QAP obj=68]: the master played $axb3$; under μ our $Qg5$ plays the role of that game’s $Pa2$ (a **cross-type** correspondence); cross-type roles: our $Rc3$ as their $Pf2$, our $pe5$ as their $nb3$; transferred move $g5c1$. Synthesis: 2 retrieved analogs transferred to Rd5 (summed QAP objective 87, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



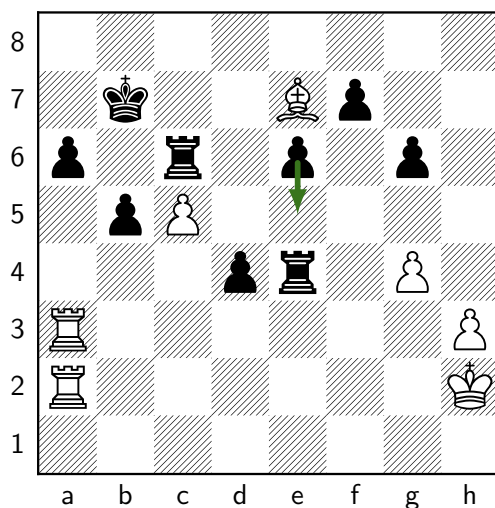
(a)



(b)



(c)



(d)

Figure 33: **Move 39, White: f4.** (a) Our position; Melanie plays f4 (f3f4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#3848` (ply 75) [structural distance $d=57$, QAP obj=25]: the master played f5; under μ our $Pf3$ plays the role of that game’s $Pf2$; cross-type roles: our $Qg5$ as their $Ra7$, our $Rc3$ as their $Pe6$; transferred move f3f4 (**endorses this move**). (c) Retrieved from `ccr1_top_p0001#1748` (ply 68) [structural distance $d=56$, QAP obj=8]: the master played Nc3; under μ our $Pf3$ plays the role of that game’s $Ne2$ (a **cross-type** correspondence); cross-type roles: our $pe5$ as their $Rf1$, our $Rd5$ as their $Pb3$; transferred move f3f4 (**endorses this move**). (d) Retrieved from `ccr1_top_p0003#174` (ply 93) [structural distance $d=54$, QAP obj=39]: the master played e5; under μ our $Rd5$ plays the role of that game’s $Pe3$ (a **cross-type** correspondence); cross-type roles: our $Rc3$ as their $be7$, our $be2$ as their $Rc6$; transferred move d5d6. Synthesis: 2 retrieved analogs transferred to f4 (summed QAP objective 34, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.

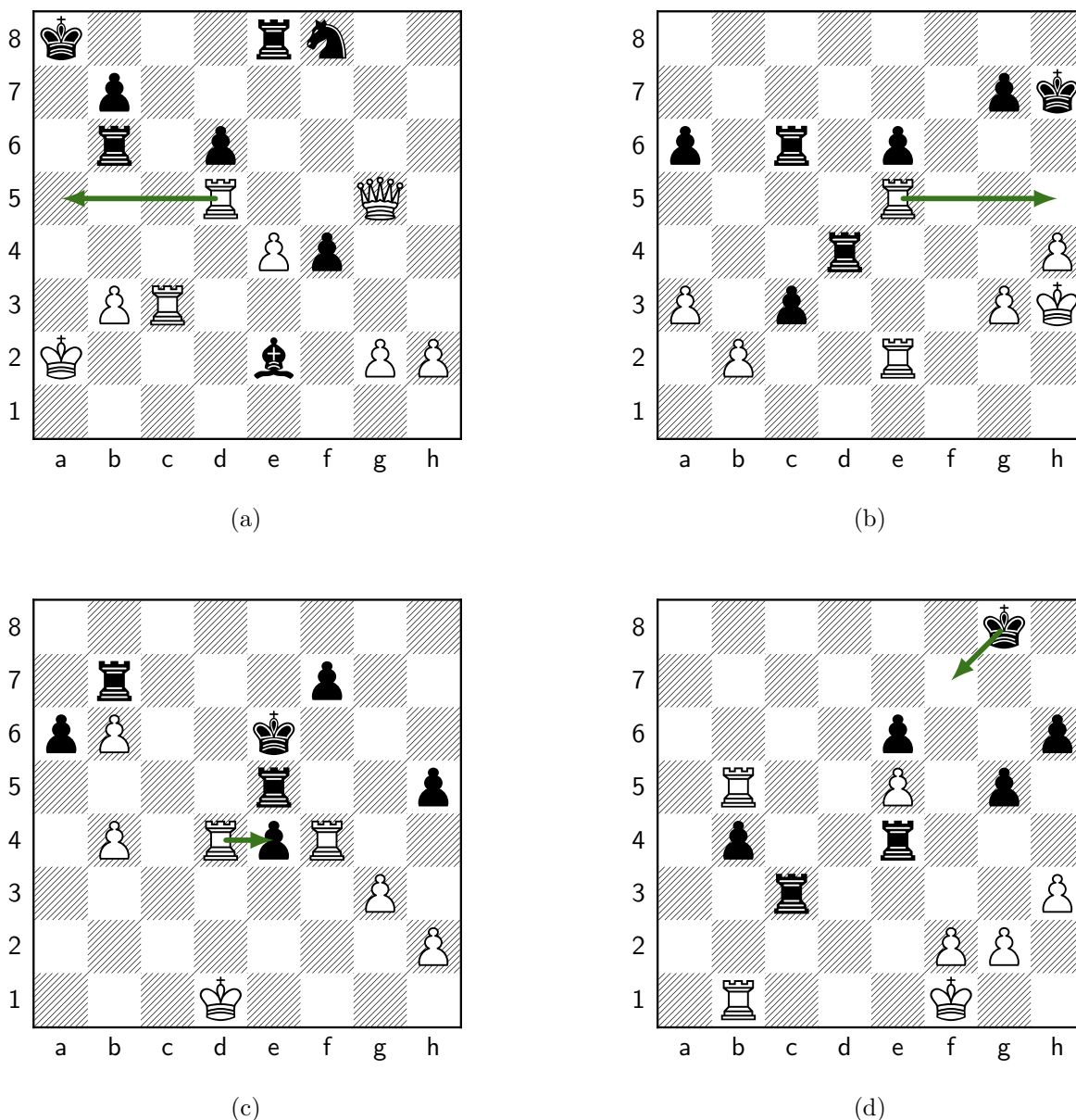
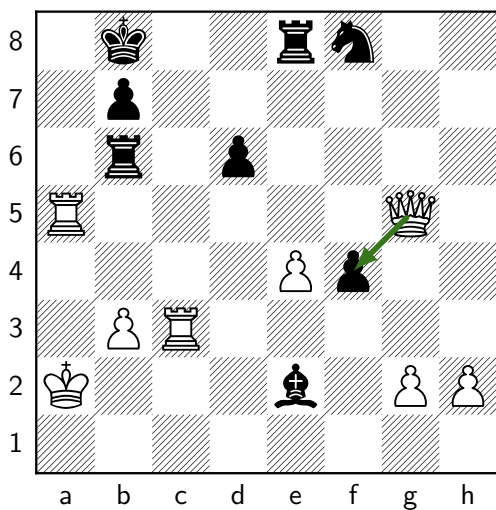
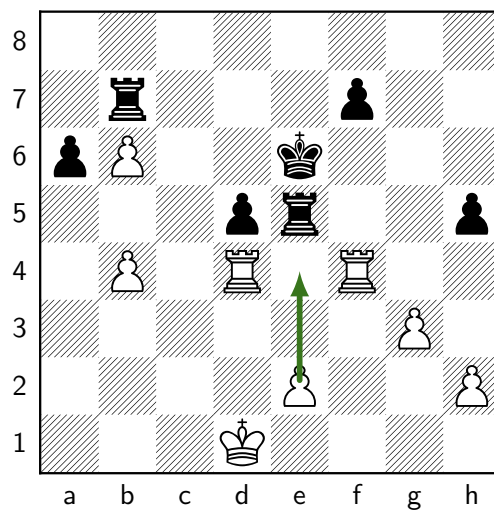


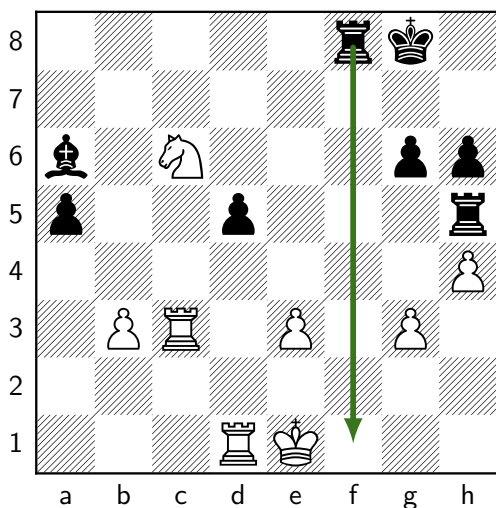
Figure 34: **Move 40, White: Ra5+.** (a) Our position; Melanie plays Ra5+ (d5a5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0001#2149` (ply 86) [structural distance $d=56$, QAP obj=44]: the master played Rh5+; under μ our Rd5 plays the role of that game’s Re5; cross-type roles: our Qg5 as their Pb2, our be2 as their pa6; transferred move d5a5 (**endorses this move**). (c) Retrieved from `ccrl_top_p0000#294` (ply 82) [structural distance $d=56$, QAP obj=54]: the master played Rdx e4; under μ our Rd5 plays the role of that game’s Rd4; cross-type roles: our be2 as their ph5, our Rc3 as their pa6; transferred move d5c5. (d) Retrieved from `ccrl_top_p0003#949` (ply 65) [structural distance $d=53$, QAP obj=39]: the master played Kf7; under μ our Ka2 plays the role of that game’s Kg1; cross-type roles: our Rc3 as their Pg5, our Qg5 as their Re4; transferred move a2a3. Synthesis: 1 retrieved analog transferred to Ra5+ (summed QAP objective 44, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



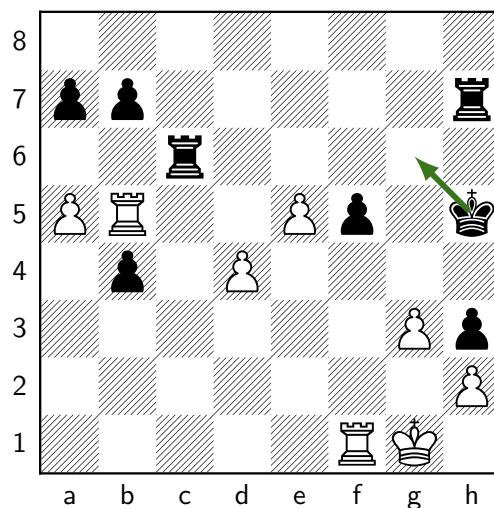
(a)



(b)

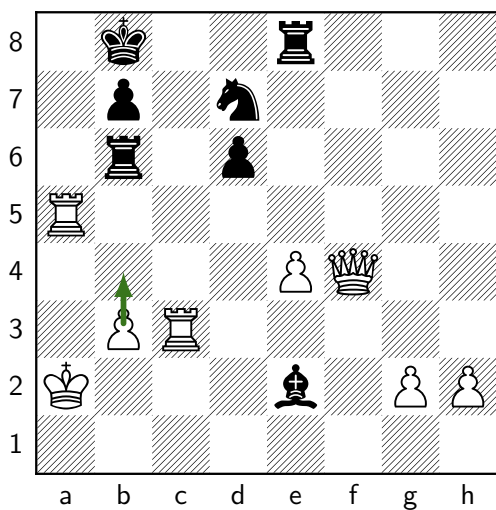


(c)

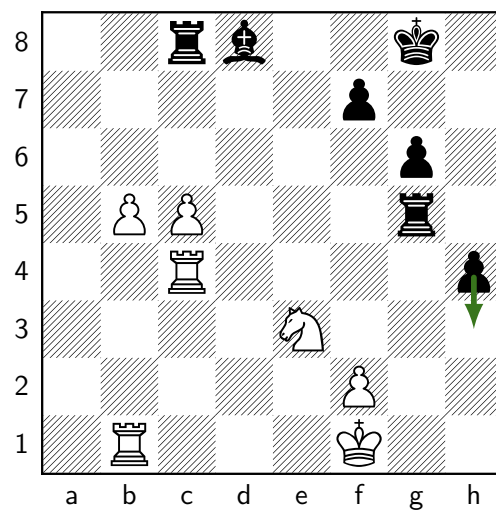


(d)

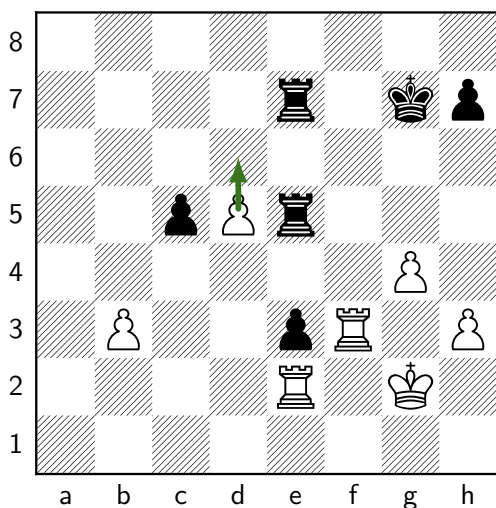
Figure 35: **Move 41, White: Qxf4.** (a) Our position; Melanie plays Qxf4 (g5f4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#294` (ply 80) [structural distance $d=62$, QAP obj=36]: the master played e4; under μ our *Pe4* plays the role of that game’s *Pe2*; cross-type roles: our *be2* as their *Pg3*, our *Qg5* as their *Rd4*; transferred move e4e5. (c) Retrieved from `ccr1_top_p0002#1243` (ply 65) [structural distance $d=49$, QAP obj=34]: the master played Rf1+; under μ our *Ra5* plays the role of that game’s *Rf1*; cross-type roles: our *Rc3* as their *Ba6*, our *rb6* as their *nc6*; transferred move a5a8. (d) Retrieved from `ccr1_top_p0000#955` (ply 75) [structural distance $d=69$, QAP obj=32]: the master played Kg6; under μ our *Ka2* plays the role of that game’s *Kh4*; cross-type roles: our *Qg5* as their *rb5*, our *be2* as their *pe5*; transferred move a2a3. Synthesis: no retrieved analog transferred to Qxf4 – the search selector chose it on value (material / mate); panels (b)–(d) show the strongest nearby analogies.



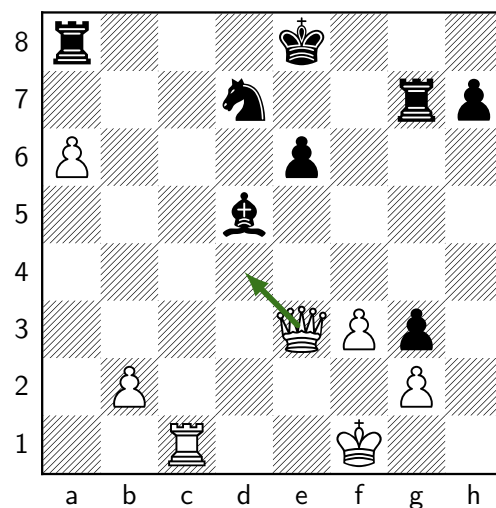
(a)



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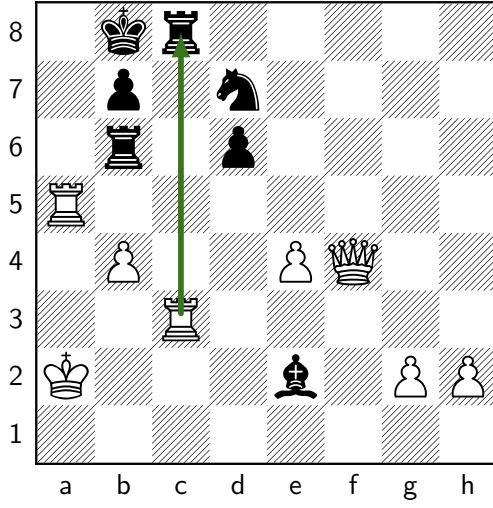


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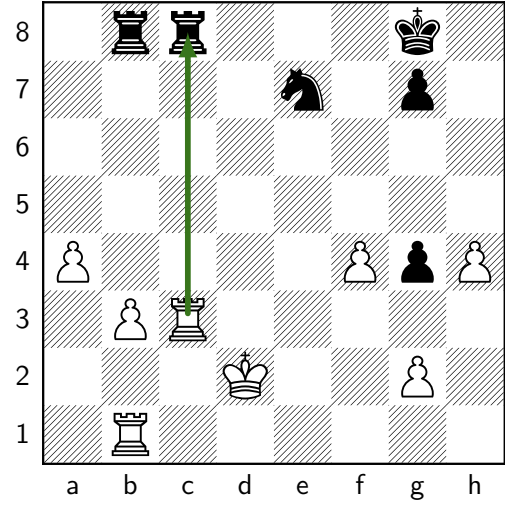


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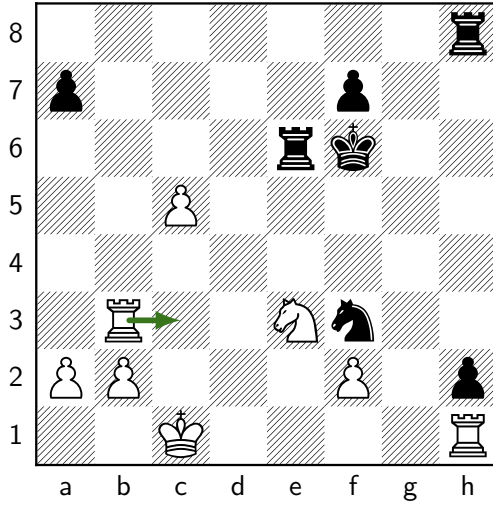
Figure 36: **Move 42, White: b4.** (a) Our position; Melanie plays b4 (b3b4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0002#2111` (ply 87) [structural distance $d=58$, QAP obj=47]: the master played h3; under μ our $Pb3$ plays the role of that game’s $Ph5$; cross-type roles: our $Qf4$ as their $Rc8$, our $Pe4$ as their $Bd8$; transferred move b3b4 (**endorses this move**). (c) Retrieved from `ccrl_top_p0000#854` (ply 96) [structural distance $d=67$, QAP obj=37]: the master played d6; under μ our $Pb3$ plays the role of that game’s $Pd5$; cross-type roles: our $Qf4$ as their $Re2$, our $be2$ as their $Pb3$; transferred move b3b4 (**endorses this move**). (d) Retrieved from `ccrl_top_p0002#3674` (ply 72) [structural distance $d=68$, QAP obj=55]: the master played Qd4; under μ our $Qf4$ plays the role of that game’s $Qe3$; cross-type roles: our $Rc3$ as their $Pg2$, our $be2$ as their $pg3$; transferred move f4e3. Synthesis: 2 retrieved analogs transferred to b4 (summed QAP objective 85, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



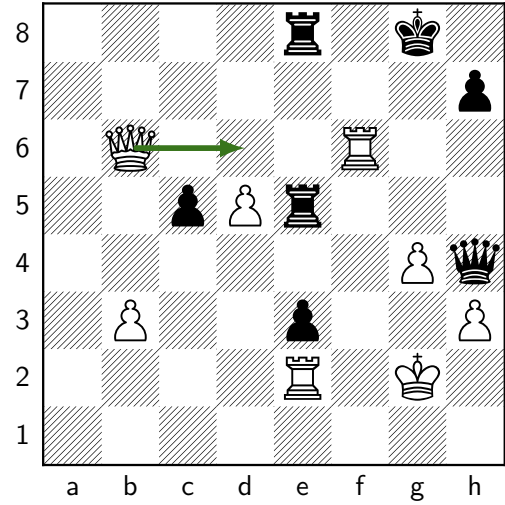
(a)



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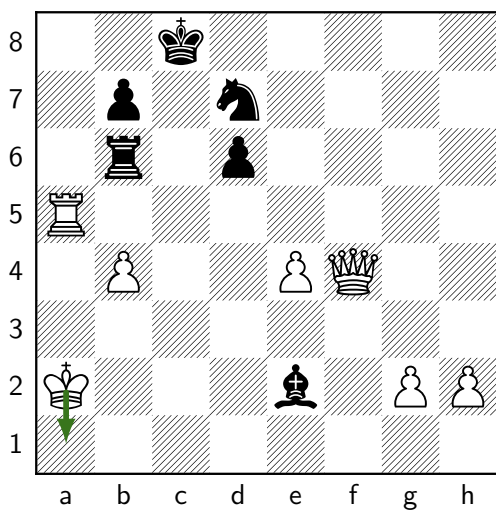


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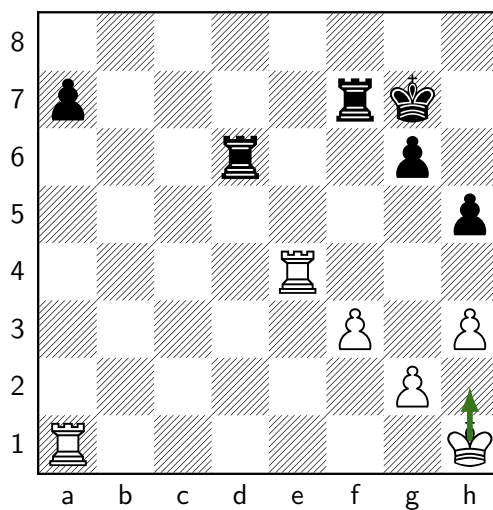


(d)

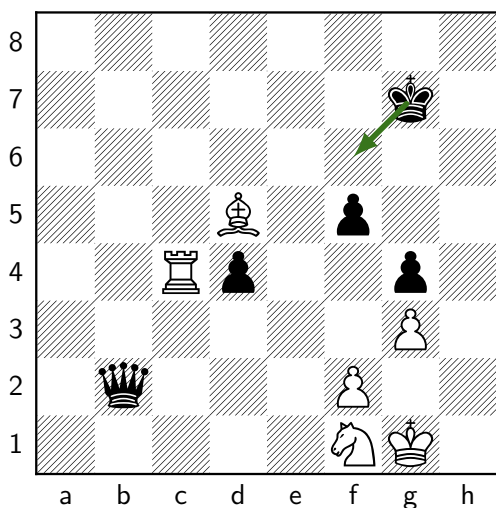
Figure 37: **Move 43, White: Rxc8+.** (a) Our position; Melanie plays Rxc8+ (c3c8). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0003#1891` (ply 74) [structural distance $d=57$, QAP obj=52]: the master played Rxc8+; under μ our *Rc3* plays the role of that game’s *Rc3*; cross-type roles: our *Ra5* as their *Pg2*, our *be2* as their *Pa4*; transferred move **c3c8 (endorses this move)**. (c) Retrieved from `ccr1_top_p0000#822` (ply 68) [structural distance $d=65$, QAP obj=45]: the master played Rc3; under μ our *Ra5* plays the role of that game’s *Rb3*; cross-type roles: our *Qf4* as their *Rh1*, our *be2* as their *Pb2*; transferred move **a5d5**. (d) Retrieved from `ccr1_top_p0000#854` (ply 90) [structural distance $d=69$, QAP obj=28]: the master played Qd6; under μ our *Ra5* plays the role of that game’s *Qb6* (a **cross-type** correspondence); cross-type roles: our *Qf4* as their *Re2*, our *be2* as their *Pb3*; transferred move **a5d5**. Synthesis: 1 retrieved analog transferred to Rxc8+ (summed QAP objective 52, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



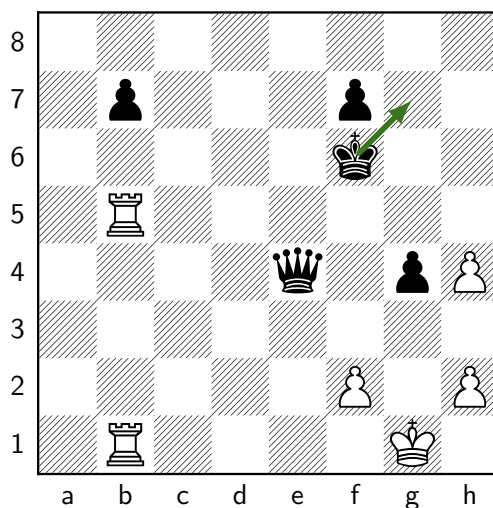
(a)



(b)

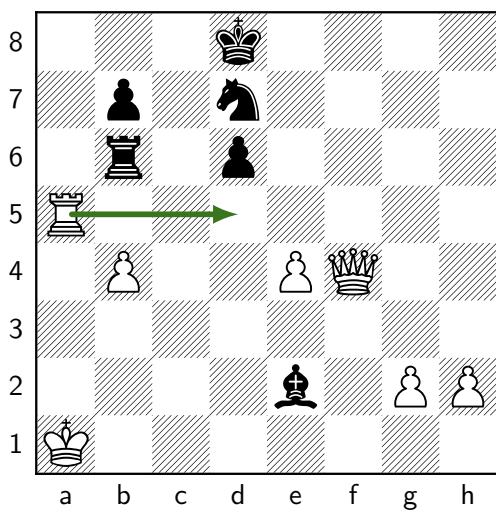


(c)

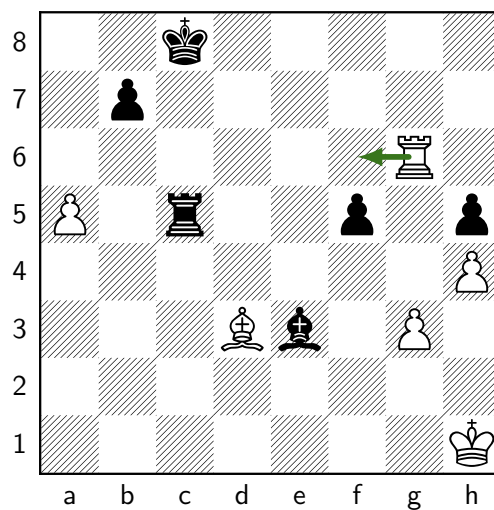


(d)

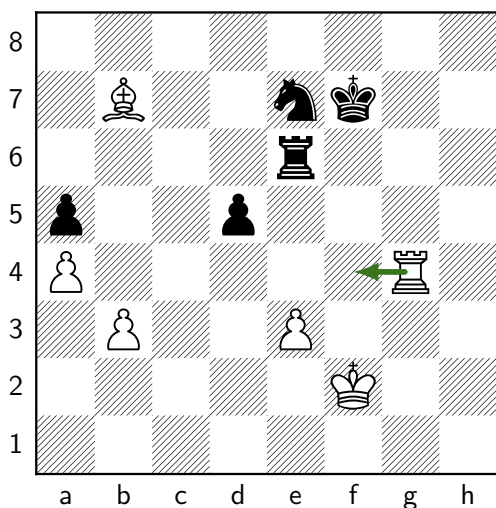
Figure 38: **Move 44, White: Ka1.** (a) Our position; Melanie plays Ka1 (a2a1). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#423` (ply 80) [structural distance $d=46$, QAP obj=18]: the master played Kh2; under μ our Ka2 plays the role of that game's Kh1; cross-type roles: our Qf4 as their Ra1, our nd7 as their ph5; transferred move a2a1 (**endorses this move**). (c) Retrieved from `ccr1_top_p0003#1060` (ply 87) [structural distance $d=43$, QAP obj=18]: the master played Kf6; under μ our Ka2 plays the role of that game's Kg2; cross-type roles: our pb7 as their bd5, our Qf4 as their rc4; transferred move a2a1 (**endorses this move**). (d) Retrieved from `ccr1_top_p0003#97` (ply 91) [structural distance $d=47$, QAP obj=14]: the master played Kg7; under μ our Ka2 plays the role of that game's Kf3; cross-type roles: our Ra5 as their Pf7, our be2 as their ph4; transferred move a2a1 (**endorses this move**). Synthesis: 4 retrieved analogs transferred to Ka1 (summed QAP objective 54, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



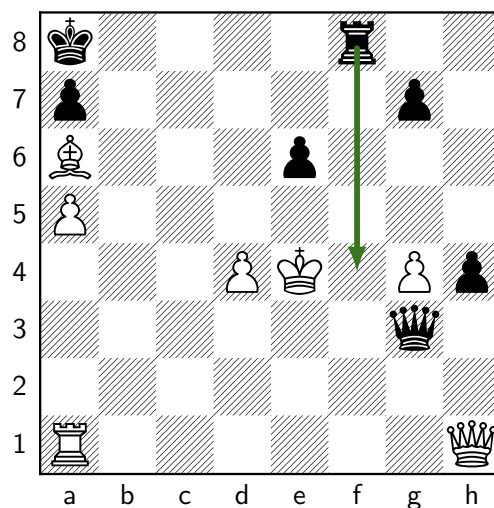
(a)



(b)

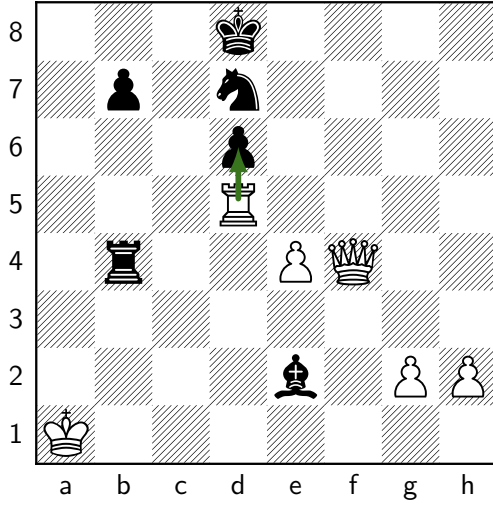


(c)

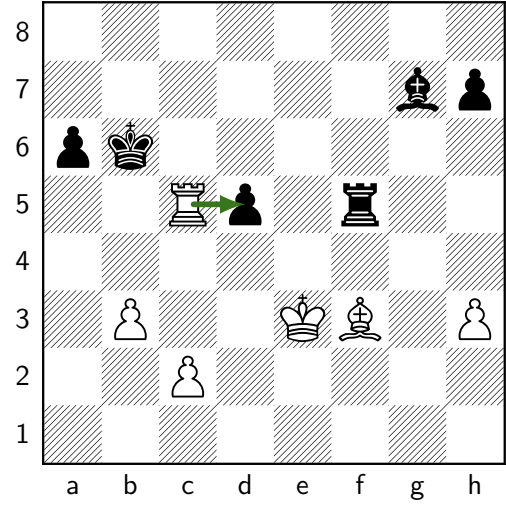


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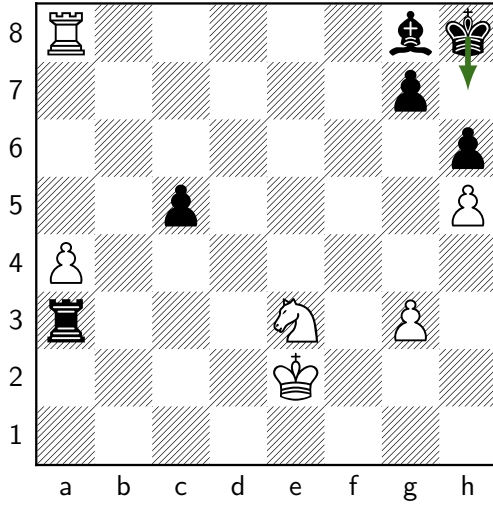
Figure 39: **Move 45, White: Rd5.** (a) Our position; Melanie plays Rd5 (a5d5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#648` (ply 86) [structural distance $d=43$, QAP obj=40]: the master played Rf6; under μ our Ra5 plays the role of that game’s Rg6; cross-type roles: our Qf4 as their Bd3, our nd7 as their be3; transferred move a5d5 (**endorses this move**). (c) Retrieved from `ccr1_top_p0001#3370` (ply 82) [structural distance $d=43$, QAP obj=33]: the master played Rf4+; under μ our Ra5 plays the role of that game’s Rg4; cross-type roles: our be2 as their Pa4, our nd7 as their pa5; transferred move a5d5 (**endorses this move**). (d) Retrieved from `ccr1_top_p0002#2396` (ply 79) [structural distance $d=44$, QAP obj=18]: the master played Rf4+; under μ our Ra5 plays the role of that game’s Rf1; cross-type roles: our nd7 as their Pe6, our rb6 as their Qg3; transferred move a5d5 (**endorses this move**). Synthesis: 3 retrieved analogs transferred to Rd5 (summed QAP objective 90, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



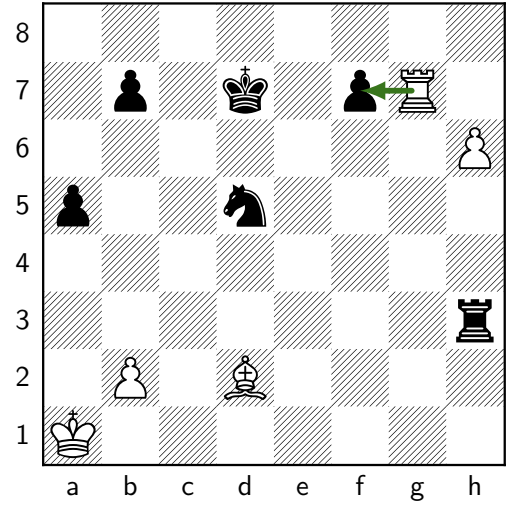
(a)



(b)

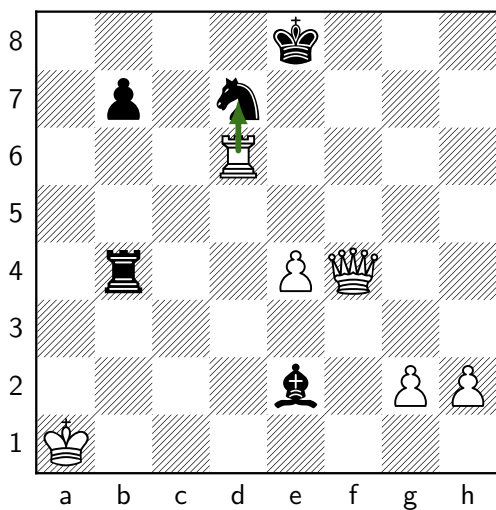


(c)

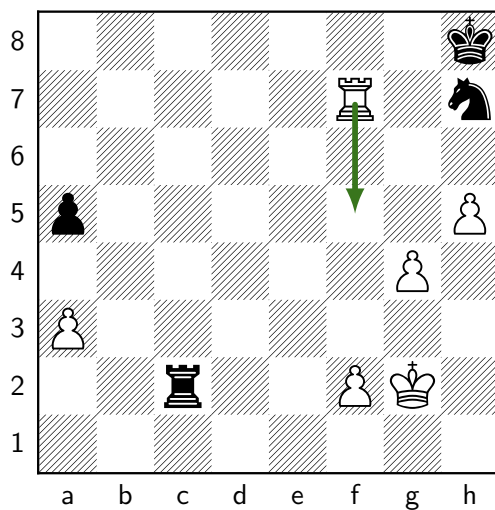


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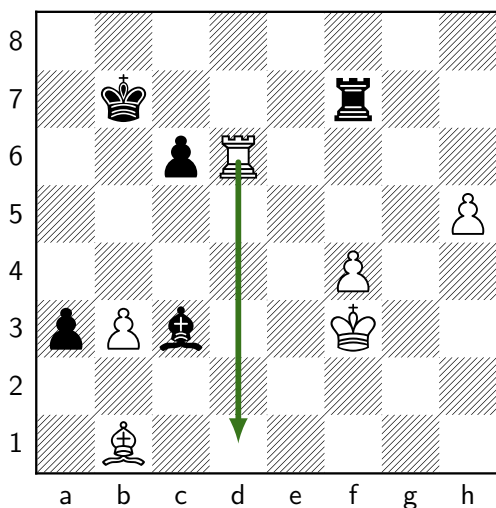
Figure 40: **Move 46, White: Rxd6**. (a) Our position; Melanie plays Rxd6 (d5d6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0003#2426` (ply 74) [structural distance $d=39$, QAP obj=25]: the master played Rxd5; under μ our *Rd5* plays the role of that game’s *Rc5*; cross-type roles: our *Qf4* as their *Bf3*, our *nd7* as their *pa6*; transferred move d5d6 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#1161` (ply 75) [structural distance $d=40$, QAP obj=60]: the master played Kh7; under μ our *Ka1* plays the role of that game’s *Kh1*; cross-type roles: our *Pe4* as their *Bg8*, our *be2* as their *Pg7*; transferred move a1a2. (d) Retrieved from `ccr1_top_p0000#236` (ply 104) [structural distance $d=40$, QAP obj=51]: the master played Rxf7+; under μ our *Rd5* plays the role of that game’s *Rg7*; cross-type roles: our *Qf4* as their *Bd2*, our *be2* as their *nd5*; transferred move d5h5. Synthesis: 1 retrieved analog transferred to Rxd6 (summed QAP objective 25, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



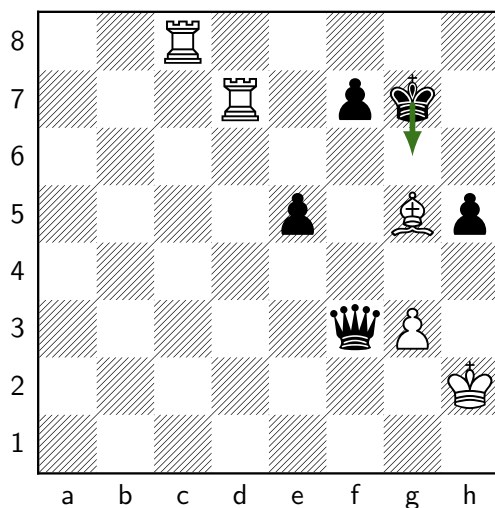
(a)



(b)



(c)



(d)

Figure 41: **Move 47, White: Rxd7.** (a) Our position; Melanie plays Rxd7 (d6d7). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0003#1089` (ply 80) [structural distance $d=41$, QAP obj=32]: the master played Rf5; under μ our *Rd6* plays the role of that game’s *Rf7*; cross-type roles: our *be2* as their *Pa3*, our *Qf4* as their *Pg4*; transferred move d6d5. (c) Retrieved from `ccr1_top_p0000#661` (ply 100) [structural distance $d=43$, QAP obj=28]: the master played Rd1; under μ our *Rd6* plays the role of that game’s *Rd6*; cross-type roles: our *Qf4* as their *Bb1*, our *nd7* as their *pc6*; transferred move d6d1. (d) Retrieved from `ccr1_top_p0002#3505` (ply 85) [structural distance $d=31$, QAP obj=22]: the master played Kg6; under μ our *Ka1* plays the role of that game’s *Kg2*; cross-type roles: our *pb7* as their *rc8*, our *Qf4* as their *Pe5*; transferred move a1a2. Synthesis: no retrieved analog transferred to Rxd7 – the **search** selector chose it on value (material / mate); panels (b)–(d) show the strongest nearby analogies.

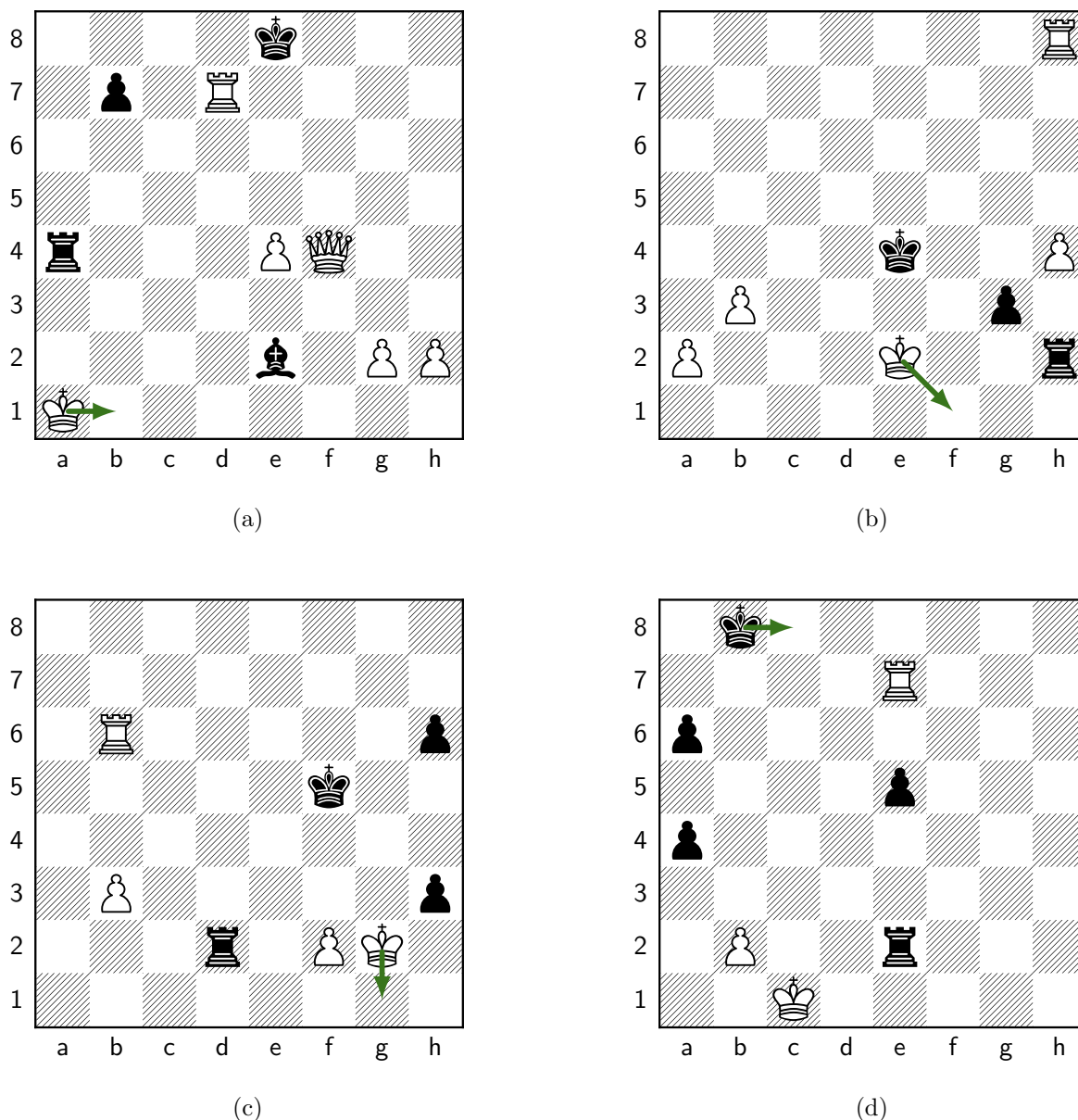
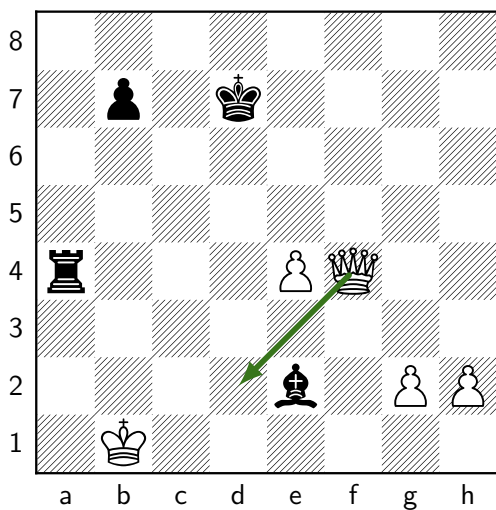
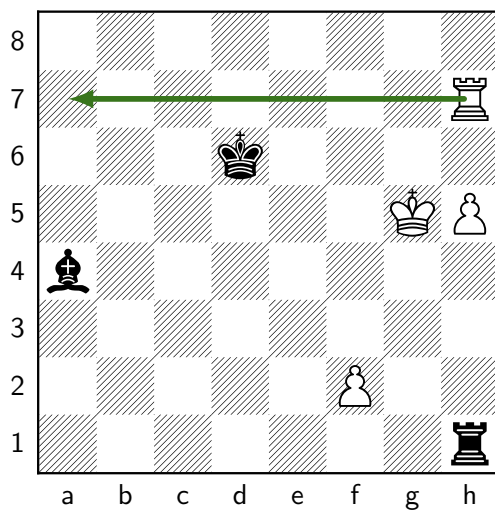


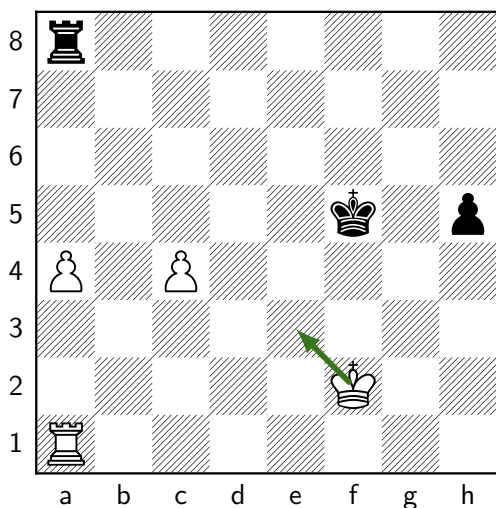
Figure 42: **Move 48, White: Kb1.** (a) Our position; Melanie plays Kb1 (a1b1). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#3580` (ply 78) [structural distance $d=33$, QAP obj=78]: the master played Kf1; under μ our *Ka1* plays the role of that game’s *Ke2*; cross-type roles: our *be2* as their *Pb3*, our *Qf4* as their *Rh8*; transferred move a1b1 (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#519` (ply 98) [structural distance $d=32$, QAP obj=28]: the master played Kg1; under μ our *Ka1* plays the role of that game’s *Kg2*; cross-type roles: our *ra4* as their *ph3*, our *Qf4* as their *Rb6*; transferred move a1b1 (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#2765` (ply 99) [structural distance $d=31$, QAP obj=14]: the master played Kc8; under μ our *Ka1* plays the role of that game’s *Kb1*; cross-type roles: our *be2* as their *Pa4*, our *Rd7* as their *pb2*; transferred move a1b1 (**endorses this move**). Synthesis: 3 retrieved analogs transferred to Kb1 (summed QAP objective 121, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



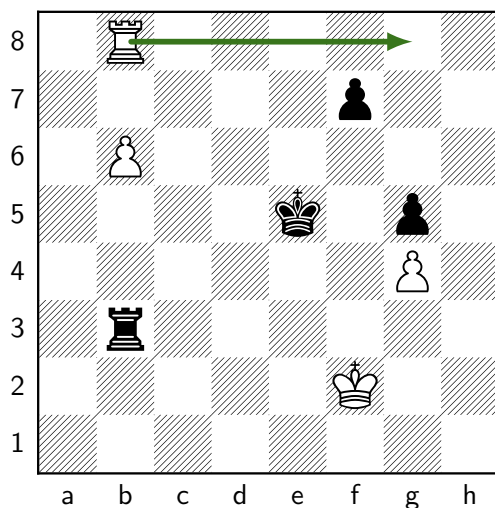
(a)



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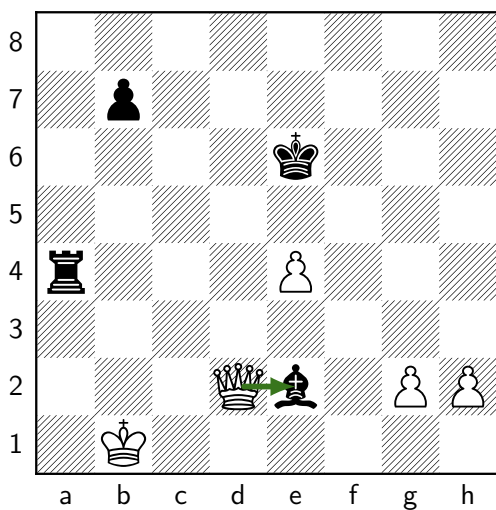


(c)

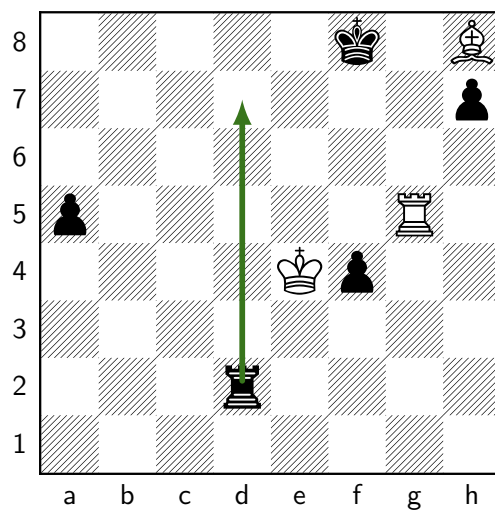


(d)

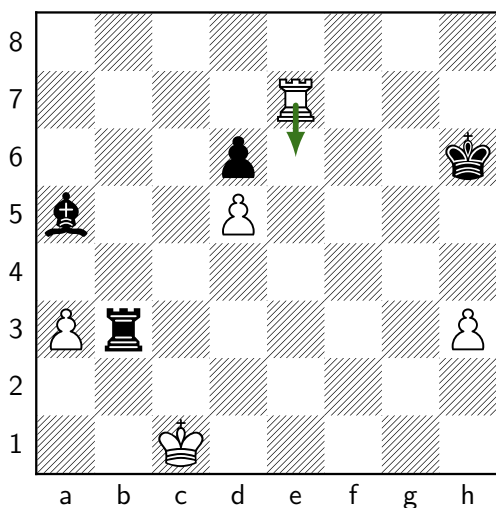
Figure 43: **Move 49, White: Qd2+.** (a) Our position; Melanie plays Qd2+ (f4d2). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#3899` (ply 108) [structural distance $d=23$, QAP obj=43]: the master played Ra7; under μ our Qf4 plays the role of that game's Rh7 (a **cross-type** correspondence); transferred move f4f7. (c) Retrieved from `ccr1_top_p0002#912` (ply 102) [structural distance $d=22$, QAP obj=41]: the master played Ke3; under μ our Kb1 plays the role of that game's Kf2; cross-type roles: our Qf4 as their Ra1, our be2 as their Pc4; transferred move b1b2. (d) Retrieved from `ccr1_top_p0003#2271` (ply 90) [structural distance $d=20$, QAP obj=35]: the master played Rg8; under μ our Qf4 plays the role of that game's Rb8 (a **cross-type** correspondence); cross-type roles: our be2 as their Pg4; transferred move f4f8. Synthesis: no retrieved analog transferred to Qd2+ – the **search** selector chose it on value (material / mate); panels (b)–(d) show the strongest nearby analogies.



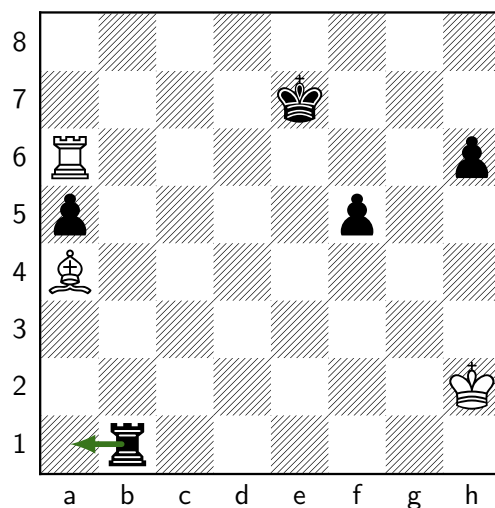
(a)



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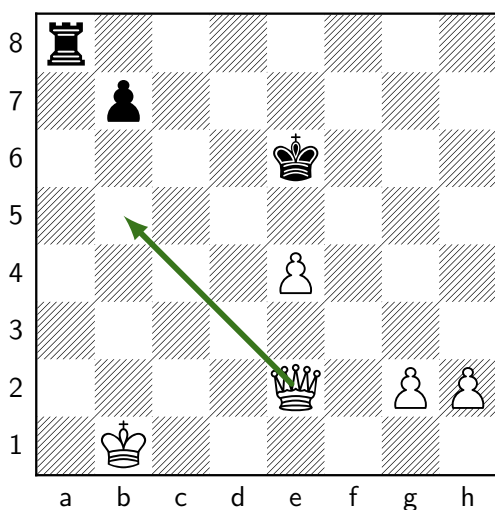


(c)

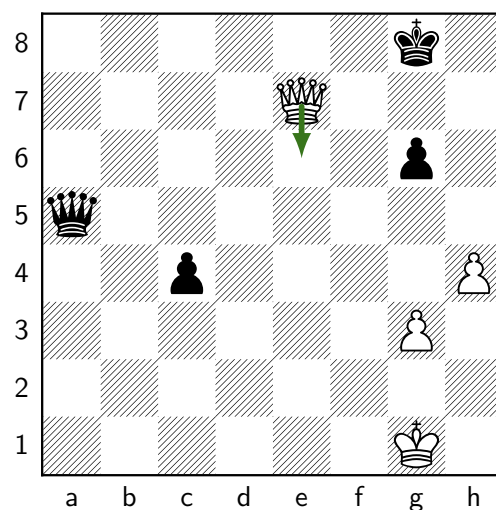


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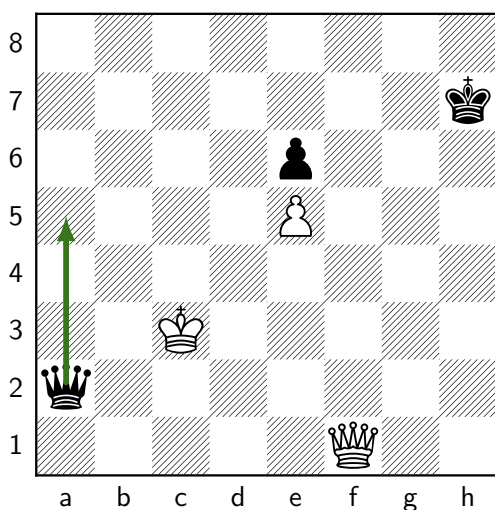
Figure 44: **Move 50, White: Qxe2.** (a) Our position; Melanie plays **Qxe2** (d2e2). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0003#2050` (ply 87) [structural distance $d=11$, QAP obj=1]: the master played **Rd7**; under μ our **Qd2** plays the role of that game's **Rd7** (a **cross-type** correspondence); transferred move **d2e2** (**endorses this move**). (c) Retrieved from `ccr1_top_p0001#2538` (ply 86) [structural distance $d=11$, QAP obj=6]: the master played **Re6+**; under μ our **Pe4** plays the role of that game's **Re7** (a **cross-type** correspondence); cross-type roles: our **be2** as their **Pa3**, our **Qd2** as their **rb3**; transferred move **e4e5**. (d) Retrieved from `ccr1_top_p0002#1094` (ply 109) [structural distance $d=13$, QAP obj=6]: the master played **Ra1**; under μ our **Ph2** plays the role of that game's **Rb8** (a **cross-type** correspondence); cross-type roles: our **Qd2** as their **ra6**, our **be2** as their **Pa5**; transferred move **h2h4**. Synthesis: 1 retrieved analog transferred to **Qxe2** (summed QAP objective 1, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



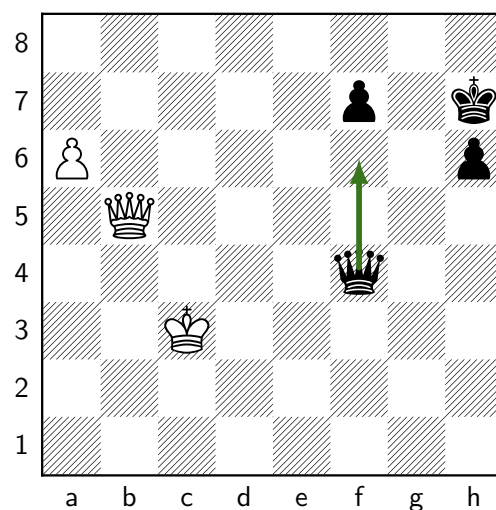
(a)



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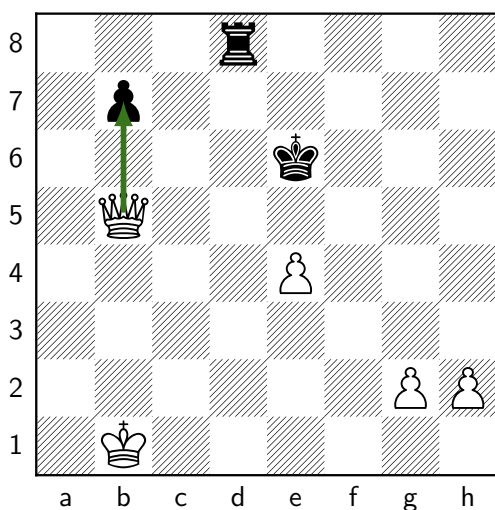


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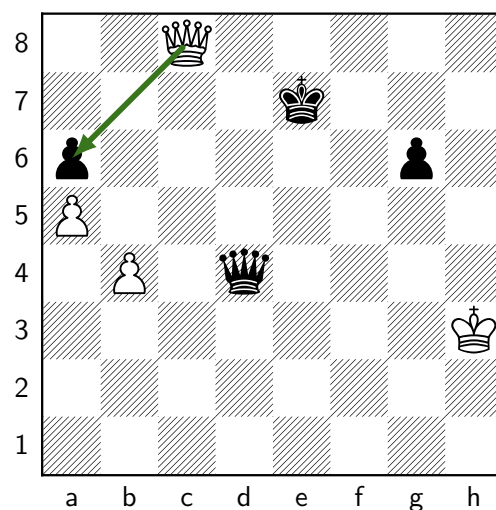


(d)

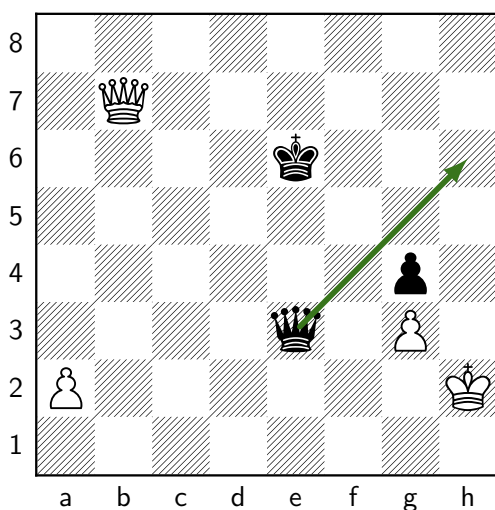
Figure 45: **Move 51, White: Qb5.** (a) Our position; Melanie plays Qb5 (e2b5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#1442` (ply 116) [structural distance $d=14$, QAP obj=1]: the master played Qe6+; under μ our Qe2 plays the role of that game’s Qe7; cross-type roles: our ra8 as their qa5; transferred move e2b5 (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#1990` (ply 97) [structural distance $d=17$, QAP obj=1]: the master played Qa5+; under μ our Qe2 plays the role of that game’s Qa7; cross-type roles: our ra8 as their qf1; transferred move e2b5 (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#426` (ply 111) [structural distance $d=16$, QAP obj=2]: the master played Qf6+; under μ our Qe2 plays the role of that game’s Qf5; cross-type roles: our pb7 as their qb5; transferred move e2e3. Synthesis: 2 retrieved analogs transferred to Qb5 (summed QAP objective 2, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



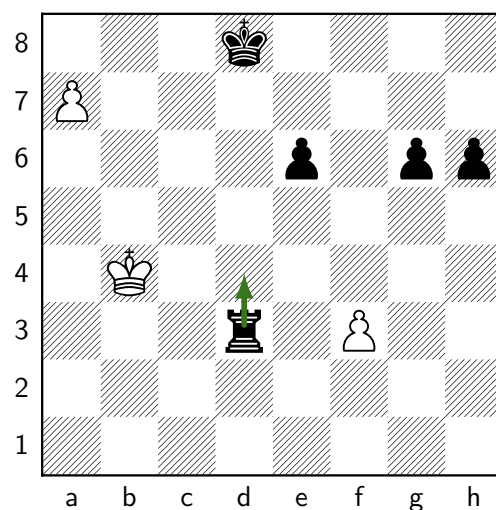
(a)



(b)

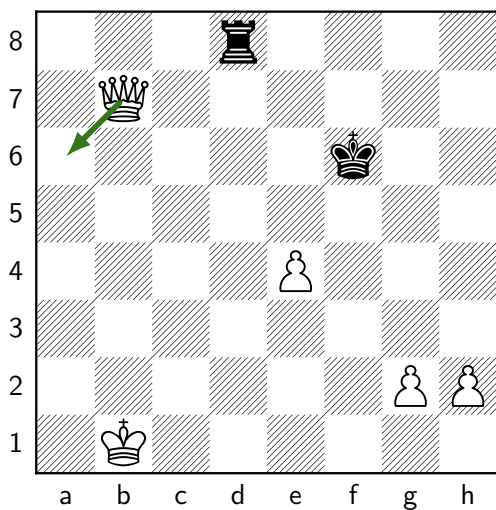


(c)

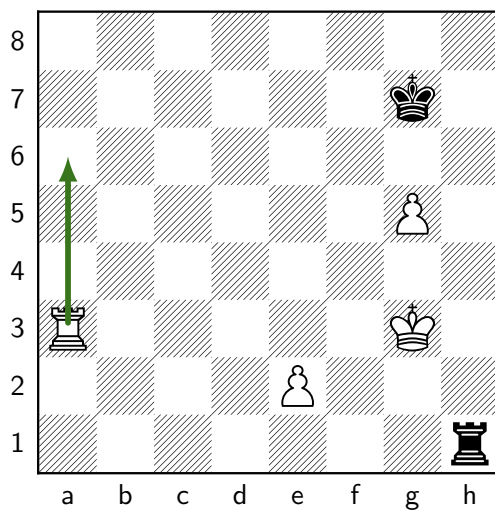


(d)

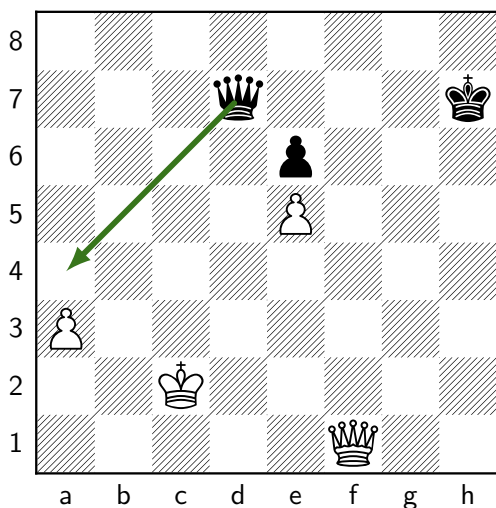
Figure 46: **Move 52, White: Qxb7.** (a) Our position; Melanie plays Qxb7 (b5b7). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#3618` (ply 106) [structural distance $d=16$, QAP obj=30]: the master played Qxa6; under μ our Qb5 plays the role of that game’s Qc8; cross-type roles: our rd8 as their Pa5; transferred move b5g5. (c) Retrieved from `ccr1_top_p0001#2494` (ply 99) [structural distance $d=13$, QAP obj=21]: the master played Qh6+; under μ our Qb5 plays the role of that game’s Qe6; cross-type roles: our rd8 as their qb7; transferred move b5a4. (d) Retrieved from `ccr1_top_p0000#279` (ply 103) [structural distance $d=18$, QAP obj=21]: the master played Rd4+; under μ our Qb5 plays the role of that game’s Rd6 (a **cross-type** correspondence); cross-type roles: our rd8 as their Ph6; transferred move b5h5. Synthesis: no retrieved analog transferred to Qxb7 – the **search** selector chose it on value (material / mate); panels (b)–(d) show the strongest nearby analogies.



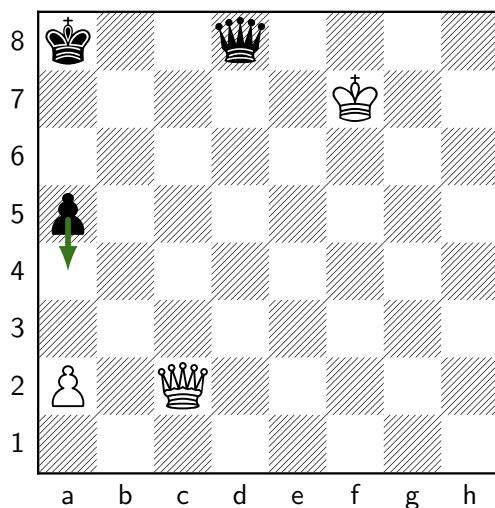
(a)



(b)

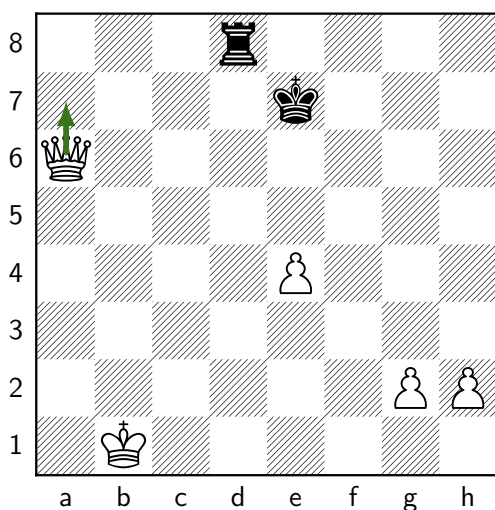


(c)

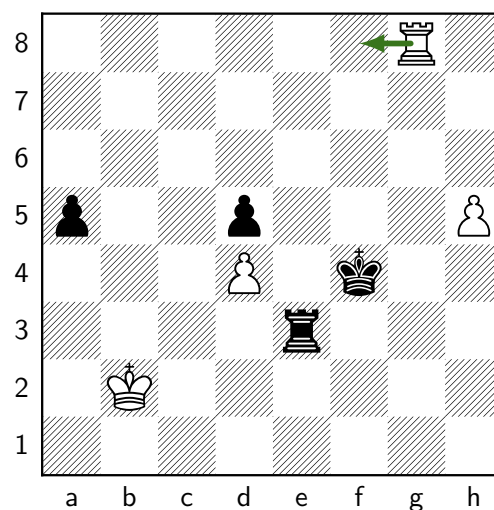


(d)

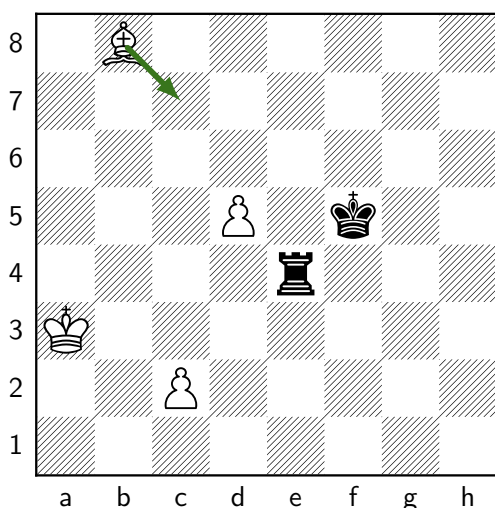
Figure 47: **Move 53, White: Qa6+.** (a) Our position; Melanie plays Qa6+ (b7a6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#2741` (ply 108) [structural distance $d=17$, QAP obj=2]: the master played Ra6; under μ our Qb7 plays the role of that game's Ra3 (a **cross-type** correspondence); transferred move b7a6 (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#1990` (ply 89) [structural distance $d=15$, QAP obj=2]: the master played Qa4+; under μ our Qb7 plays the role of that game's Qd2; cross-type roles: our rd8 as their qf1; transferred move b7a6 (**endorses this move**). (d) Retrieved from `ccr1_top_p0003#2049` (ply 105) [structural distance $d=14$, QAP obj=2]: the master played a4; under μ our Pe4 plays the role of that game's Pa4; cross-type roles: our rd8 as their qc2; transferred move e4e5. Synthesis: 3 retrieved analogs transferred to Qa6+ (summed QAP objective 5, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



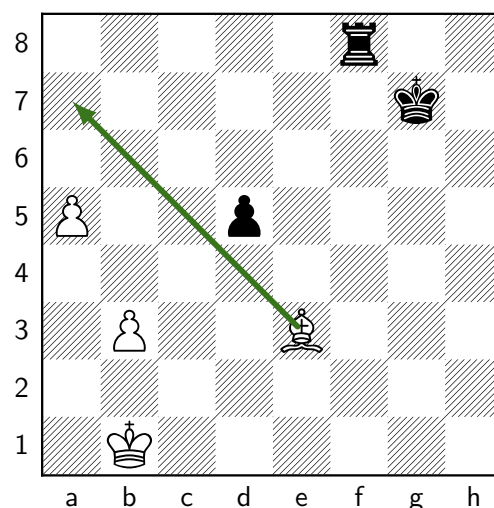
(a)



(b)



(c)



(d)

Figure 48: **Move 54, White: Qa7+.** (a) Our position; Melanie plays Qa7+ (a6a7). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#1638` (ply 116) [structural distance $d=10$, QAP obj=3]: the master played Rf8+; under μ our Qa6 plays the role of that game's Rg8 (a **cross-type** correspondence); transferred move a6a7 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#3159` (ply 94) [structural distance $d=6$, QAP obj=3]: the master played Bc7; under μ our Qa6 plays the role of that game's Bb8 (a **cross-type** correspondence); transferred move a6a7 (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#2870` (ply 112) [structural distance $d=6$, QAP obj=3]: the master played Ba7; under μ our Qa6 plays the role of that game's Be3 (a **cross-type** correspondence); transferred move a6a7 (**endorses this move**). Synthesis: 5 retrieved analogs transferred to Qa7+ (summed QAP objective 14, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.

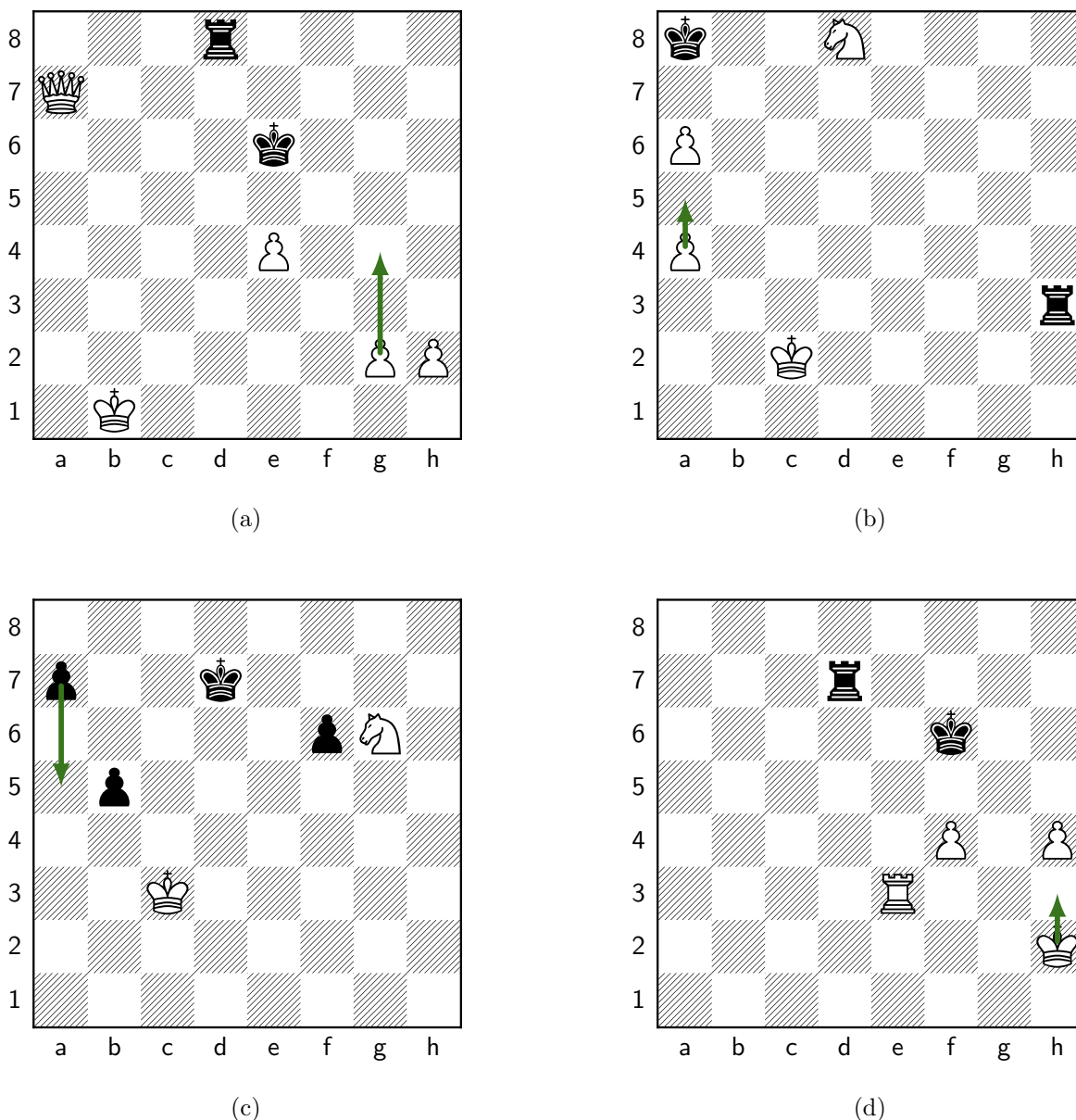
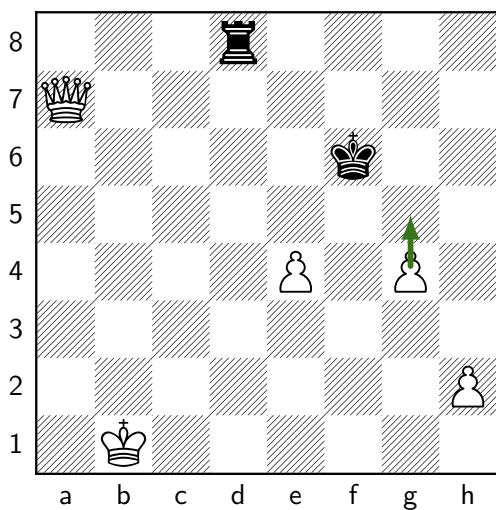
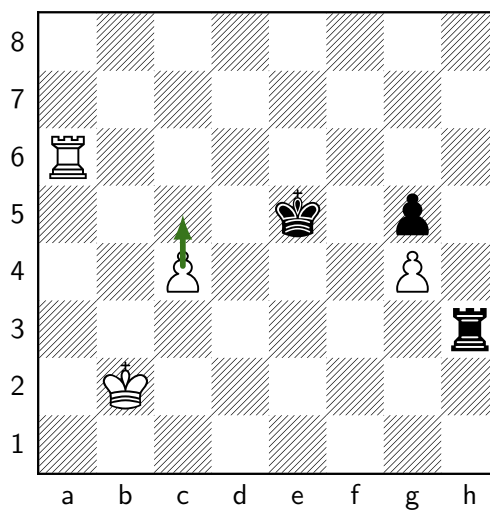


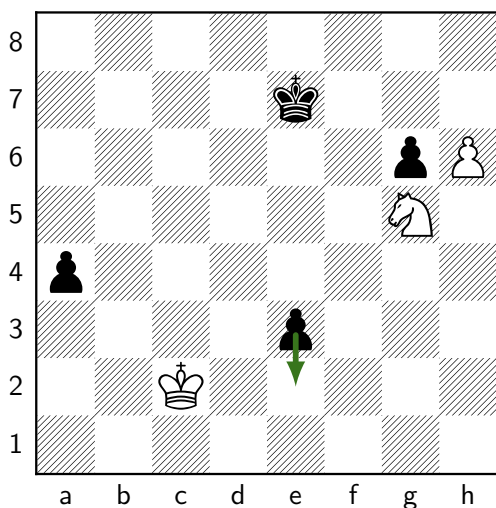
Figure 49: **Move 55, White: g4**. (a) Our position; Melanie plays g4 (g2g4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#2996` (ply 94) [structural distance $d=5$, QAP obj=1]: the master played a5; under μ our *Pg2* plays the role of that game’s *Pa4*; cross-type roles: our *Ph2* as their *rh3*, our *Qa7* as their *Nd8*; transferred move g2g4 (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#39` (ply 97) [structural distance $d=6$, QAP obj=0]: the master played a5; under μ our *Pg2* plays the role of that game’s *Pa2*; cross-type roles: our *rd8* as their *Pf6*, our *Qa7* as their *ng6*; transferred move g2g4 (**endorses this move**). (d) Retrieved from `ccr1_top_p0003#548` (ply 120) [structural distance $d=6$, QAP obj=1]: the master played Kh3; under μ our *Kb1* plays the role of that game’s *Kh2*; cross-type roles: our *Qa7* as their *Re3*; transferred move b1a2. Synthesis: 2 retrieved analogs transferred to g4 (summed QAP objective 1, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



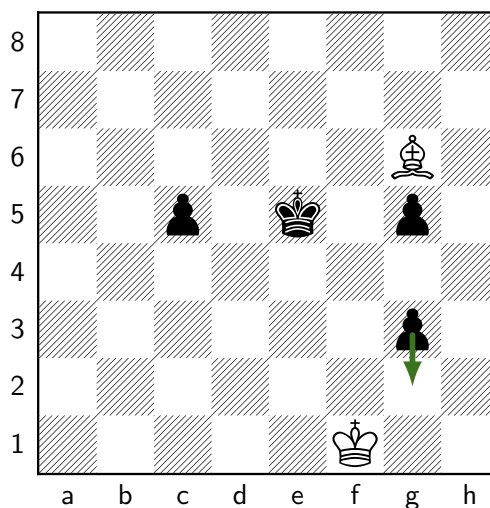
(a)



(b)

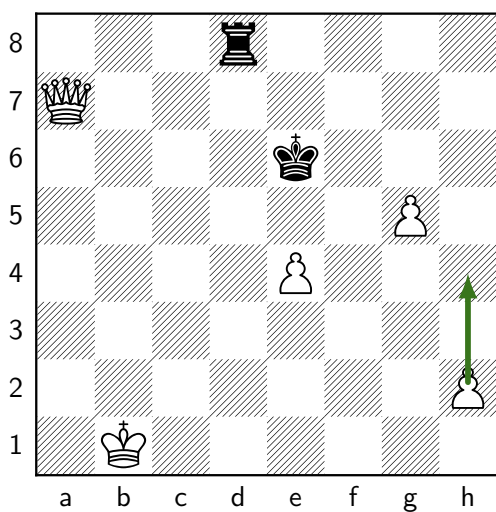


(c)

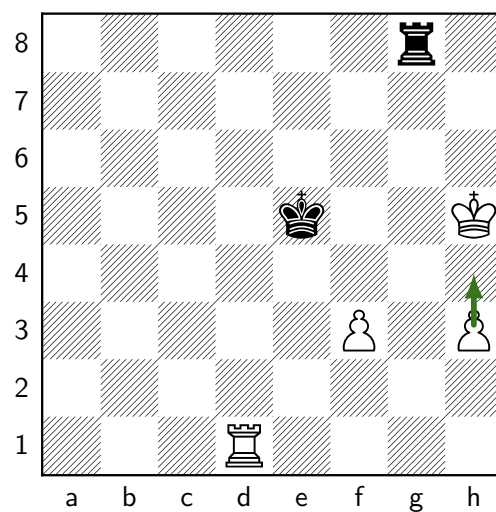


(d)

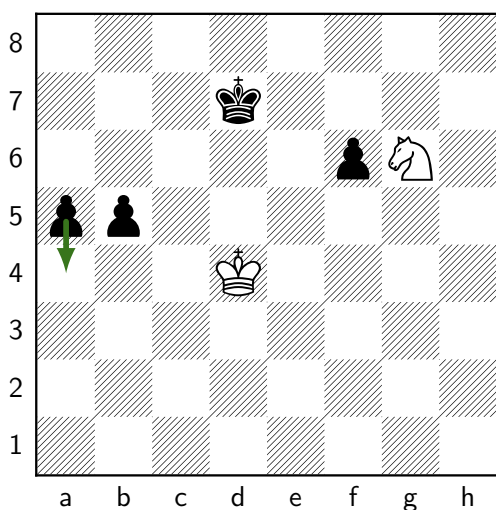
Figure 50: **Move 56, White: g5+ (g4g5)**. (a) Our position; Melanie plays g5+ (g4g5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#685` (ply 122) [structural distance $d=6$, QAP obj=1]: the master played c5; under μ our $Pg4$ plays the role of that game’s $Pc4$; cross-type roles: our $Qa7$ as their $Ra6$; transferred move g4g5 (**endorses this move**). (c) Retrieved from `ccr1_top_p0001#2189` (ply 95) [structural distance $d=6$, QAP obj=1]: the master played e2; under μ our $Pg4$ plays the role of that game’s $Pe6$; cross-type roles: our $rd8$ as their $Pg6$, our $Qa7$ as their $ng5$; transferred move g4g5 (**endorses this move**). (d) Retrieved from `ccr1_top_p0001#3683` (ply 103) [structural distance $d=7$, QAP obj=0]: the master played g2+; under μ our $Pg4$ plays the role of that game’s $Pg6$; cross-type roles: our $Ph2$ as their $bg6$, our $Qa7$ as their $Pg5$; transferred move g4g5 (**endorses this move**). Synthesis: 4 retrieved analogs transferred to g5+ (summed QAP objective 3, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



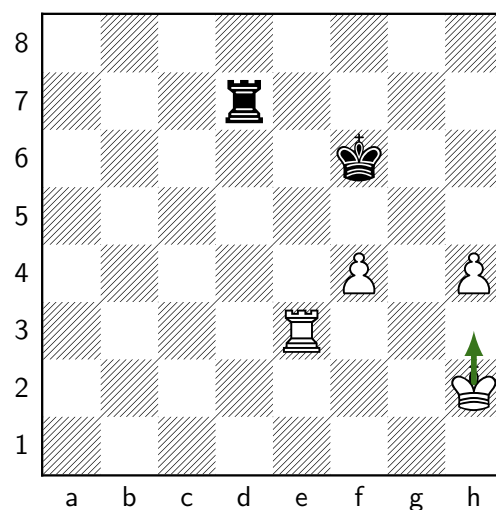
(a)



(b)

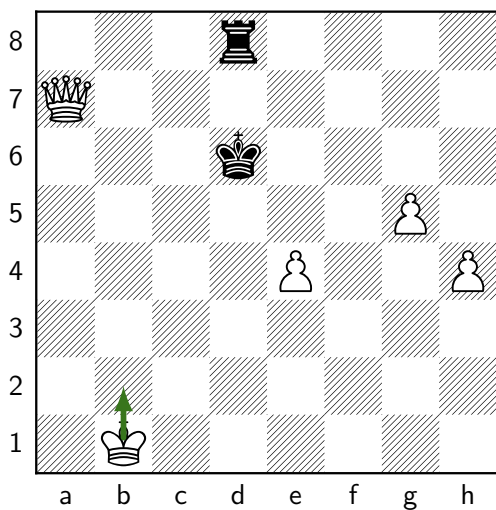


(c)

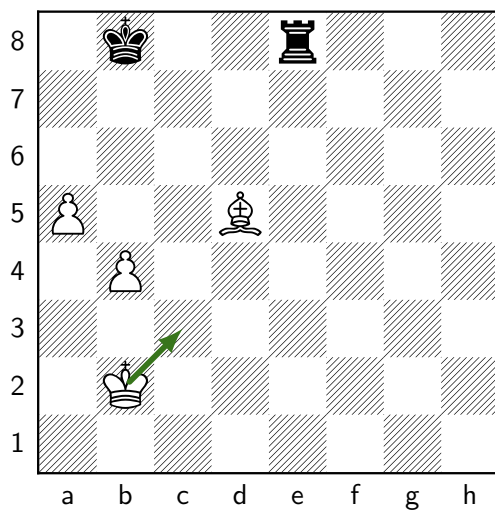


(d)

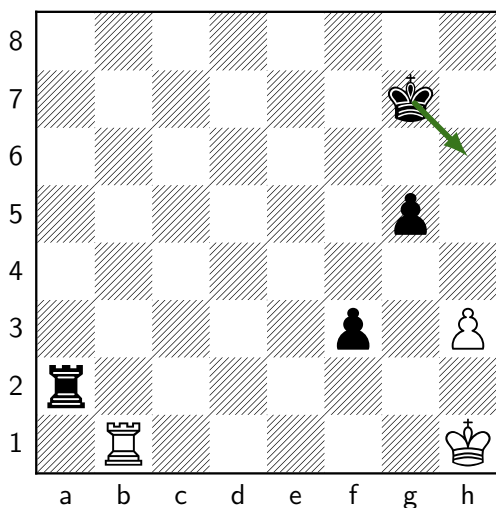
Figure 51: **Move 57, White: h4.** (a) Our position; Melanie plays **h4** (**h2h4**). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#1726` (ply 110) [structural distance $d=7$, QAP obj=1]: the master played **h4**; under μ our *Ph2* plays the role of that game’s *Ph3*; cross-type roles: our *Qa7* as their *Rd1*; transferred move **h2h4** (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#39` (ply 99) [structural distance $d=7$, QAP obj=0]: the master played **a4**; under μ our *Ph2* plays the role of that game’s *Pa4*; cross-type roles: our *rd8* as their *ng6*; transferred move **h2h4** (**endorses this move**). (d) Retrieved from `ccr1_top_p0003#548` (ply 120) [structural distance $d=5$, QAP obj=1]: the master played **Kh3**; under μ our *Kb1* plays the role of that game’s *Kh2*; cross-type roles: our *Qa7* as their *Re3*; transferred move **b1a2**. Synthesis: 3 retrieved analogs transferred to **h4** (summed QAP objective 2, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



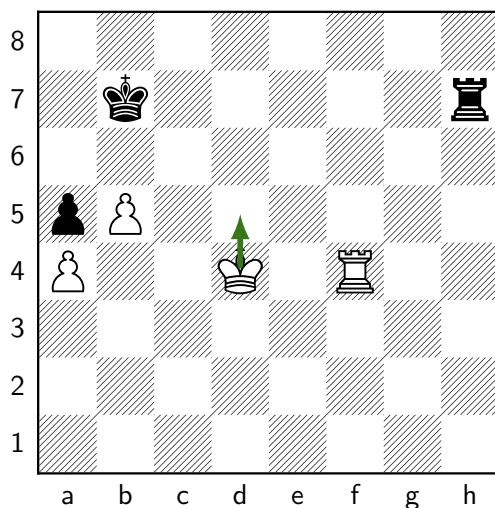
(a)



(b)

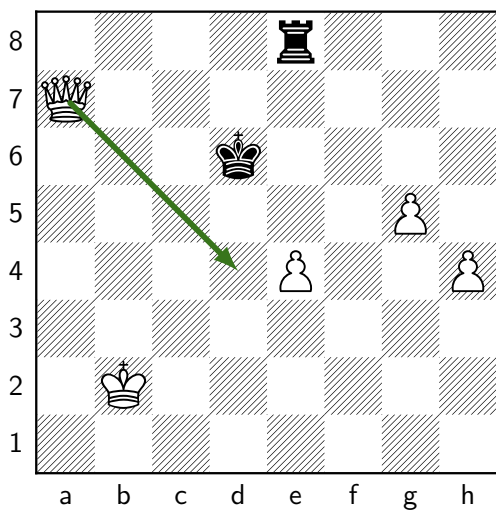


(c)

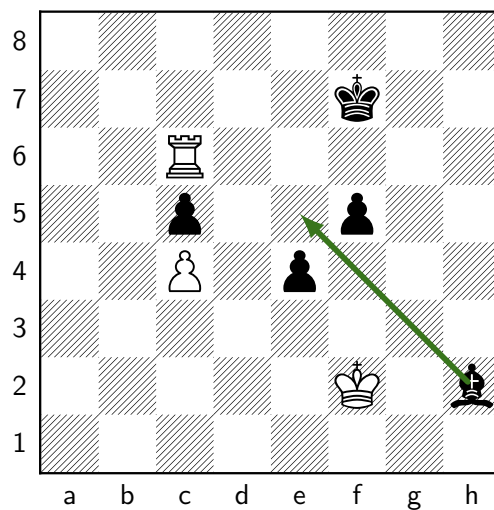


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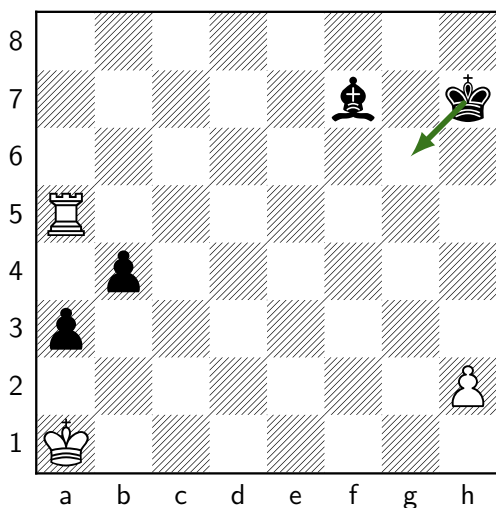
Figure 52: **Move 58, White: Kb2**. (a) Our position; Melanie plays Kb2 (b1b2). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0003#1726` (ply 98) [structural distance $d=7$, QAP obj=2]: the master played Kc3; under μ our *Kb1* plays the role of that game’s *Kb2*; cross-type roles: our *Qa7* as their *Bd5*; transferred move b1b2 (**endorses this move**). (c) Retrieved from `ccr1_top_p0003#2281` (ply 115) [structural distance $d=8$, QAP obj=2]: the master played Kh6; under μ our *Kb1* plays the role of that game’s *Kg2*; cross-type roles: our *Qa7* as their *Ra2*; transferred move b1b2 (**endorses this move**). (d) Retrieved from `ccr1_top_p0003#1458` (ply 102) [structural distance $d=11$, QAP obj=1]: the master played Kd5; under μ our *Kb1* plays the role of that game’s *Kd4*; cross-type roles: our *Qa7* as their *Rf4*; transferred move b1b2 (**endorses this move**). Synthesis: 4 retrieved analogs transferred to Kb2 (summed QAP objective 6, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



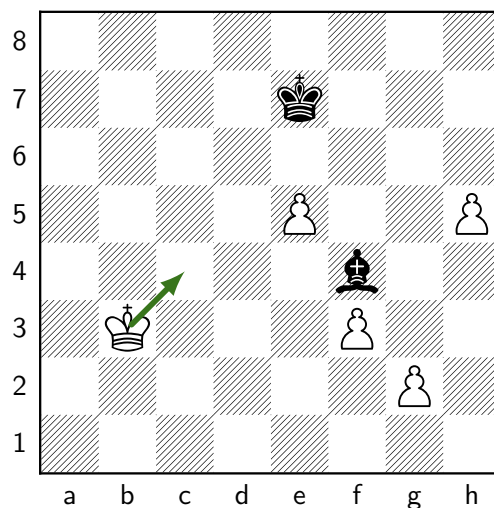
(a)



(b)

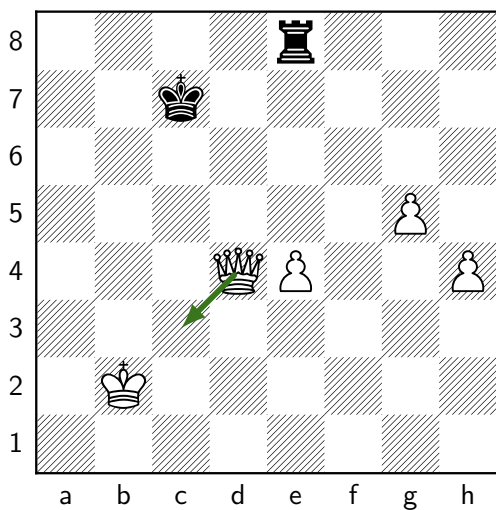


(c)

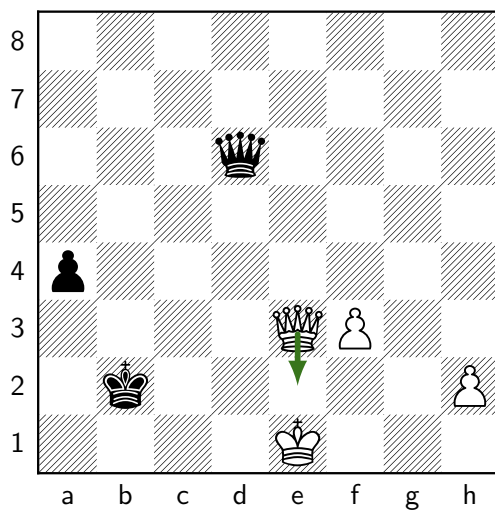


(d)

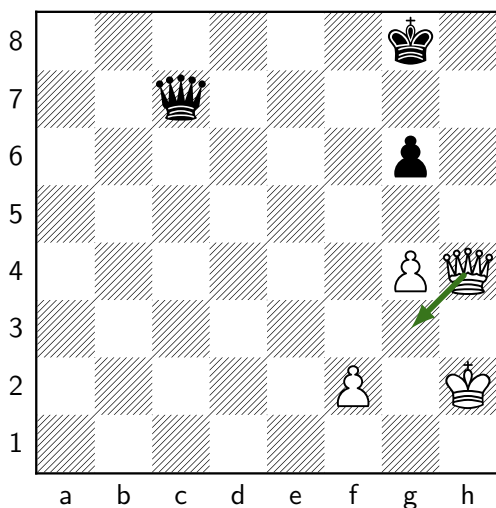
Figure 53: **Move 59, White: Qd4+.** (a) Our position; Melanie plays Qd4+ (a7d4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from ccr1_top_p0000#221 (ply 113) [structural distance $d=6$, QAP obj=6]: the master played Be5; under μ our Qa7 plays the role of that game's Bh7 (a **cross-type** correspondence); transferred move a7d4 (**endorses this move**). (c) Retrieved from ccr1_top_p0003#1646 (ply 131) [structural distance $d=5$, QAP obj=10]: the master played Kg6; under μ our Kb2 plays the role of that game's Kh2; cross-type roles: our Qa7 as their Bf7; transferred move b2a3. (d) Retrieved from ccr1_top_p0003#2424 (ply 102) [structural distance $d=10$, QAP obj=6]: the master played Kc4; under μ our Kb2 plays the role of that game's Kb3; cross-type roles: our re8 as their bf4, our Qa7 as their Ph5; transferred move b2c3. Synthesis: 1 retrieved analog transferred to Qd4+ (summed QAP objective 6, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



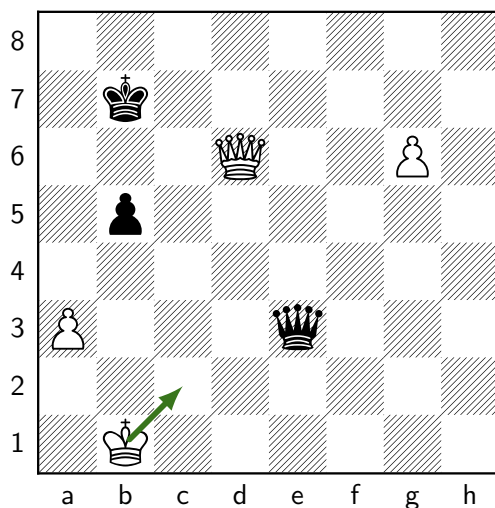
(a)



(b)



(c)



(d)

Figure 54: **Move 60, White: Qc3+.** (a) Our position; Melanie plays Qc3+ (d4c3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0003#2192` (ply 124) [structural distance $d=14$, QAP obj=21]: the master played Qe2+; under μ our Qd4 plays the role of that game's Qe3; cross-type roles: our re8 as their qd6; transferred move d4c3 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#1226` (ply 104) [structural distance $d=16$, QAP obj=4]: the master played Qg3; under μ our Qd4 plays the role of that game's Qh4; cross-type roles: our re8 as their qc7; transferred move d4c3 (**endorses this move**). (d) Retrieved from `ccr1_top_p0002#2065` (ply 116) [structural distance $d=12$, QAP obj=20]: the master played Kc2; under μ our Kb2 plays the role of that game's Kb1; cross-type roles: our re8 as their qe3; transferred move b2c2. Synthesis: 3 retrieved analogs transferred to Qc3+ (summed QAP objective 45, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.

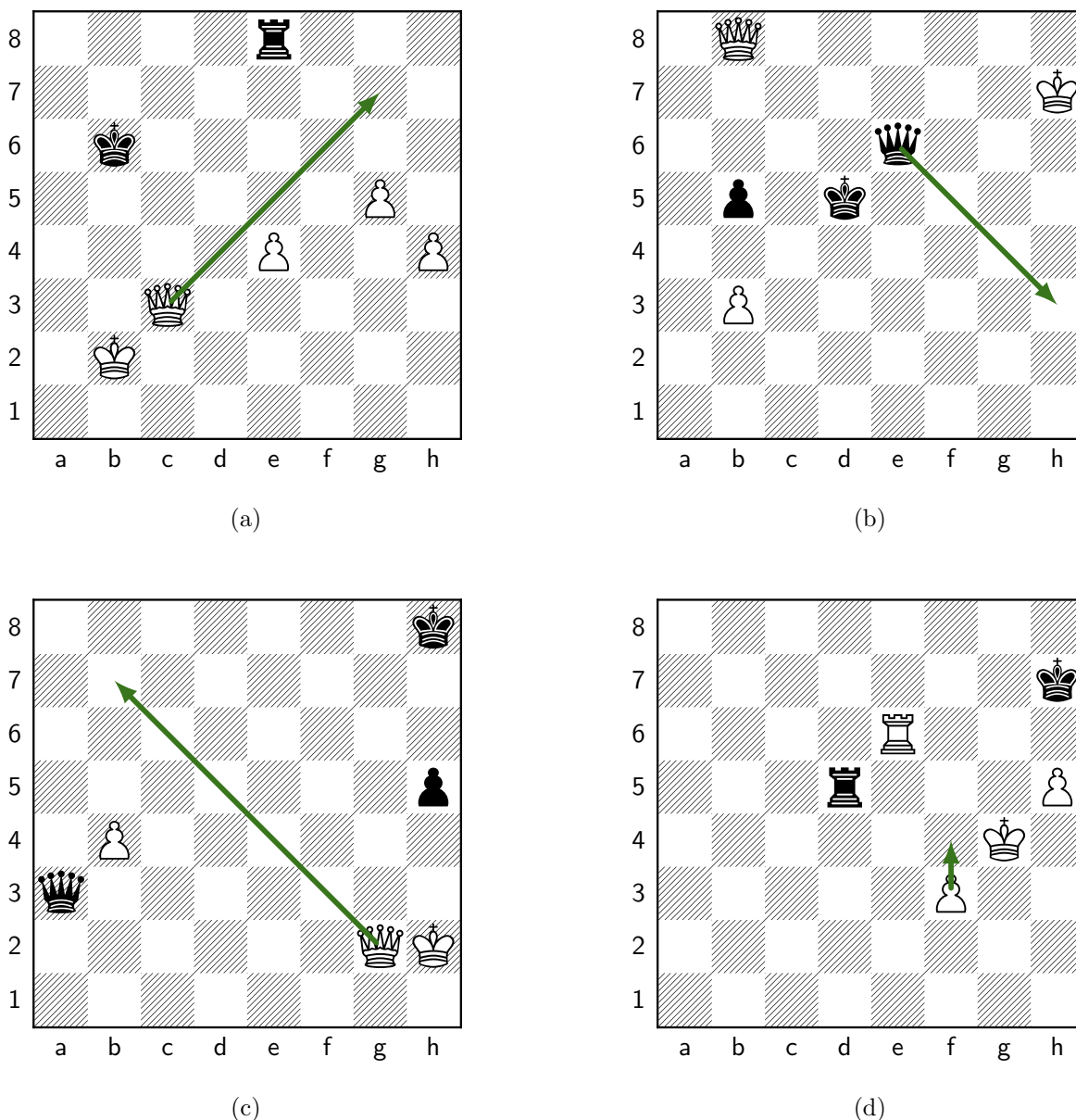


Figure 55: **Move 61, White: Qg7.** (a) Our position; Melanie plays Qg7 (c3g7). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#1905` (ply 111) [structural distance $d=15$, QAP obj=16]: the master played Qh3+; under μ our Qc3 plays the role of that game’s Qe3; cross-type roles: our re8 as their qb8; transferred move c3g7 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#1845` (ply 130) [structural distance $d=15$, QAP obj=7]: the master played Qb7; under μ our Qc3 plays the role of that game’s Qg2; cross-type roles: our re8 as their qa3; transferred move c3g7 (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#2972` (ply 112) [structural distance $d=15$, QAP obj=19]: the master played f4; under μ our Qc3 plays the role of that game’s Pf3 (a **cross-type** correspondence); cross-type roles: our Ph4 as their Re6; transferred move c3a3. Synthesis: 2 retrieved analogs transferred to Qg7 (summed QAP objective 23, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.

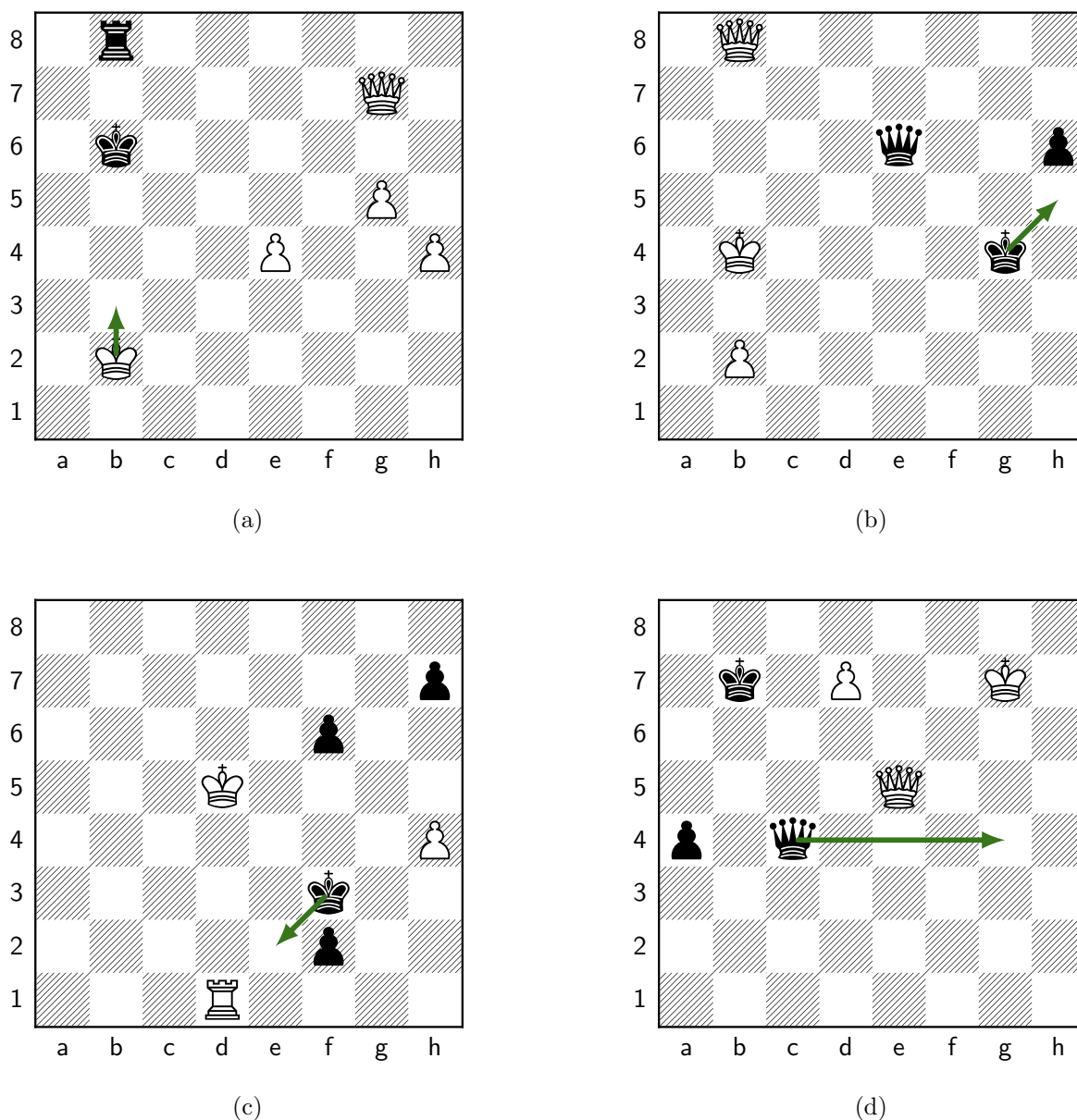


Figure 56: **Move 62, White: Kb3**. (a) Our position; Melanie plays Kb3 (b2b3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0003#2683` (ply 117) [structural distance $d=18$, QAP obj=3]: the master played Kh5; under μ our Kb2 plays the role of that game’s Kg5; cross-type roles: our rb8 as their qb8; transferred move b2b3 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#712` (ply 129) [structural distance $d=20$, QAP obj=1]: the master played Ke2; under μ our Kb2 plays the role of that game’s Kf6; cross-type roles: our Qg7 as their Pf2; transferred move b2b3 (**endorses this move**). (d) Retrieved from `ccr1_top_p0001#3920` (ply 129) [structural distance $d=20$, QAP obj=3]: the master played Qg4+; under μ our Qg7 plays the role of that game’s Qc5; cross-type roles: our rb8 as their qe5; transferred move g7g6. Synthesis: 2 retrieved analogs transferred to Kb3 (summed QAP objective 4, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.

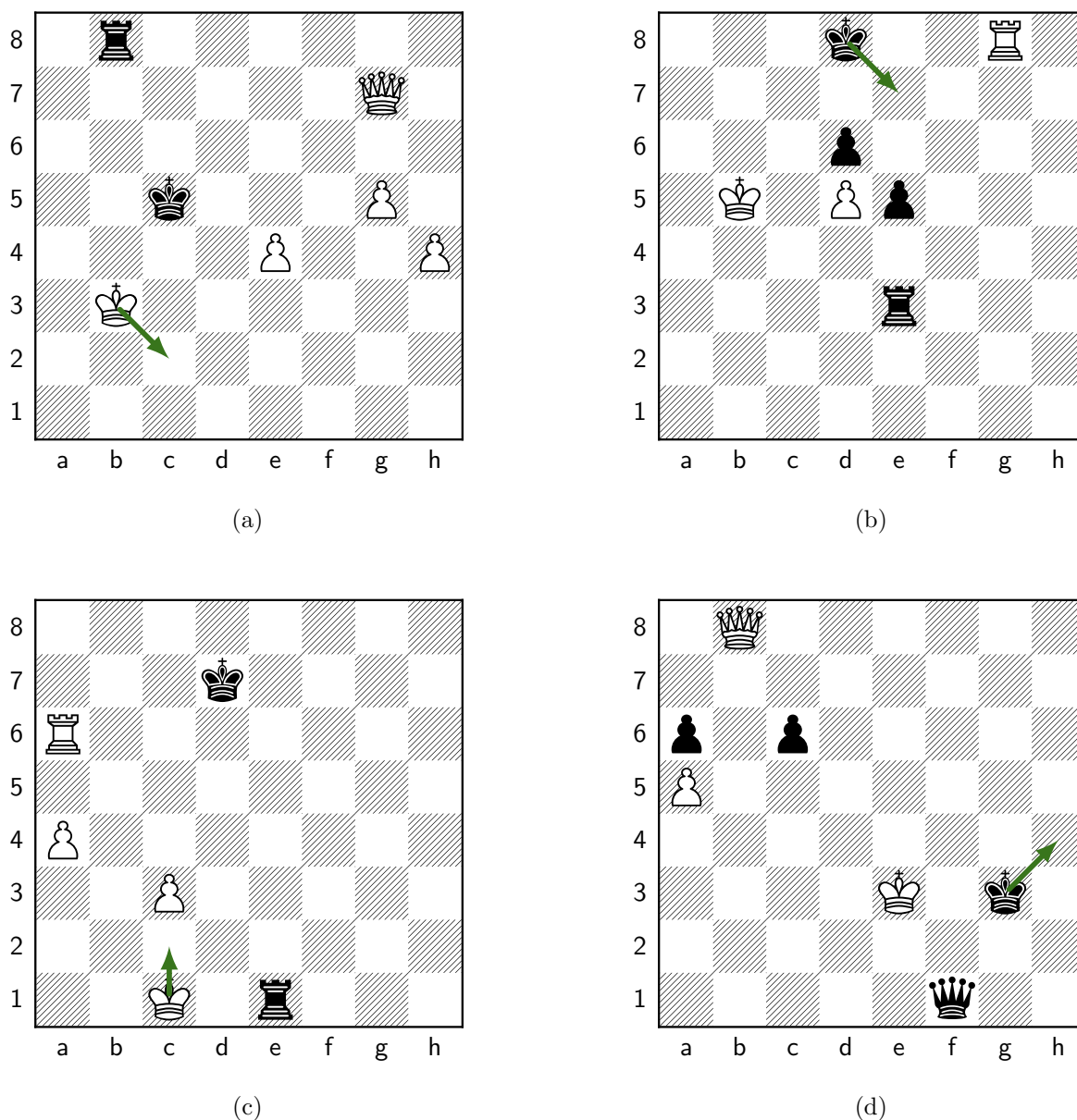
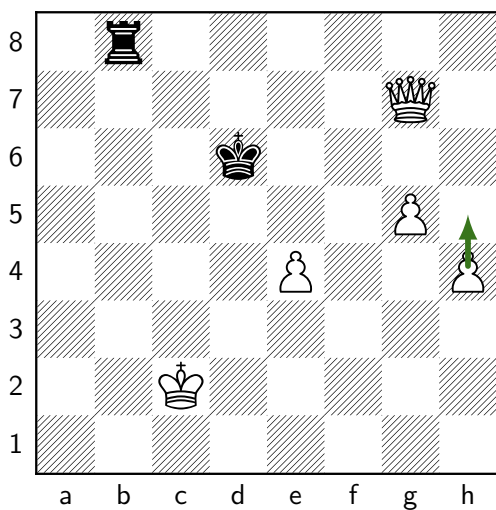
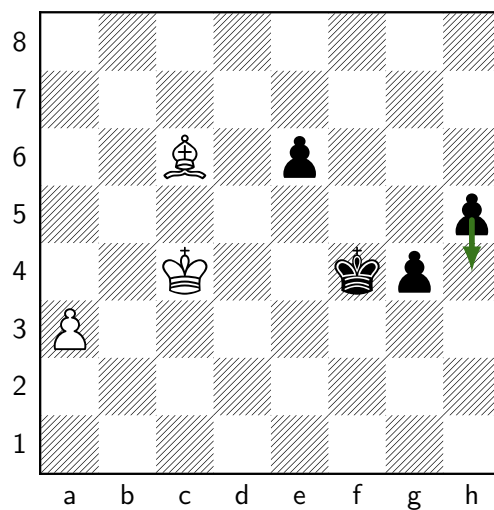


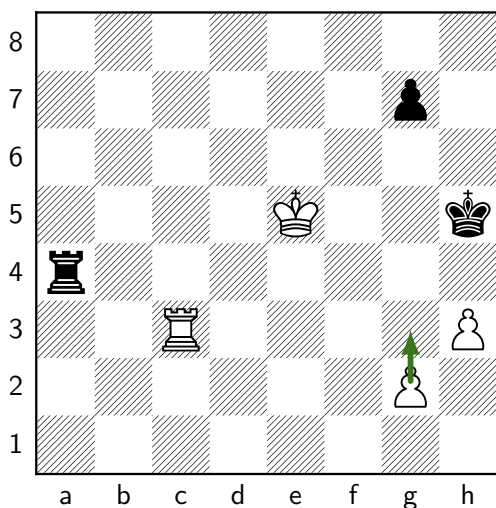
Figure 57: **Move 63, White: Kc2.** (a) Our position; Melanie plays Kc2 (b3c2). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#790` (ply 121) [structural distance $d=9$, QAP obj=43]: the master played Ke7; under μ our *Kb3* plays the role of that game’s *Kd1*; cross-type roles: our *Qg7* as their *Re3*; transferred move b3c2 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#2555` (ply 122) [structural distance $d=12$, QAP obj=29]: the master played Kc2; under μ our *Kb3* plays the role of that game’s *Kc1*; cross-type roles: our *Qg7* as their *Ra6*; transferred move b3c2 (**endorses this move**). (d) Retrieved from `ccr1_top_p0002#348` (ply 127) [structural distance $d=12$, QAP obj=68]: the master played Kh4; under μ our *Kb3* plays the role of that game’s *Kg6*; cross-type roles: our *rb8* as their *qb8*; transferred move b3a4. Synthesis: 2 retrieved analogs transferred to Kc2 (summed QAP objective 71, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



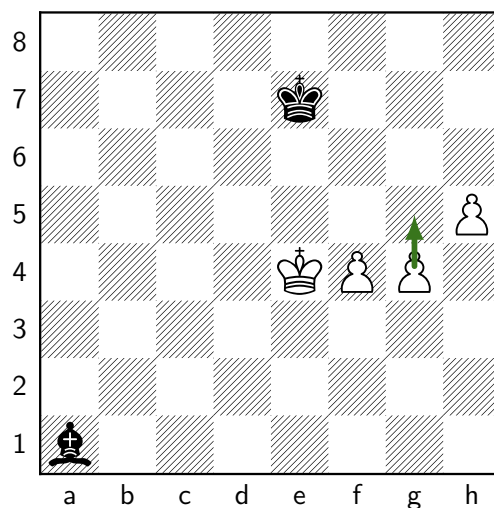
(a)



(b)



(c)



(d)

Figure 58: **Move 64, White: h5.** (a) Our position; Melanie plays h5 (h4h5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0003#2414` (ply 127) [structural distance $d=11$, QAP obj=2]: the master played h4; under μ our Ph_4 plays the role of that game’s Ph_4 ; cross-type roles: our rb_8 as their bc_6 , our Qg_7 as their Pe_6 ; transferred move h4h5 (**endorses this move**). (c) Retrieved from `ccr1_top_p0003#2910` (ply 142) [structural distance $d=12$, QAP obj=2]: the master played g3; under μ our Ph_4 plays the role of that game’s Pg_2 ; cross-type roles: our Qg_7 as their Rc_3 ; transferred move h4h5 (**endorses this move**). (d) Retrieved from `ccr1_top_p0003#2424` (ply 112) [structural distance $d=11$, QAP obj=2]: the master played g5; under μ our Ph_4 plays the role of that game’s Pg_4 ; cross-type roles: our rb_8 as their ba_1 , our Qg_7 as their Pf_4 ; transferred move h4h5 (**endorses this move**). Synthesis: 3 retrieved analogs transferred to h5 (summed QAP objective 5, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.

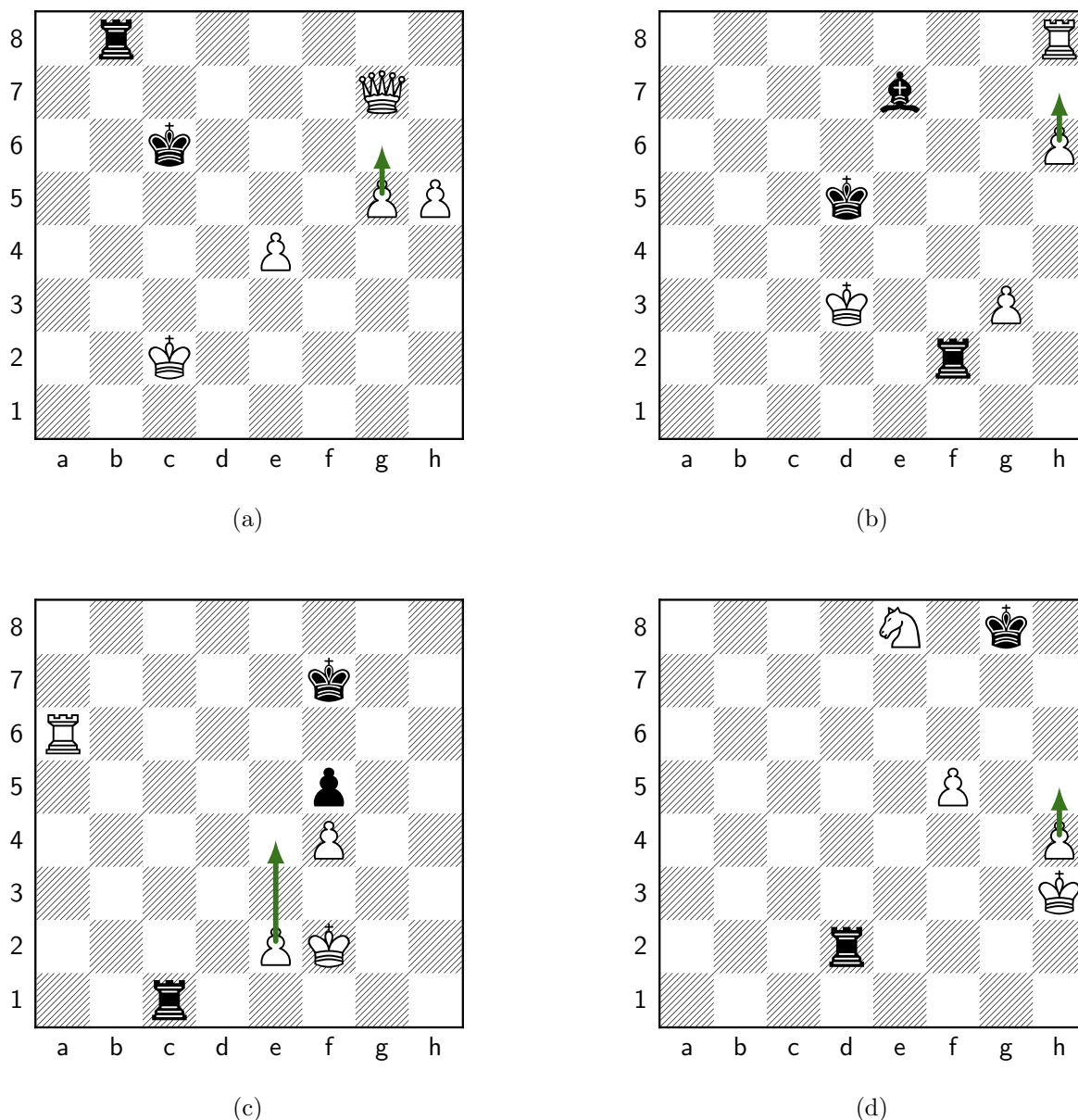
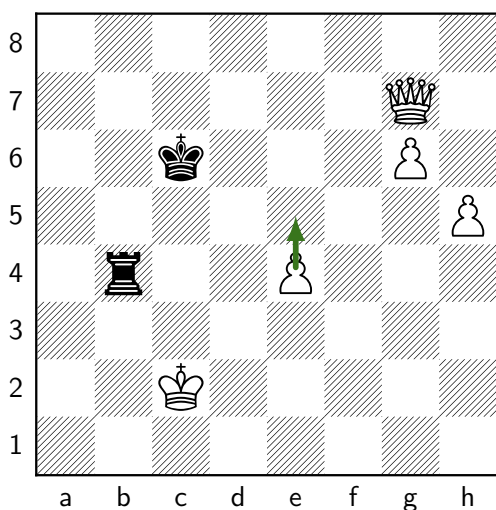
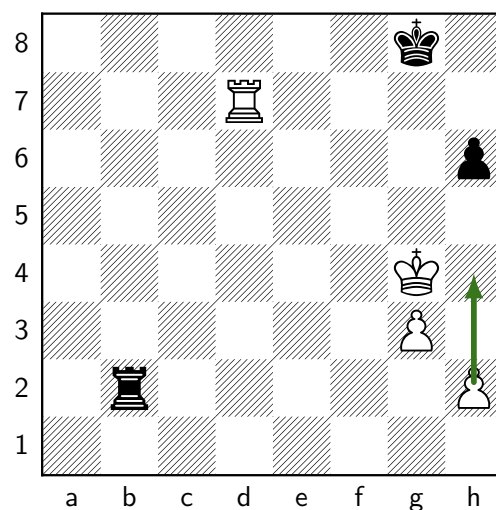


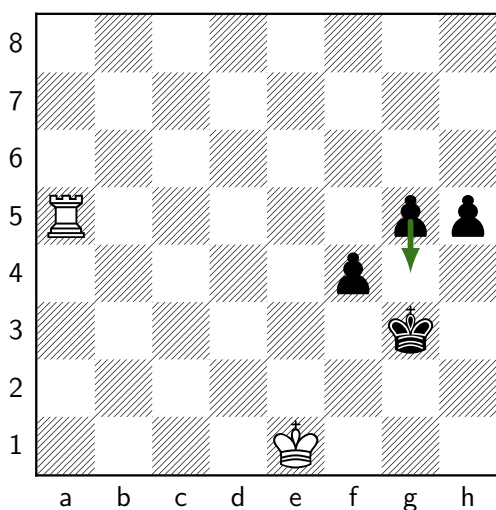
Figure 59: **Move 65, White: g6.** (a) Our position; Melanie plays g6 (g5g6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#1899` (ply 118) [structural distance $d=10$, QAP obj=3]: the master played h7; under μ our $Pg5$ plays the role of that game’s $Ph6$; cross-type roles: our $Ph5$ as their $be7$, our $Qg7$ as their $Rh8$; transferred move g5g6 (**endorses this move**). (c) Retrieved from `ccr1_top_p0001#633` (ply 116) [structural distance $d=10$, QAP obj=2]: the master played e4; under μ our $Pg5$ plays the role of that game’s $Pe2$; cross-type roles: our $Qg7$ as their $Ra6$; transferred move g5g6 (**endorses this move**). (d) Retrieved from `ccr1_top_p0003#2289` (ply 112) [structural distance $d=9$, QAP obj=1]: the master played h5; under μ our $Pg5$ plays the role of that game’s $Ph4$; cross-type roles: our $Qg7$ as their $Ne8$; transferred move g5g6 (**endorses this move**). Synthesis: 4 retrieved analogs transferred to g6 (summed QAP objective 7, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



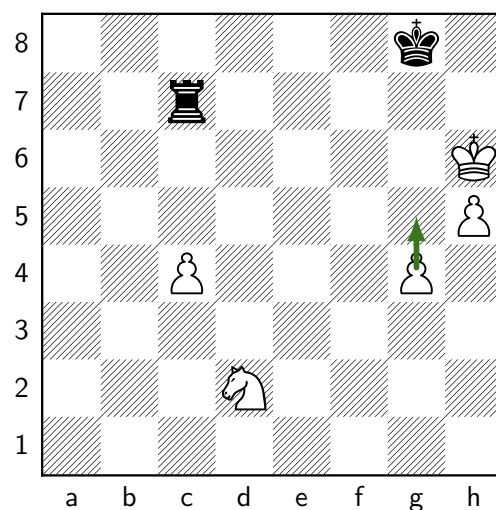
(a)



(b)

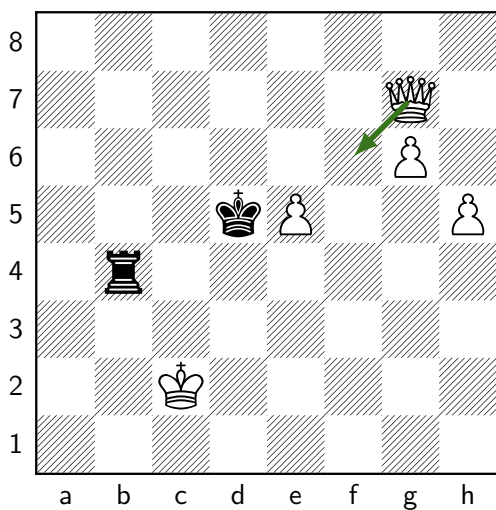


(c)

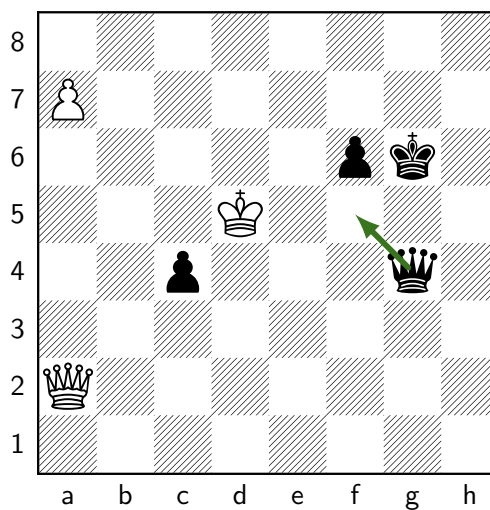


(d)

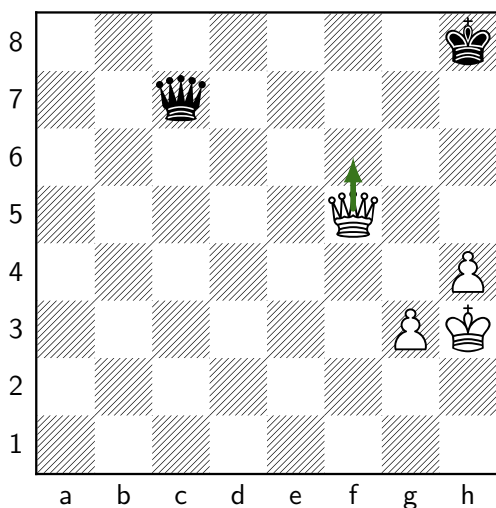
Figure 60: **Move 66, White: e5.** (a) Our position; Melanie plays e5 (e4e5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#3488` (ply 124) [structural distance $d=14$, QAP obj=9]: the master played h4; under μ our Pe_4 plays the role of that game’s Ph_2 ; cross-type roles: our Qg_7 as their Rd_7 ; transferred move e4e5 (**endorses this move**). (c) Retrieved from `ccr1_top_p0001#1138` (ply 115) [structural distance $d=12$, QAP obj=9]: the master played g4; under μ our Pe_4 plays the role of that game’s Pg_4 ; cross-type roles: our Qg_7 as their Ph_5 ; transferred move e4e5 (**endorses this move**). (d) Retrieved from `ccr1_top_p0001#2264` (ply 126) [structural distance $d=14$, QAP obj=20]: the master played g5; under μ our Ph_5 plays the role of that game’s Pg_4 ; cross-type roles: our Qg_7 as their Nd_2 ; transferred move h5h6. Synthesis: 2 retrieved analogs transferred to e5 (summed QAP objective 18, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



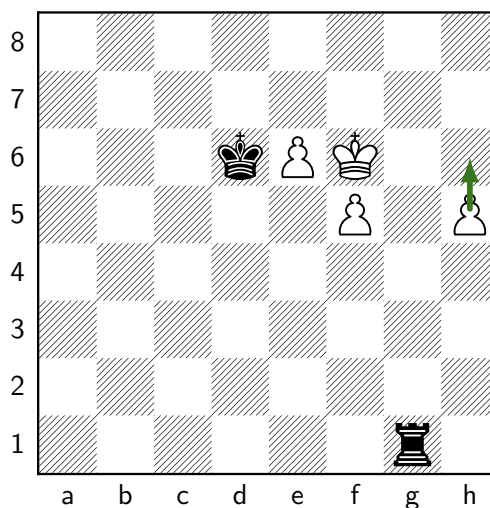
(a)



(b)



(c)



(d)

Figure 61: **Move 67, White: Qf6.** (a) Our position; Melanie plays Qf6 (g7f6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0001#194` (ply 117) [structural distance $d=21$, QAP obj=25]: the master played Qf5+; under μ our Qg7 plays the role of that game’s Qg5; cross-type roles: our rb4 as their pa7, our kd5 as their qa2; transferred move g7f6 (**endorses this move**). (c) Retrieved from `ccrl_top_p0000#2205` (ply 118) [structural distance $d=21$, QAP obj=1]: the master played Qf6+; under μ our Qg7 plays the role of that game’s Qf5; cross-type roles: our rb4 as their Ph4, our Pg6 as their qc7; transferred move g7f6 (**endorses this move**). (d) Retrieved from `ccrl_top_p0002#936` (ply 116) [structural distance $d=21$, QAP obj=20]: the master played h6; under μ our Ph5 plays the role of that game’s Ph5; cross-type roles: our Qg7 as their Pf5; transferred move h5h6. Synthesis: 2 retrieved analogs transferred to Qf6 (summed QAP objective 25, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.

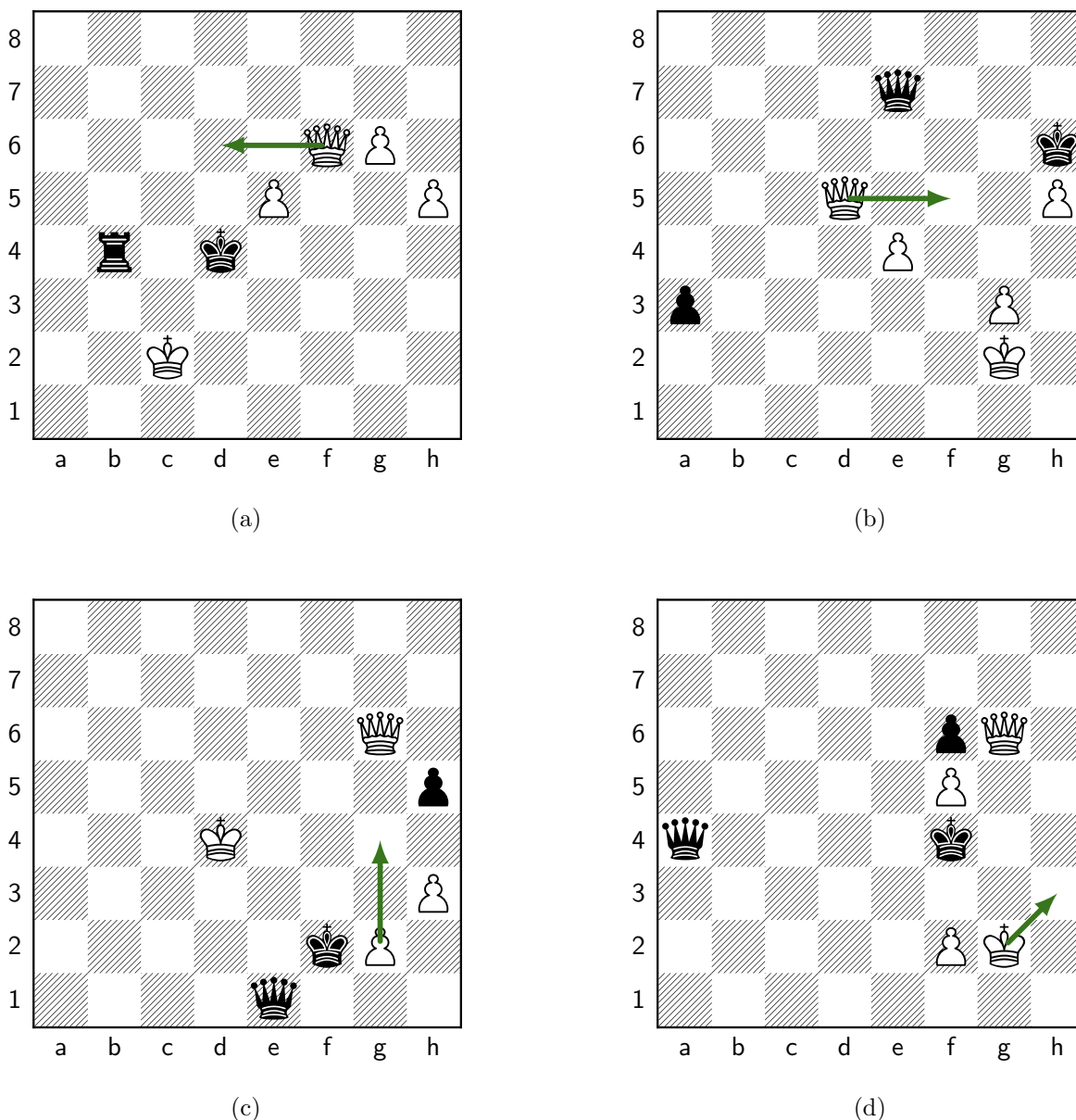
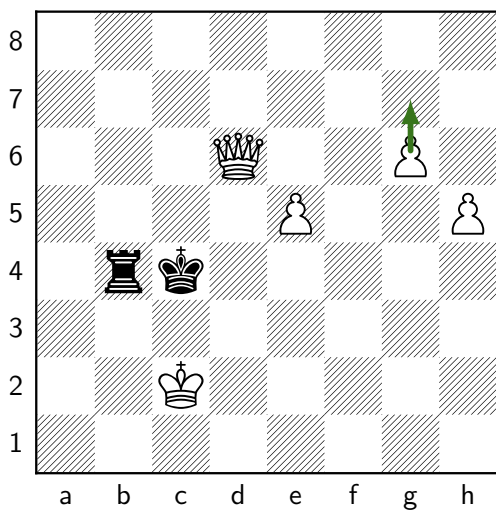
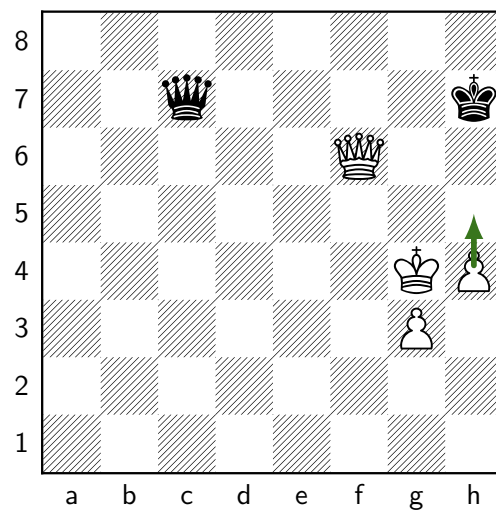


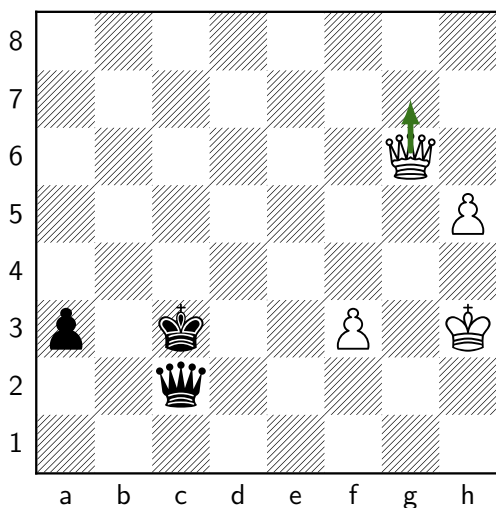
Figure 62: **Move 68, White: Qd6+.** (a) Our position; Melanie plays Qd6+ (f6d6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top.p0002#3329` (ply 136) [structural distance $d=24$, QAP obj=19]: the master played Qf5; under μ our Qf6 plays the role of that game’s Qd5; cross-type roles: our rb4 as their Pe4, our Pg6 as their qe7; transferred move f6d6 (**endorses this move**). (c) Retrieved from `ccr1_top.p0001#2168` (ply 148) [structural distance $d=25$, QAP obj=30]: the master played g4; under μ our Pe5 plays the role of that game’s Pg2; cross-type roles: our rb4 as their qe1, our Qf6 as their Ph3; transferred move e5e6. (d) Retrieved from `ccr1_top.p0001#3243` (ply 130) [structural distance $d=22$, QAP obj=28]: the master played Kh3; under μ our Kc2 plays the role of that game’s Kg2; cross-type roles: our rb4 as their qa4, our kd4 as their Qg6; transferred move c2d2. Synthesis: 1 retrieved analog transferred to Qd6+ (summed QAP objective 19, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



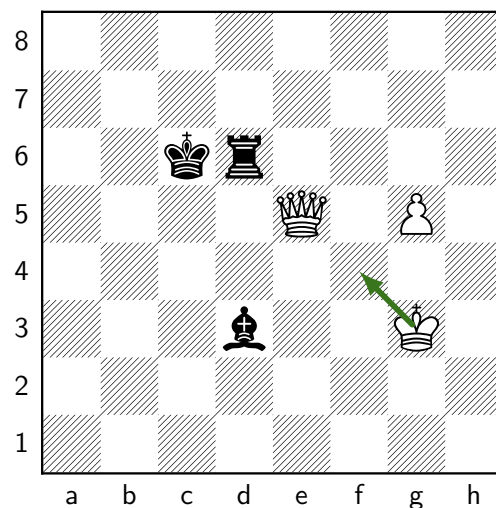
(a)



(b)

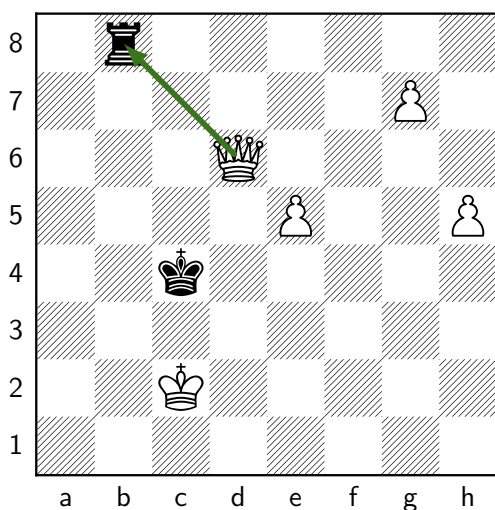


(c)

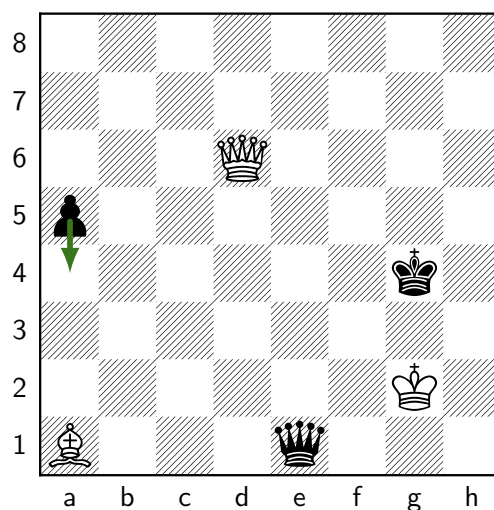


(d)

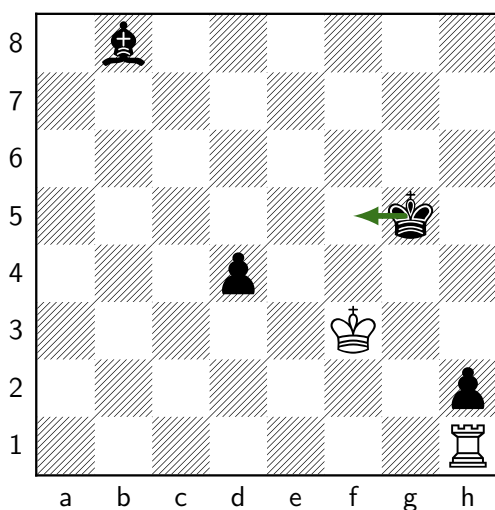
Figure 63: **Move 69, White: g7.** (a) Our position; Melanie plays g7 (g6g7). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#2205` (ply 122) [structural distance $d=29$, QAP obj=1]: the master played h5; under μ our $Pg6$ plays the role of that game’s $Ph4$; cross-type roles: our $rb4$ as their $qc7$; transferred move g6g7 (**endorses this move**). (c) Retrieved from `ccr1_top_p0003#1520` (ply 144) [structural distance $d=28$, QAP obj=22]: the master played Qg7+; under μ our $Qd6$ plays the role of that game’s $Qg6$; cross-type roles: our $rb4$ as their $qc2$; transferred move d6d7. (d) Retrieved from `ccr1_top_p0000#2792` (ply 152) [structural distance $d=25$, QAP obj=16]: the master played Kf4; under μ our $Kc2$ plays the role of that game’s $Kg3$; cross-type roles: our $Pg6$ as their $bd3$; transferred move c2d2. Synthesis: 1 retrieved analog transferred to g7 (summed QAP objective 1, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



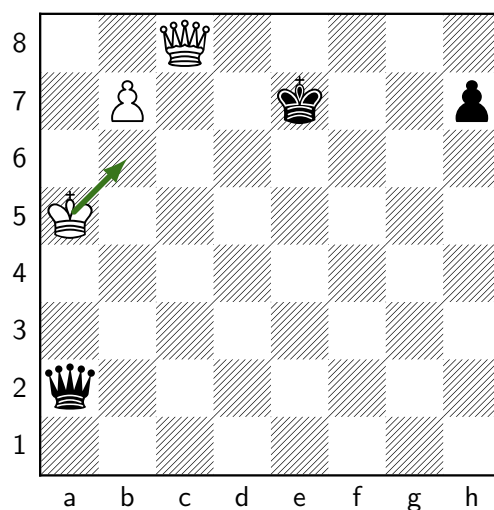
(a)



(b)



(c)



(d)

Figure 64: **Move 70, White: Qxb8**. (a) Our position; Melanie plays Qxb8 (d6b8). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top.p0002#1019` (ply 143) [structural distance $d=17$, QAP obj=8]: the master played a4; under μ our *Pe5* plays the role of that game’s *Pa4*; cross-type roles: our *Ph5* as their *qd6*, our *rb8* as their *ba1*; transferred move e5e6. (c) Retrieved from `ccr1_top.p0002#3412` (ply 133) [structural distance $d=17$, QAP obj=8]: the master played Kf5; under μ our *Kc2* plays the role of that game’s *Kg4*; cross-type roles: our *Pe5* as their *Bb8*, our *rb8* as their *Ph2*; transferred move c2c1. (d) Retrieved from `ccr1_top.p0002#1006` (ply 124) [structural distance $d=17$, QAP obj=3]: the master played Kb6; under μ our *Kc2* plays the role of that game’s *Ka5*; cross-type roles: our *rb8* as their *qa2*; transferred move c2c1. Synthesis: no retrieved analog transferred to Qxb8 – the search selector chose it on value (material / mate); panels (b)–(d) show the strongest nearby analogies.

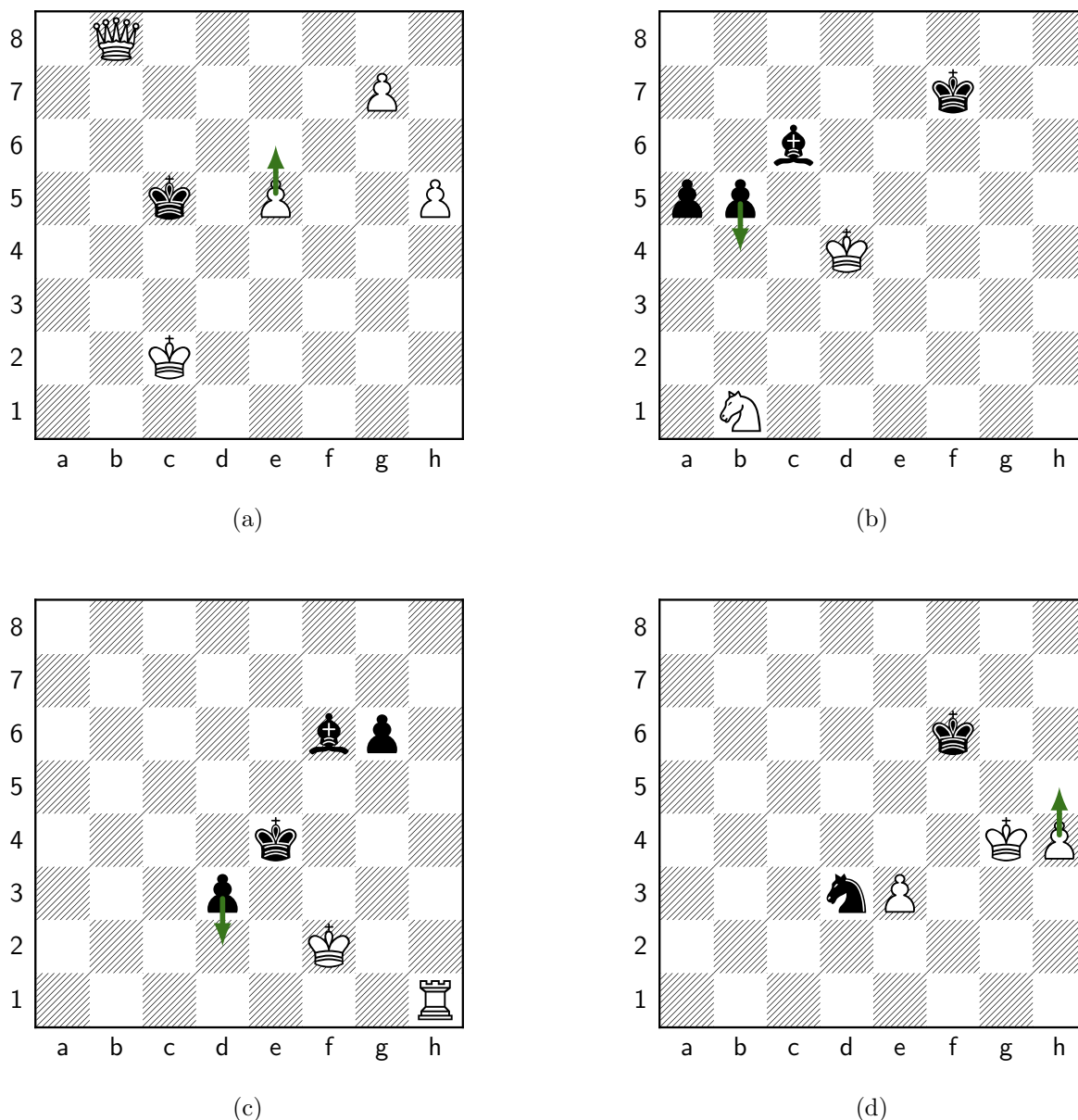
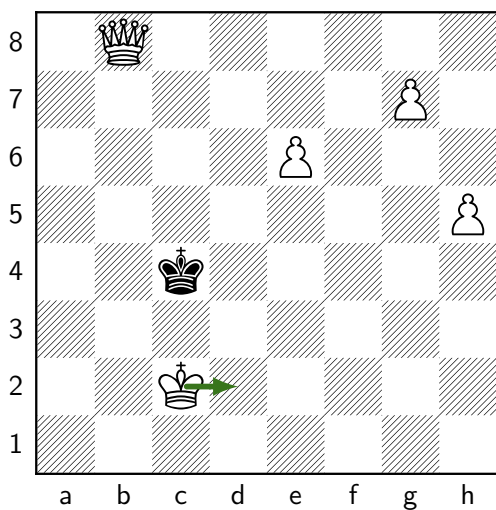
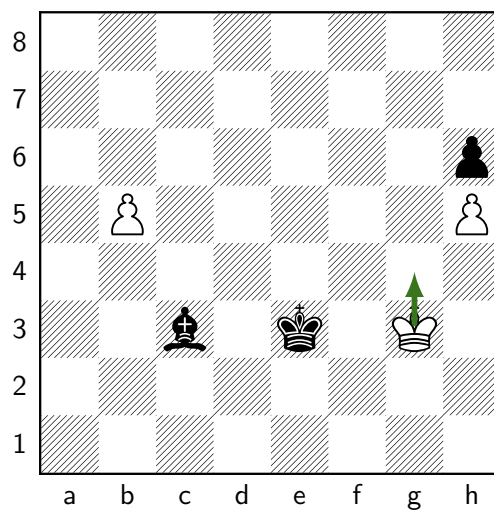


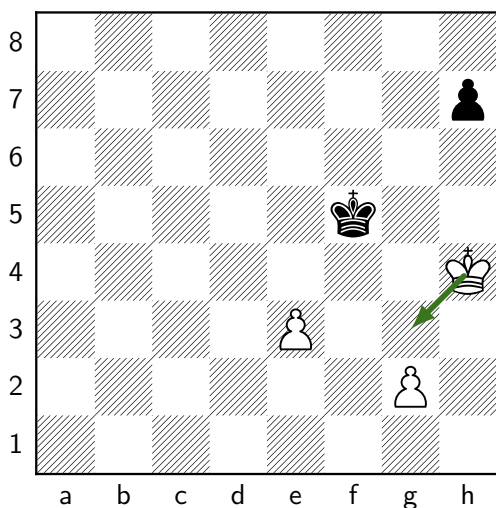
Figure 65: **Move 71, White: e6.** (a) Our position; Melanie plays **e6** (**e5e6**). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0003#319` (ply 145) [structural distance $d=9$, QAP obj=2]: the master played **b4**; under μ our *Pe5* plays the role of that game’s *Pb4*; cross-type roles: our *Qb8* as their *Bc6*, our *Pg7* as their *nb1*; transferred move **e5e6** (**endorses this move**). (c) Retrieved from `ccrl_top_p0003#1253` (ply 149) [structural distance $d=9$, QAP obj=0]: the master played **d2**; under μ our *Pe5* plays the role of that game’s *Pd6*; cross-type roles: our *Qb8* as their *Bf6*, our *Pg7* as their *rh1*; transferred move **e5e6** (**endorses this move**). (d) Retrieved from `ccrl_top_p0001#1415` (ply 140) [structural distance $d=9$, QAP obj=0]: the master played **h5**; under μ our *Pe5* plays the role of that game’s *Ph4*; cross-type roles: our *Ph5* as their *nd3*; transferred move **e5e6** (**endorses this move**). Synthesis: 4 retrieved analogs transferred to **e6** (summed QAP objective 3, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



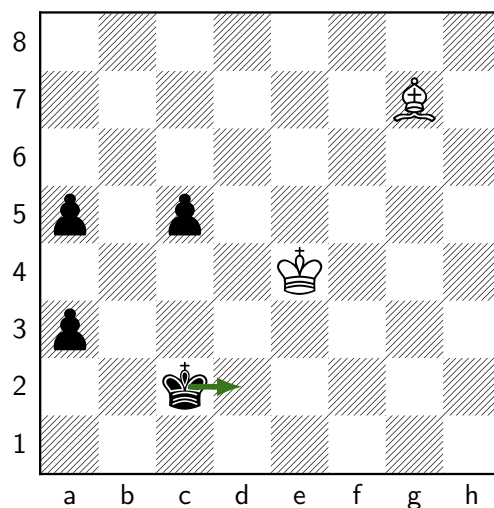
(a)



(b)

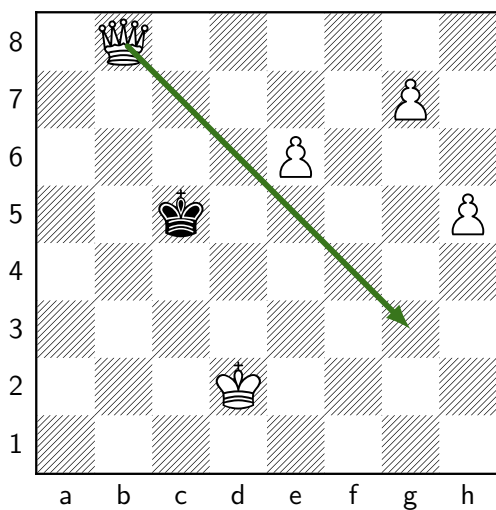


(c)

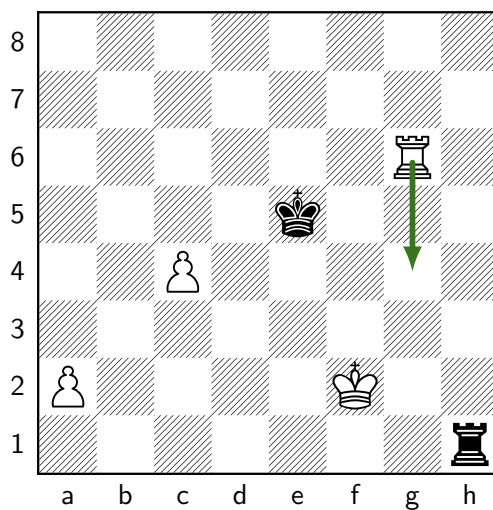


(d)

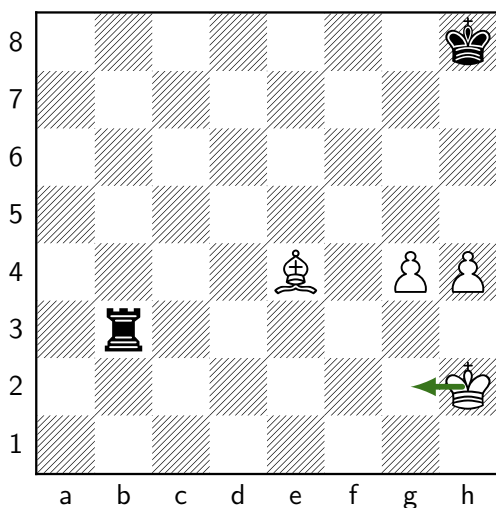
Figure 66: **Move 72, White: Kd2.** (a) Our position; Melanie plays Kd2 (c2d2). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#2485` (ply 126) [structural distance $d=5$, QAP obj=1]: the master played Kg4; under μ our *Kc2* plays the role of that game’s *Kg3*; cross-type roles: our *Pg7* as their *bc3*, our *Qb8* as their *Pb5*; transferred move c2d2 (**endorses this move**). (c) Retrieved from `ccr1_top_p0003#1379` (ply 150) [structural distance $d=5$, QAP obj=1]: the master played Kg3; under μ our *Kc2* plays the role of that game’s *Kh4*; cross-type roles: our *Qb8* as their *Pe3*; transferred move c2d2 (**endorses this move**). (d) Retrieved from `ccr1_top_p0002#99` (ply 127) [structural distance $d=5$, QAP obj=0]: the master played Kd2; under μ our *Kc2* plays the role of that game’s *Kc7*; cross-type roles: our *Pe6* as their *bg7*, our *Qb8* as their *Pc5*; transferred move c2d2 (**endorses this move**). Synthesis: 4 retrieved analogs transferred to Kd2 (summed QAP objective 2, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



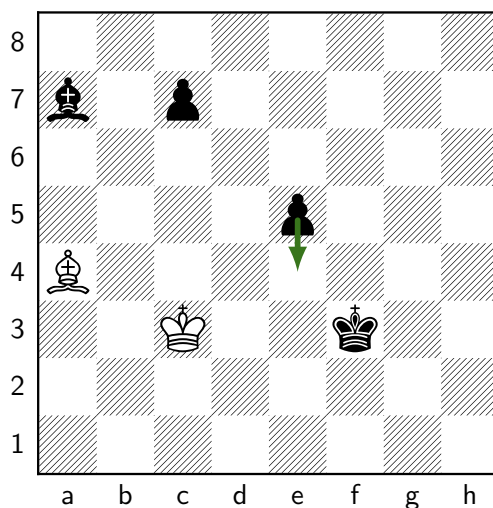
(a)



(b)

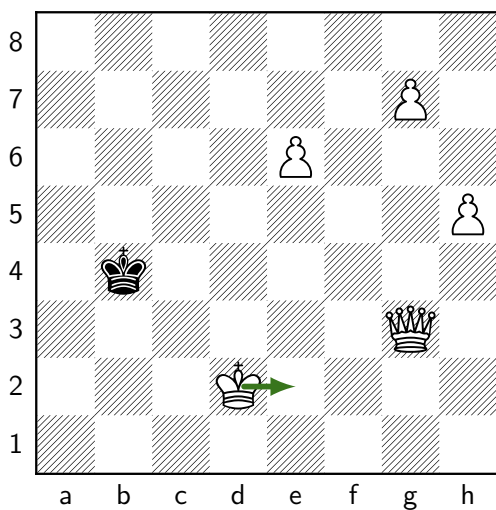


(c)

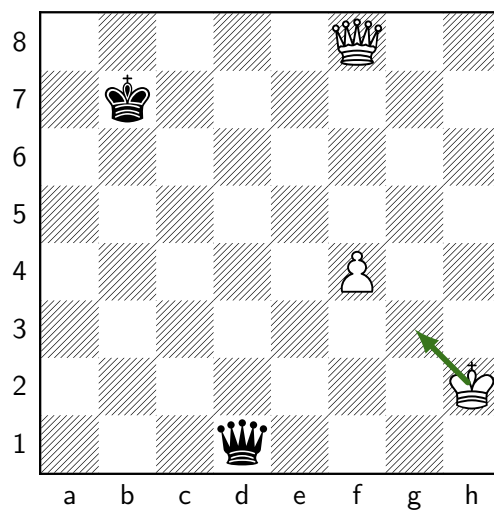


(d)

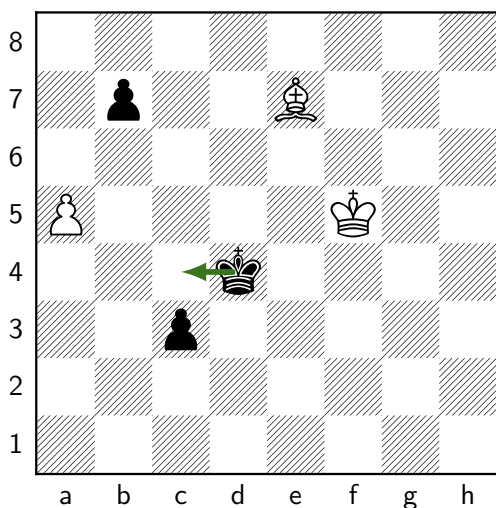
Figure 67: **Move 73, White: Qg3.** (a) Our position; Melanie plays Qg3 (b8g3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0001#157` (ply 130) [structural distance $d=6$, QAP obj=1]: the master played Rg4; under μ our Qb8 plays the role of that game’s Rg6 (a **cross-type** correspondence); cross-type roles: our Ph5 as their rh1; transferred move b8g3 (**endorses this move**). (c) Retrieved from `ccrl_top_p0000#3619` (ply 140) [structural distance $d=6$, QAP obj=0]: the master played Kg2; under μ our Kd2 plays the role of that game’s Kh2; cross-type roles: our Pe6 as their rb3, our Qb8 as their Be4; transferred move d2e2. (d) Retrieved from `ccrl_top_p0000#312` (ply 147) [structural distance $d=5$, QAP obj=0]: the master played e4; under μ our Qb8 plays the role of that game’s Pe4 (a **cross-type** correspondence); cross-type roles: our Ph5 as their Ba7, our Pg7 as their ba4; transferred move b8f4. Synthesis: 1 retrieved analog transferred to Qg3 (summed QAP objective 1, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



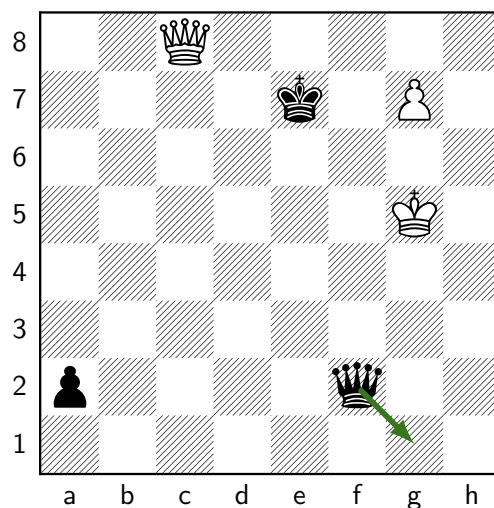
(a)



(b)

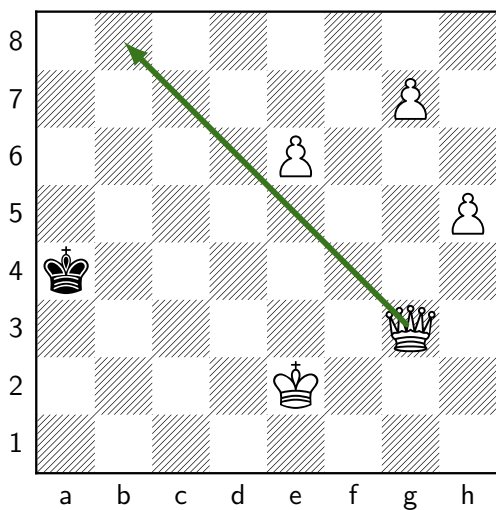


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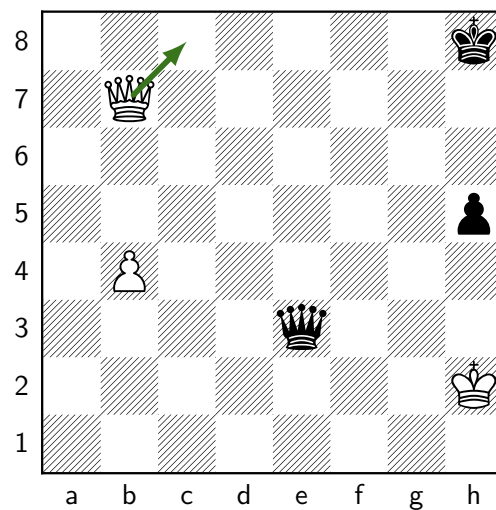


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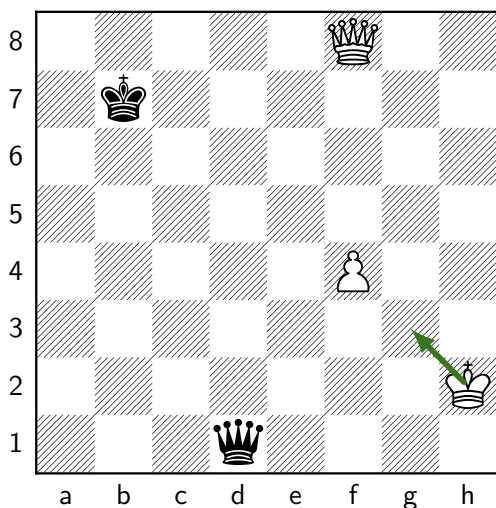
Figure 68: **Move 74, White: Ke2.** (a) Our position; Melanie plays Ke2 (d2e2). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#3594` (ply 158) [structural distance $d=7$, QAP obj=2]: the master played Kg3; under μ our *Kd2* plays the role of that game’s *Kh2*; cross-type roles: our *Ph5* as their *qd1*; transferred move d2e2 (**endorses this move**). (c) Retrieved from `ccr1_top_p0001#3303` (ply 135) [structural distance $d=9$, QAP obj=0]: the master played Kc4; under μ our *Kd2* plays the role of that game’s *Kd5*; cross-type roles: our *Qg3* as their *Pb7*, our *Ph5* as their *be7*; transferred move d2e2 (**endorses this move**). (d) Retrieved from `ccr1_top_p0003#2429` (ply 131) [structural distance $d=9$, QAP obj=2]: the master played Qg1+; under μ our *Qg3* plays the role of that game’s *Qf7*; cross-type roles: our *Ph5* as their *qc8*; transferred move g3b8. Synthesis: 2 retrieved analogs transferred to Ke2 (summed QAP objective 3, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



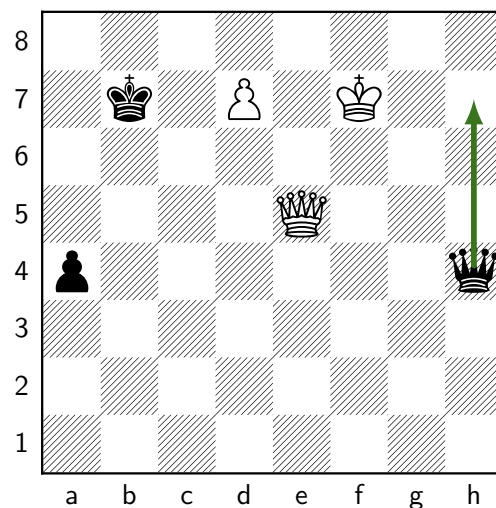
(a)



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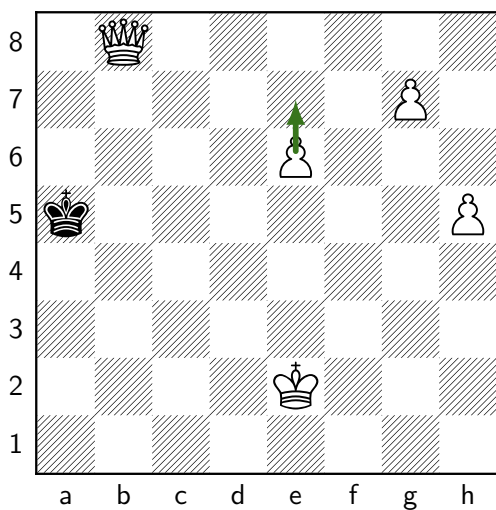


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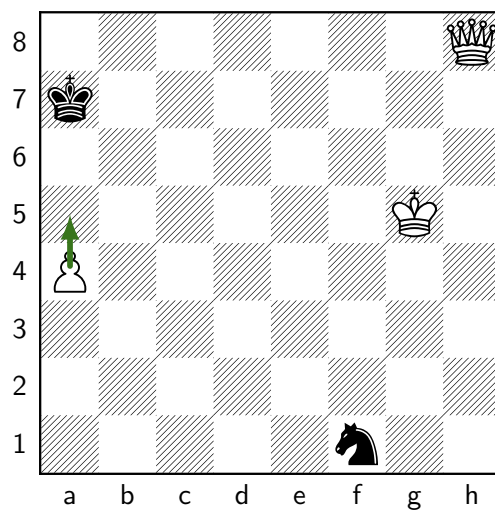


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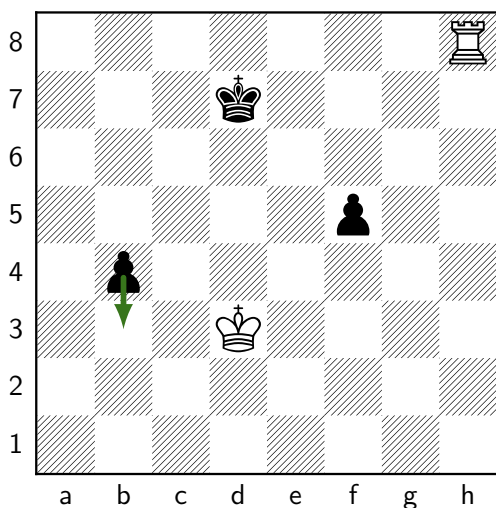
Figure 69: **Move 75, White: Qb8.** (a) Our position; Melanie plays Qb8 (g3b8). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#1845` (ply 132) [structural distance $d=8$, QAP obj=2]: the master played Qc8+; under μ our Qg3 plays the role of that game's Qb7; cross-type roles: our Ph5 as their qe3; transferred move g3b8 (**endorses this move**). (c) Retrieved from `ccr1_top_p0001#3594` (ply 158) [structural distance $d=7$, QAP obj=2]: the master played Kg3; under μ our Ke2 plays the role of that game's Kh2; cross-type roles: our Ph5 as their qd1; transferred move e2d2. (d) Retrieved from `ccr1_top_p0001#3920` (ply 133) [structural distance $d=8$, QAP obj=1]: the master played Qh7+; under μ our Qg3 plays the role of that game's Qh5; cross-type roles: our Pe6 as their qe5; transferred move g3g1. Synthesis: 1 retrieved analog transferred to Qb8 (summed QAP objective 2, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



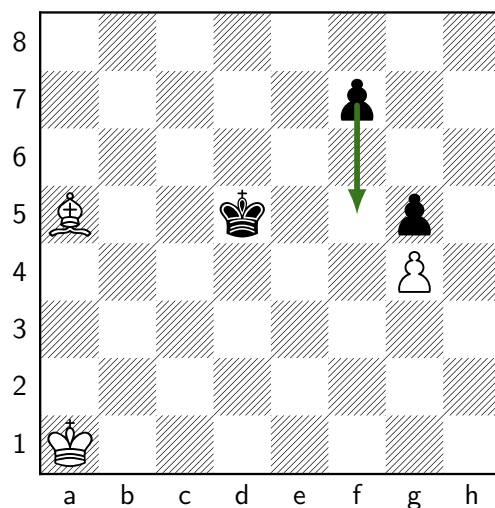
(a)



(b)

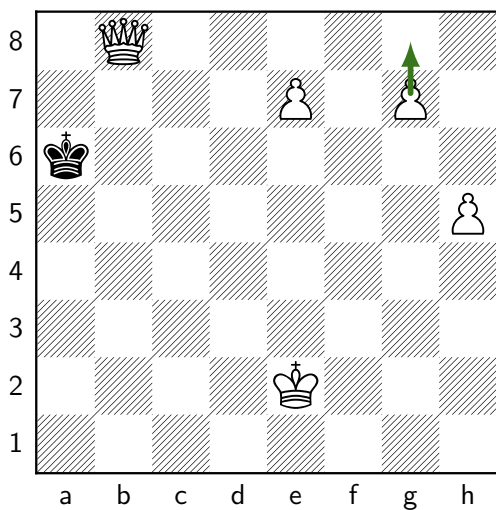


(c)

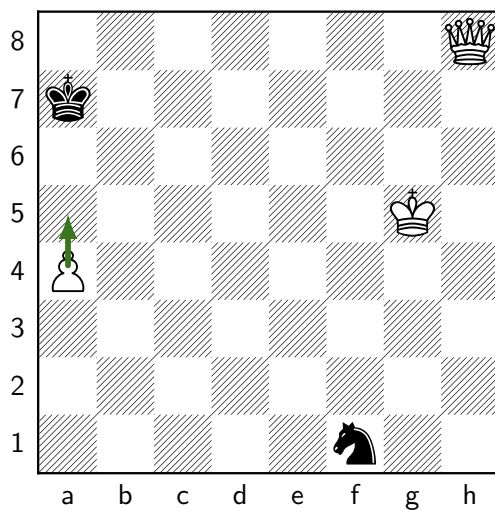


(d)

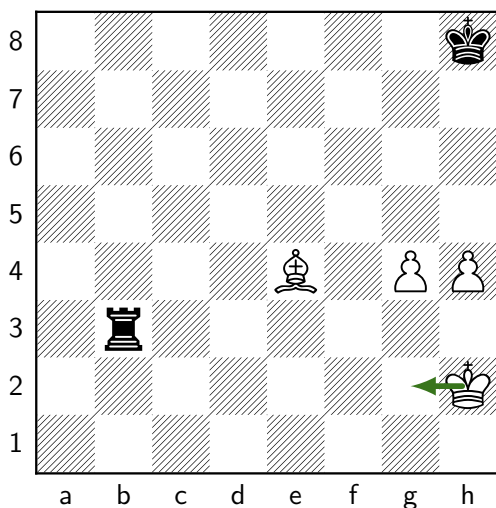
Figure 70: **Move 76, White: e7.** (a) Our position; Melanie plays **e7** (**e6e7**). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0000#914` (ply 140) [structural distance $d=6$, QAP obj=0]: the master played **a5**; under μ our *Pe6* plays the role of that game’s *Pa4*; cross-type roles: our *Ph5* as their *nf1*; transferred move **e6e7** (**endorses this move**). (c) Retrieved from `ccrl_top_p0003#1394` (ply 147) [structural distance $d=6$, QAP obj=0]: the master played **b3**; under μ our *Pe6* plays the role of that game’s *Pb5*; cross-type roles: our *Ph5* as their *rh8*, our *Qb8* as their *Pf5*; transferred move **e6e7** (**endorses this move**). (d) Retrieved from `ccrl_top_p0001#1531` (ply 161) [structural distance $d=5$, QAP obj=0]: the master played **f5**; under μ our *Pe6* plays the role of that game’s *Pf2*; cross-type roles: our *Pg7* as their *ba5*, our *Qb8* as their *Pg5*; transferred move **e6e7** (**endorses this move**). Synthesis: 3 retrieved analogs transferred to **e7** (summed QAP objective 0, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



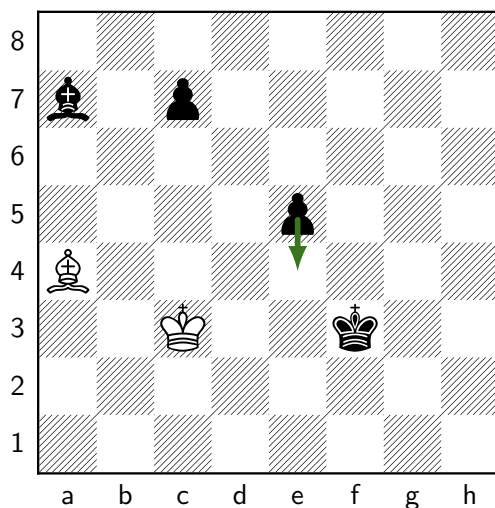
(a)



(b)

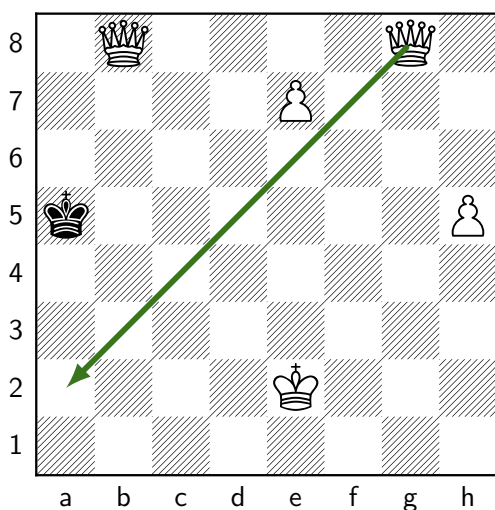


(c)

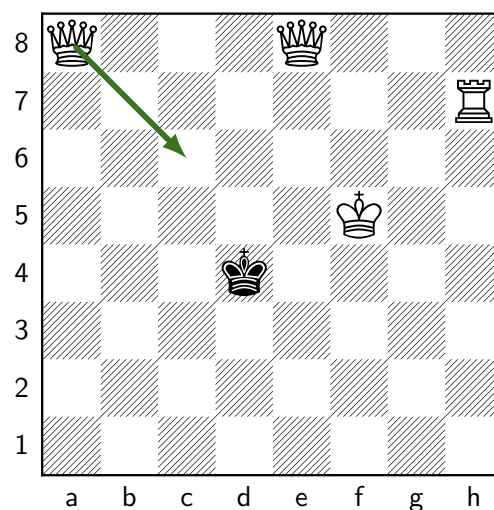


(d)

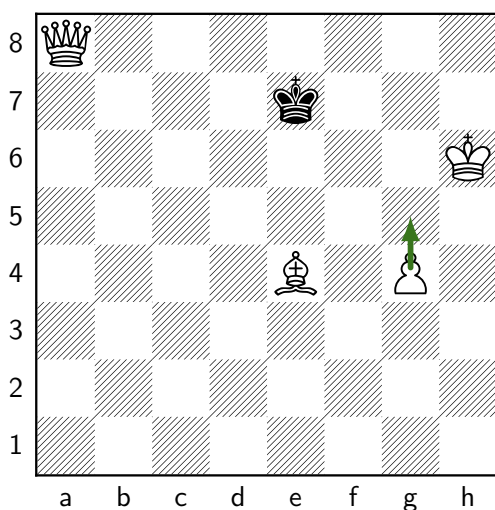
Figure 71: **Move 77, White: g8=Q.** (a) Our position; Melanie plays g8=Q (g7g8q). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from ccr1_top_p0000#914 (ply 140) [structural distance $d=6$, QAP obj=0]: the master played a5; under μ our Ph5 plays the role of that game’s Pa4; cross-type roles: our Pe7 as their nf1; transferred move h5h6. (c) Retrieved from ccr1_top_p0000#3619 (ply 140) [structural distance $d=6$, QAP obj=0]: the master played Kg2; under μ our Ke2 plays the role of that game’s Kh2; cross-type roles: our Ph5 as their rb3, our Qb8 as their Be4; transferred move e2f2. (d) Retrieved from ccr1_top_p0000#312 (ply 147) [structural distance $d=6$, QAP obj=0]: the master played e4; under μ our Qb8 plays the role of that game’s Pe4 (a **cross-type** correspondence); cross-type roles: our Ph5 as their Ba7, our Pg7 as their ba4; transferred move b8f4. Synthesis: no retrieved analog transferred to g8=Q – the search selector chose it on value (material / mate); panels (b)–(d) show the strongest nearby analogies.



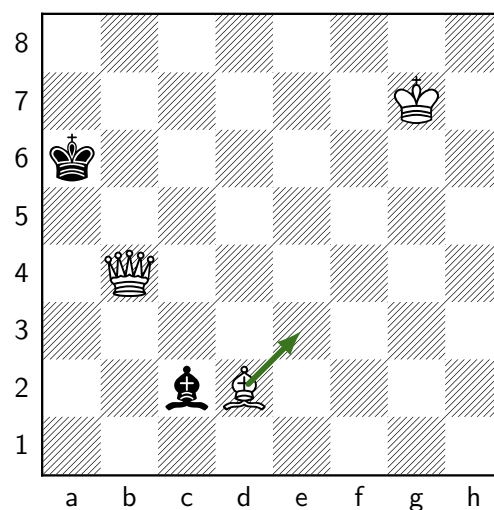
(a)



(b)



(c)



(d)

Figure 72: **Move 78, White: Qa2#**. (a) Our position; Melanie plays Qa2# (g8a2). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0003#323` (ply 156) [structural distance $d=6$, QAP obj=3]: the master played Qac6; under μ our Qg8 plays the role of that game's Qa8; cross-type roles: our Ph5 as their Rh7; transferred move g8c8. (c) Retrieved from `ccr1_top_p0000#3574` (ply 170) [structural distance $d=12$, QAP obj=3]: the master played g5; under μ our Ph5 plays the role of that game's Pg4; cross-type roles: our Qg8 as their Be4; transferred move h5h6. (d) Retrieved from `ccr1_top_p0003#38` (ply 146) [structural distance $d=17$, QAP obj=2]: the master played Be3; under μ our Qg8 plays the role of that game's Bd2 (a **cross-type** correspondence); cross-type roles: our Ph5 as their bc2; transferred move g8c4. Synthesis: no retrieved analog transferred to Qa2# – the search selector chose it on value (material / mate); panels (b)–(d) show the strongest nearby analogies.